Informatica® Business 360 Console
October 2026

Configure Data Quality

Informatica Business 360 Console Configure Data Quality
October 2026

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Preface

Follow the instructions in *Configure Data Quality* to associate data quality rules with the business entity fields to validate the field values.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

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You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Configuring data quality

You can configure data quality rule associations in MDM SaaS to validate, cleanse, transform, standardize, and enrich records in batch or real time. This process detects, assesses, and improves data quality by addressing issues, such as inaccuracies, inconsistencies, and missing values.

You can create the following types of rule associations:

- Chapter 3, “Basic rule associations” on page 13
- Chapter 4, “Advanced rule associations” on page 18
- Chapter 6, “DaaS rule associations” on page 23

You can create basic rule associations to validate and transform data based on predefined functions in MDM SaaS. For example, you can create a basic rule association to check if an Employee ID has the specified number of digits.

You can create advanced rule associations to use a predefined or custom rule specification from Cloud Data Quality. You can map the input and output fields of your business entity to the input and output fields of the rule specification. For example, you can create an advanced rule association to validate Social Security Numbers against industry standards using the predefined DUNS SSN Validation rule specification.

You can standardize and enrich addresses in batch or real time using DaaS rule associations. To standardize addresses in batch, MDM SaaS uses the predefined address verifier asset from Cloud Data Quality. To standardize addresses in real time, MDM SaaS uses the Informatica Address Verification as the DaaS data provider.

You can standardize and enrich email addresses and phone numbers in real time. For more information about configuring DaaS rule associations for email addresses and phone numbers, see “DaaS rule associations for email and phone verification” on page 28.

If you have complex scenarios that requires multiple rule associations to run in sequence, you can use the Enrichment and Validation Orchestrator. You can create rule associations that are equivalent to basic, advanced, and DaaS rule associations in an objective group, and orchestrate them to validate, transform, enrich, and cleanse and standardize data.

MDM SaaS categorizes these rule associations into different plugin categories, such as Validation, Transformation, Enrichment, and Cleansing and Standardization.

For more information about configuring objectives and adding rule associations for complex use cases, see Chapter 9, “Orchestrating data enrichment and validations” on page 34.

You can also enrich records with data from external data providers.

For more information about enriching data with external data providers, see Chapter 7, “Setting up external data providers for data enrichment” on page 30.

Additionally, you can also add integrity checks to validate business entity record fields.

For more information about configuring integrity checks, see Chapter 10, “Integrity checks” on page 144.

Configuring rule associations using objective groups

You can use Enrichment and Validation Orchestrator to configure data quality rule associations that are equivalent to the basic, advanced, and DaaS rule associations in an objective group through plugins.

The plugins are grouped into categories, such as Validation, Transformation, Enrichment, and Cleansing and Standardization. For example, to validate a date range and transform an input field value into upper case, you can use the Date Range validation plugin and the Uppercase transformation plugin.

MDM SaaS provides predefined data enhancement rules for creating rule associations that are equivalent to basic, advanced, and DaaS rule associations.

Note: Ensure that you don't edit or delete predefined data enhancement rules.

For example, your goal is to validate the first name field to check whether the value contains only the specified allowed characters. After validation, you want to transform the first name to uppercase and then verify and standardize the address values of records. To run all these steps sequentially, you can configure an objective group, add an objective, and include rule associations specific to your requirements in the desired order.

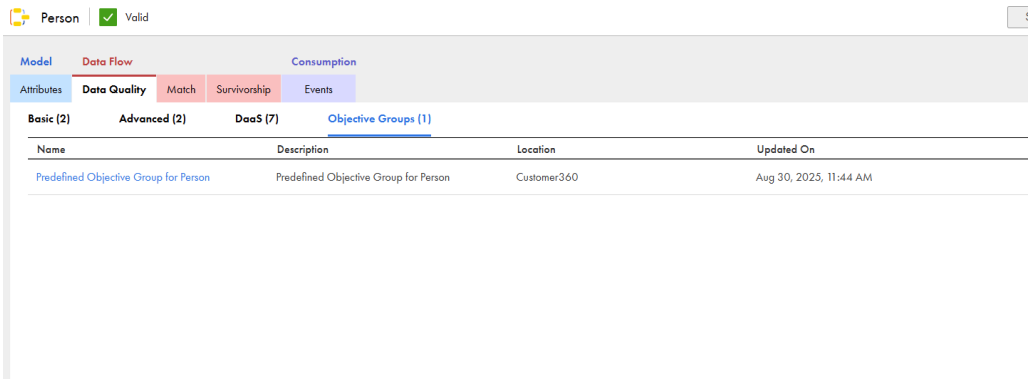
To achieve this goal, you can use the predefined data enhancement rules, such as Allowed Characters, Uppercase, and Address Verification. When MDM SaaS runs the objective, the rule associations are run in sequence.

Additionally, you can also configure objectives to trigger when you import records in bulk or add records in real time through business applications or REST APIs.

For more information about configuring objective groups, objectives, and adding rule associations to an objective, see Chapter 9, “Orchestrating data enrichment and validations” on page 34.

Consider the following differences in data quality rule associations between organizations created before and after the October 2025 release:

- MDM SaaS doesn't impact existing basic, advanced, and DaaS rule associations in organizations that were created before the October 2025 release. Additionally, you can also configure objective groups, objectives, and rule associations through plugins.
  If you create rule associations using objective groups, MDM SaaS uses Enrichment and Validation Orchestrator to run rule associations.
- When you create basic, advanced, and DaaS rule associations in organizations created after the October 2025 release, MDM SaaS uses Enrichment and Validation Orchestrator to run rule associations. MDM SaaS creates a Predefined Objective Group for <BusinessEntity_Name> on save of the business entity. The predefined objective group contains objectives and rule associations that are equivalent to the basic, advanced, and DaaS rule associations.
  The following image shows the predefined objective group for the Person business entity:



You can view the predefined objective group on the **Explore** page. You can't open, add, update, or delete the predefined objective group.

- When you create basic and advanced rule associations in organizations created after the October 2025 release, MDM SaaS creates objectives and rule associations using Enrichment and Validation Orchestrator. To delete these basic and advanced rule associations, you must be assigned the Designer or MDM Designer user roles.
- To create objectives and rule associations that are equivalent to the basic, advanced, and DaaS rule associations, ensure that you provide create and update permissions for Enrichment Provider, Enrichment Provider Mapping, and Enrichment Scenario Group assets in Administrator.
  Additionally, ensure that the folder that stores the business entity and rule associations have the required permission.

CHAPTER 2

Rules and guidelines for configuring data quality rule associations

Consider the following rules and guidelines when you configure data quality rule associations:

- When you configure data quality rules for picklists, MDM SaaS runs the data quality rules on the code values of the selected reference data. The rule validation considers only the code and not the name of the code value.
  For example, if you configure a basic rule association with a lowercase function for the Country picklist, MDM SaaS runs the rule association on the code IN. It doesn't run the rule association on the name India.
- When you delete or restore records through business applications, ingress, file import jobs, or REST APIs, MDM SaaS doesn't apply data quality rule associations on the field values.
- When you update a data quality rule association, MDM SaaS doesn't reprocess the existing records to apply the updated data quality rule association. To apply the updated rule association, edit the record.
- When you configure a data quality rule association for a field and the validation fails, you can downgrade the trust score of the field, and that helps determine the confidence level of the source data.
- Ensure that production and non-production keys are specified according to usage. The system enforces limited transactions and consumes based on usage.

CHAPTER 3

Basic rule associations

A basic rule association is an association between a business entity field and a simple condition-based rule. When you create basic rule associations, you configure conditions, which are predefined functions, such as Regular Match and Concatenate. Use basic rule associations for business entities involved in high concurrent cleansing transactions.

For example, Smith Store wants to master data from a newly acquired business. Smith Store wants to combine the first name and last name of each customer to generate the full name to print on receipts. Smith Store can associate a basic rule with the Full Name field and concatenate the first name and last name values.

You can also create basic rule associations to validate input fields that are empty. These rule associations aren't applicable if configured for required fields.

When you configure basic rule associations, you configure conditions that transform or validate field values. To set the conditions, you can use predefined validation functions or predefined transformation functions. A validation function validates whether data matches the specified condition. A transformation function transforms source data.

The following table describes the validation functions that you can use:

Function	Description
Maximum Length	Validates whether the length of a string is less than or equal to the specified length.
Minimum Length	Validates whether the length of a string is greater than or equal to the specified length.
Starts With	Validates whether a string starts with the specified string.
Ends With	Validates whether a string ends with the specified string.
No Spaces	Validates whether a string does not contain spaces.
Contains	Validates whether a value contains the specified string in the exact order.
Does Not Contain	Validates whether a value does not contain the specified string in the exact order.
Allowed Characters	Validates whether a value contains the specified allowed characters.
Not Allowed Characters	Validates whether a value contains the specified characters that are not allowed.
Regular Match	Validates whether a string matches the specified regular expression pattern. Business 360 Console follows the ECMA-262 standard for regular expression.

Function	Description
External Validation	Validates the field data based on an external validation rule that is associated to a Cloud Application Integration process. The Cloud Application Integration process uses external APIs to validate the data. Basic rule associations with external validation functions validate the field values of records that you create or update in real time through the user interface and REST API. For more information about creating a process, see *Cloud Application Integration*.
Date Range	Validates whether a date is in between two specified dates.
Greater Than Date	Validates whether a date is greater than the specified date.
Lesser Than Date	Validates whether a date is less than the specified date.
No Future Date	Validates whether a date is not a future date.
Maximum Decimal Scale	Validates whether the number of decimal places is less than or equal to the specified value.
Minimum Value	Validates whether a numerical value is greater than or equal to the specified value.
Maximum Value	Validates whether a numerical value is less than or equal to the specified value.
Maximum Precision	Validates whether the total number of digits in a decimal number is less than or equal to the specified value.

The following table describes the transformation functions that you can use:

Function	Description
Concatenate	Concatenates two or more strings.
Concatenate With Spaces	Concatenates two or more strings with spaces in between.
Uppercase	Converts lowercase string characters to uppercase.
Lowercase	Converts uppercase string characters to lowercase.
Title Case	Capitalizes the first character in each word of a string and converts all other characters to lowercase.
Left Trim	Removes spaces from the beginning of a string.
Right Trim	Removes spaces at the end of a string.
Replace	Replaces a character set in a string with another character set.

Rules and guidelines for basic rule associations

Consider the following rules and guidelines to create basic rule associations:

- Basic rule associations with a maximum or minimum value consider only the first 16 digits of numeric values.
  For example, if you specify 20 as the maximum value and the input is 20.09123456904589098743, the rule considers 20.09123456904589 and ignores the remaining digits.
- You can't create basic rule associations for boolean fields and dynamic fields.
- You can't create multiple basic rule associations for a field with the same validation or transformation function.
- When users enter value for a field and switch to a different field in a business application, basic rule associations are run.
- A basic rule association with an external validation function must use a validation rule associated with the Cloud Application Integration process. The validation rule must contain param<n> as the input field name, where n is the position of the input field. Otherwise, the rule association fails.
  For example, when you configure an external validation function for fields, such as First Name, Middle Name, and Last Name, the validation rule must contain param1, param2, and param3 as the input field names.
- When a field with an external validation function is updated in a business application, MDM SaaS sends the value of the field as the value of param1. For example, if the value of the First Name field with a basic rule association that uses an external validation function is updated, MDM SaaS sends the updated value as the value of param1.
- When you configure basic rule associations with functions, such as maximum length, minimum length, maximum precision, and maximum decimal scale, ensure that fields don't contain null or negative values. If fields contain null or negative values, MDM SaaS doesn't apply rule associations to these fields.
- When you configure basic rule associations with functions, such as maximum value, minimum value, data range, and regular match, and select business entity fields for validation, ensure that these fields aren't null.
  If inputs fields are null, MDM SaaS doesn't apply basic rule associations to these fields.
- Basic rule associations run sequentially in the order they are listed on the **Data Quality** tab of a business entity.

Adding basic rule associations

When you use a validation predefined function, ensure that you configure the validation message.

1. On the business entity page, select a field to associate rules.
2. In the **Properties** section, click **Data Quality** > **Basic**.
3. Click **Add Basic Rule Association**.

   A form appears.
4. In the **General** section, enter a unique name for the rule association.
5. Optionally, enter a description.
6. To validate inputs fields that are empty, enable **Run rule association even if all input fields are empty**.

7. On the **Condition** tab, perform the following tasks:
   a. Select a function.
   b. For the input field, select an option from the list.
   c. If you select **Select a Field**, select a business entity field from the list.
   d. If you select **Enter a Value**, type a value for validation.
8. If you select a validation function, on the **Validation** tab, perform the following tasks:
   a. Select one of the following severity levels for the validation failure:
      - Information. For minor data quality issues. Users can save the data.
      - Warning. For less severe data quality issues. Users can save the data.
      - Error. For severe data quality issues. Users can't save the data.

      If you enable error remediation for the business entity, users can save the records with validation errors irrespective of the severity level. However, the trust score is downgraded based on the trust score configuration that you specify.

      For more information about error remediation, see Business entity properties.

      Note: When you ingress or import data, the trust score is downgraded if the rule validation fails, irrespective of error remediation or severity levels.
   b. Enter an error message to display.
   c. Enter the percentage of trust score to downgrade when the validation fails.

      Note: Severity of the validation failure and trust score downgrade are applicable only when you run an ingress or file import job to import data.
9. To verify the created rule, perform the following tasks:
   a. Click **Test**.

      The **Test** dialog box appears.
   b. Enter values in the **Input** column, and click **Run**.

      The validation result appears in the **Output** column.
   c. Click **Close**.
10. Click **Save**.

Changing the execution order of basic rule associations

When you have multiple basic rule associations for a business entity field, set their order of execution to obtain clean and correct data. For example, if you want to concatenate the first name and last name of a person, you might need to validate the length of the strings first. In this case, you first run the rule association to validate the length of the names, and then run the rule association to concatenate the names. If a business entity field contains basic and advanced rule associations, the basic rule associations run first.

1. On the **Basic** tab, click **Change Order**.

   The **Change Order** dialog box appears.
2. Use the up and down arrows to change the order of the rule associations.

3. Click **OK**.

   The list of rule associations with the updated order of execution appears.

CHAPTER 4

Advanced rule associations

An advanced rule association is an association between business entity fields and a Data Quality rule specification.

You can use the predefined rule specifications, or create custom rule specifications in Data Quality and use them. When you create advanced rule associations, you must map the input and output fields of the rule specification with the business entity fields.

The supported predefined rule specification is DUNS SSN Validation that validates the DUNS and Social Security numbers.

You can create advanced rule associations to validate input fields that are empty. Based on the rule specification, the advanced rule association validates the input fields in the business application. If the input fields are empty, the rule returns a value for the empty fields.

For more information about creating or editing rule specifications in Cloud Data Quality, see *Rule specification assets* in *Cloud Data Quality*.

Note: When you add advanced rule associations to a field, ensure that you add the field to the custom pages that you use in your business application.

Rules and guidelines for advanced rule associations

Consider the following rules and guidelines to create advanced rule associations:

- Use simple advanced data quality rule associations instead of complex data quality rule associations.
- In an advanced rule association, don't use a rule that contains mapplet in the rule logic. The data quality rule association might not work.
- To ensure the success of the business entity field comparison operation in Business 360 Console, use integer or string values in the rule statement output in Cloud Data Quality.
- When you configure an advanced rule association with a transformation function, set rule statement output values for all the conditions in Cloud Data Quality to generate values in the output field.
- When you modify a rule specification in Cloud Data Quality, select the rule specification again or reset the fields of the advanced rule association in Business 360 Console. Otherwise, the changes to the rule specification aren't effective when the rule is executed.
- When you configure advanced rule associations for Date and Time fields, ensure that you select rule specifications that contain the `sysdate` expression.
  If you use a rule specification that contains the `current_timestamp` expression, MDM SaaS returns an older date and time.
- Enable DTM caching in mappings so that DTM doesn't initialize mappings every time.

- MDM SaaS runs the advanced rule associations concurrently after the basic rule associations are run.
- When you configure advanced rule associations for boolean fields with rule specifications in basic mode in Cloud Data Quality, ensure that the rule logic contains 0 and 1 as the boolean values within single quotes.
- When you add advanced rule associations that can run on input fields with no values, the rule associations run even if all the input fields that you add or update in real time, import through the ingress job, and those that you map and import through the file import job in MDM SaaS business applications are empty.
- If the existing advanced rule associations with transformation function don't run on the input fields with no values, you must enable the rule association even if all the input fields are empty.
- When you run advanced rule associations with validation function on input fields with no values, MDM SaaS displays errors in the transform step of the ingress and file import jobs. If you don't want to run the rule association even if all the input fields are empty, disable the rule associations to run on input fields with no values.
- When you configure advanced rule associations for boolean fields with rule specifications in advanced mode in Cloud Data Quality, ensure that the rule logic contains true and false as boolean values without single quotes.
- After you modify rule specifications in Cloud Data Quality, ensure that you redeploy the existing advanced rule associations that use the updated rule specifications. If you don't redeploy the advanced rule associations in Business 360 Console, they might not work as expected in real time.
  For more information about redeploying the advanced rule associations, see "Redeploying advanced rule associations" on page 21.
- In Cloud Data Quality, when you add a rule specification that contains an expression with the ISNULL function, the advanced rule associations that use the rule specification in Business 360 Console doesn't work as expected.
  To validate the input fields that have null values, use an expression in the following format:

  ```
  IIF((<Input_Field_Name>)=''OR ISNULL(<Input_Field_Name>),'INVALID','VALID')
  ```
- Advanced rule associations run on the fields even if the field contains only empty space.
- When you overwrite permissions for a rule specification or a folder that contains the rule specification, you can't add rule associations. The existing rule associations that are based on the rule specification might fail.
  Ensure that you revert the permissions for the rule specification asset or the folder that contains the rule specification asset.
- If you configure an advanced rule association for a field and run an ingress or file import job to update the other field values, the rule association runs on the field even though you don't modify the field value.

  For example, when you configure advanced rule associations for the first name field and run an ingress or file import job to update the middle name field value, the rule associations run on the first name field.

  However, if you update the value of the middle name field in real time or through REST APIs, the rule association doesn't run on the first name field.
- When users enter details for a particular section of a record and save the section in a business application, advanced rule associations are run.
- MDM SaaS applies advanced rule associations with transformation functions after you submit a record. However, MDM SaaS applies advanced rule associations with validation functions after you apply changes to each section of a record.
- You can't create advanced rule associations for dynamic fields.

Adding advanced rule associations

Use Cloud Data Quality rule specifications to add advanced rule associations for business entity fields.

1. On the business entity page, select a field to associate rules.
2. In the **Properties** section, click **Data Quality** > **Advanced**.
3. Click **Add Advanced Rule Association**.

   A form appears.
4. To validate inputs fields that are empty, enable **Run rule association even if all input fields are empty**.
5. To select a rule specification, perform the following tasks:
   a. Click the asset picker.

      The **Select an Asset** page appears.
   b. Select a rule specification, and click **Select**.
6. Enter a unique name for the rule association.
7. Optionally, enter a description.
8. On the **Input and Output Fields** tab, perform the following tasks:
   a. In the **Input Fields** section, select a business entity field or enter an input value for each rule specification field.
   b. In the **Output Fields** section, select a business entity field for each rule specification field.
9. On the **Validation** tab, perform the following tasks:
   a. In the **Rule Status** section, select a rule specification field that contains the validation status.
   b. Enter a validation status that indicates successful validation.

      Note: Ensure that the validation status matches the rule specification output value for successful validation.
   c. In the **Error Status** section, select one of the following options to define the severity of the validation failure:
      - Information. For minor data quality issues. Users can save the data.
      - Warning. For less severe data quality issues. Users can save the data.
      - Error. For severe data quality issues. Users can't save the data.

      If you enable error remediation for the business entity, users can save the records with validation errors irrespective of the severity level. However, the trust score is downgraded based on the trust score configuration that you specify.

      For more information about error remediation, see Business entity properties.

      Note: When you ingress or import data, the trust score is downgraded if the rule validation fails, irrespective of error remediation or severity levels.
   d. To configure the error message, select one of the following options:
      - From Rule Specification Field. Select a rule specification field that contains error message.
      - Custom Message. Enter a custom error message.
   e. Select a field to use to downgrade the trust score.

      Note: Trust score downgrade is applicable only when you ingress data.
   f. Enter the percentage of trust score to downgrade when the validation fails.
   g. Click **Save**.

Redeploying advanced rule associations

After you modify rule specifications in Cloud Data Quality, you must redeploy the existing advanced rule associations that use the updated rule specifications. If you don't redeploy the advanced rule associations in Business 360 Console, they might not work as expected in real time.

1. On the business entity page, select a field that's configured with an advanced rule association.
2. In the **Properties** section, click **Data Quality** > **Advanced**.
3. Click the advanced rule association.
4. To redeploy the rule association, perform one of the following tasks:
   - On the **Input and Output Fields** tab, remap input or output fields to a different business entity field, and save the rule association.
   - Edit the rule association to revert the input or output fields to the previous mapping, and save the rule association.
   - On the **Validation** tab, update a field value.
   - In the **Properties** section, hover over the rule association and, click **Delete**.
     After you delete the rule association, add the rule association again.
5. Save the business entity.

CHAPTER 5

Deleting rule associations

You can delete basic or advanced rule associations that you no longer need. You can delete rule associations for fields in the **Properties** section or on the **Data Quality** tab.

Note: You can't delete predefined basic and advanced rule associations.

1. On the business entity page, select a field that has rule associations.
2. To delete a basic rule association, click **Data Quality** > **Basic** in the **Properties** section.
3. Hover over the basic rule association to delete, and click **Delete**.
4. To delete a advanced rule association, click **Data Quality** > **Advanced** in the **Properties** section.
5. Hover over the advanced rule association to delete, and click **Delete**.

CHAPTER 6

DaaS rule associations

A DaaS rule association is an association between a field group and Informatica Data as a Service (DaaS). You can configure DaaS rule associations in batch or real-time. You can use DaaS rule associations to validate and enrich postal addresses in batches and real time. You can also use DaaS rule associations to validate email addresses and phone numbers in real time. After users add email addresses and phone numbers and submit the record, DaaS rule associations for real-time email and phone verification are run.

To validate and enrich addresses in batches, a DaaS rule association uses an address verifier asset from Cloud Data Quality. To validate and enrich addresses in real time, a DaaS rule association uses Informatica Address Verification. To validate email addresses in real time, a DaaS rule association uses Informatica Email Verification. To validate phone numbers in real time, a DaaS rule association uses Informatica Global Phone Validation.

Note: Ensure that you don't rename license file name in the Batch Address Verification section of Global Settings.

For more information about adding license details, see *Configure Global Settings*.

DaaS rule association for batch processing

To validate and enrich the address values in the records that you import in bulk, use the predefined address verifier asset from Cloud Data Quality. The predefined address verifier asset contains a predefined mapping of the input and output fields.

Note: You can't add a DaaS rule association to validate and enrich an email, phone, and custom field groups in batches.

Adding DaaS rule association to address field groups

You can verify and enrich the postal addresses in the records with the Cloud Data Quality address verifier. To validate the address data that you import in bulk, add DaaS rule associations to the field groups that store postal addresses.

1. On the business entity page, select a field group that stores postal addresses.
2. In the **Properties** section, click **Data Quality** > **DaaS Batch**.
3. To enable DaaS validation, select **Enable DaaS Validation**.

   Note: When you enable DaaS validation, you can't add basic and advanced rule associations to the fields in the field group.
4. Click **Configure Address Verification**.

5. In the **Verifier** field, view the preselected Address Verifier asset.

   Note: By default, the predefined address verifier asset named Address_Verifier_OOTB is selected. Ensure that you don't modify the predefined verifier asset. For more information about the address verifier asset, see the *Informatica Cloud Data Quality Online Help*.

6. On the **Input and Output Fields** tab, view the input and output fields that are mapped to the business entity fields.

   For more information about the input and output field mappings, see "Predefined mapping of input and output fields" on page 27.

7. On the **Validation** tab, perform the following tasks:

   a. In the **Error Status** section, select the severity of the validation failure.

   b. Enter a custom error message.

   c. In the **Trust Score** section, select the fields to use to downgrade the trust score.

   d. Enter the percentage of trust score to downgrade when the validation fails.

8. Click **Save**.

DaaS rule association for real-time processing

To validate and enrich addresses in real time, MDM SaaS uses Informatica Address Verification as the default DaaS provider. The address verification DaaS provider contains predefined input and output fields to use in DaaS rule associations. You can also enrich email addresses and phone numbers in real-time.

The predefined address field group comes configured with a default DaaS rule association and a mapping of the address verification DaaS provider fields and business entity fields. You can also create a DaaS rule association for custom field groups and map the input and output fields as required.

When you add or update address values of records through REST APIs, MDM SaaS chooses how to store the enriched address values. When you add DaaS rule associations, you can choose the enrichment management option to store the address values.

Guidelines for real-time address verification

Consider the following guidelines before you create a DaaS rule association for real-time address verification:

- You can't create a DaaS rule association or edit the predefined DaaS rule association for the predefined postal address field group.
- You can add one DaaS rule association at each level within nested custom field groups.
- Before you create a DaaS rule association, ensure that you save any changes to fields of the business entity.
- Ensure that you map a business entity field to the **Country** field of the Address Verification DaaS provider.

  Note: If you don't map any of the business entity fields to the Country field, the rule can't validate or enrich the values of the address fields in real time.
- Unlike email and phone validation, if users add an incorrect address in the business application, real-time address verification doesn't display the validation error during record creation. The validation error appears after the user submits the record.
- When you add a DaaS rule association for real-time address verification to a field group, ensure that block survivorship is enabled for the field group.

- After users enter address values and submit the record in business applications, DaaS rule associations for real-time address verification are run asynchronously.
- When you map the input and output fields of a dependent picklist to the data provider fields in the **Additional Fields** section, ensure that you map fields that belong to the same data type. If data types of the mapped fields don't match, DaaS rule associations might fail.
- When a DaaS rule association for real-time address verification returns an invalid address with status codes, such as `I0`, `I1`, `I2`, `I3`, `I4`, `I5`, and `I6`, MDM SaaS downgrades the trust score to 70%.
  The status codes indicate that the Address Verification DaaS data provider can't suggest a valid address, returns a partial address, or can't correct an address due to a single match.
- When you choose to create a patch record to save the enriched address values, MDM SaaS saves the enriched address values in the patch record and updates the master record with the enriched address. However, the source record contains the address that isn't enriched. If you use the Update Source Record Field REST API and copy the enriched address values from the master record to the source record, MDM SaaS reenriches the address values because they are different in the source and master records.
  To avoid MDM SaaS from reenriching the same address values, choose to save the enriched address values in the source record or copy the same address from the source record.

Adding a DaaS rule association

You can add a DaaS rule association to field groups in your business entity.

1. On the business entity page, select a postal address field group or any custom field group that stores postal addresses.
2. For a predefined postal address field group, perform the following steps:
   a. In the **Properties** section, click **Data Quality** > **DaaS Real-time**.

      You can view the predefined **<Business_Entity> Postal Address DaaS Rule**.

   b. To view the details of the DaaS rule, click the **<Business_Entity> Postal Address DaaS Rule** mapping name.

   c. In the **Field Mapping** section, view the mapping name and internal ID.

   d. In the **Rule Properties** section, enable the rule.

      By default, the rule is disabled.

   e. In the **Enrichment Management Options** section, choose how to store enriched address values of source records when records are created or updated through REST APIs.

      You can choose from the following enrichment management options:

      - Create a patch record for the address values. Creates or updates existing patch records with enriched address values. By default, MDM SaaS uses this option for existing DaaS rule associations.
      - Save the address values in the source record. Updates source records with enriched address values. By default, MDM SaaS uses this option for new DaaS rule associations.

      When you modify the enrichment management option for existing DaaS rule associations, MDM SaaS applies the change to records that you add and doesn't affect existing records. When you update address values of existing records through the user interface, MDM SaaS uses the **Create a patch record for the address values** enrichment management option. However, when you update address values of existing records through REST APIs, MDM SaaS uses the updated enrichment management option.

f. View a predefined rule that contains a default input and output field mapping.

For more information about the predefined input and output field mapping, see "Predefined mapping of input and output fields" on page 27.

3. For a custom field group that stores postal addresses, perform the following steps:

a. In the **Properties** section, click **Data Quality** > **DaaS**.

Note: Before you add a rule, ensure that you save the business entity.

b. Click **Add Rule**.

A form appears.

c. On the **Select Data Provider** tab, enter a unique name for the rule association.

The predefined rule type is **DaaS**.

d. To enable the rule, select **Enable Rule**.

e. In the **Enrichment Management Options** section, choose how to store enriched address values of source records when records are created or updated through REST APIs.

You can choose from the following enrichment management options:

- Create a patch record for the address values. Creates or updates existing patch records with enriched address values. By default, MDM SaaS uses this option for existing DaaS rule associations.
- Save the address values in the source record. Updates source records with enriched address values. By default, MDM SaaS uses this option for new DaaS rule associations.

When you modify the enrichment management option for existing DaaS rule associations, MDM SaaS applies the change to records that you add and doesn't affect existing records. When you update address values of existing records through the user interface, MDM SaaS uses the **Create a patch record for the address values** enrichment management option. However, when you update address values of existing records through REST APIs, MDM SaaS uses the updated enrichment management option.

f. Select a data provider to enrich the address values.

The supported data provider is **Address Verification DaaS**.

g. In the **Input and Output Fields of the Data Provider** section, view the input and output fields of the data provider.

h. Click **Next**.

i. On the **Map Input and Output Fields** tab, enter a unique name for the field mapping.

j. In the **Input Field Mapping** section, to map fields, drag the business entity fields to the data provider fields.

k. In the **Output Field Mapping** section, to map fields, drag the data provider fields to the business entity fields.

l. Click **Save**.

After you save the DaaS rule association, Business 360 Console automatically saves the business entity if you don't have any other changes in the business entity.

Predefined mapping of input and output fields

You can view a predefined input and output field mapping of the Address Verification DaaS data provider and business entity fields.

The following table lists the input fields of Address Verification DaaS data provider and their mapped business entity fields:

Address Verification DaaS Data Provider Field	Field Type	Mapped Business Entity Field
Address Elements > Country	Text	Country
Postal Code > Postal Code 1	Text	Postal Code
Locality > Locality 1	Text	City
Administrative Division > Administrative Division 1 > Extended	Text	County
Preformatted data > Postal Delivery Address Lines > Address Lines 1	Text	Address Line 1
Preformatted data > Postal Delivery Address Lines > Address Lines 2	Text	Address Line 2
Preformatted data > Postal Delivery Address Lines > Address Lines 3	Text	Address Line 3
Additional Fields > Text Field 1	Text	State
Additional Fields > Text Field 2	Text	Address Type
Additional Fields > Text Field 3	Text	Usage Type
Additional Fields > Text Field 4	Text	Address Status
Additional Fields > Text Field 5	Text	Postal Code Extension
Additional Fields > Boolean Field 1	Boolean	Default Indicator
Additional Fields > DateTime Field 1	DateTime	Start Date
Additional Fields > DateTime Field 2	DateTime	End Date

The following table lists the output fields of the business entity and their mapped Address Verification DaaS data provider fields:

Business Entity Field	Field Type	Mapped Address Verification DaaS Data Provider Field
Default Indicator	Boolean	Enrichments > Additional Fields > Boolean Field 1
Address Type	Text	Enrichments > Additional Fields > Text Field 2
Usage Type	Text	Enrichments > Additional Fields > Text Field 3
Address Status	Text	Enrichments > Additional Fields > Text Field 4

Business Entity Field	Field Type	Mapped Address Verification DaaS Data Provider Field
Address Line 1	Text	Preformatted data > Postal Delivery Address Lines > Address Lines 1
Address Line 2	Text	Preformatted data > Postal Delivery Address Lines > Address Lines 2
Address Line 3	Text	Preformatted data > Postal Delivery Address Lines > Address Lines 3
Country	Text	Address Elements > Country > Code
State	Text	Enrichments > Additional Fields > Text Field 1
City	Text	Address Elements > Locality > Locality 1
County	Text	Address Elements > Administrative Division > Administrative Division 2 > Extended
Postal Code	Text	Address Elements > Postal Code > Postal Code 1
Postal Code Extension	Text	Enrichments > Additional Fields > Text Field 5
Coordinate System	Text	Enrichments > Geo Codes > Arrival Point > Coordinates
Start Date	DateTime	Enrichments > Additional Fields > DateTime Field 1
End Date	DateTime	Enrichments > Additional Fields > DateTime Field 2
Latitude	Text	Enrichments > Geo Codes > Arrival Point > Latitude
Longitude	Text	Enrichments > Geo Codes > Arrival Point > Longitude
Enriched Indicator	Text	Enrichments > Status Codes > Enriched Indicator
Verification Status Code	Text	Enrichments > Status Codes > Verification Status Code
Verification Message	Text	Enrichments > Status Codes > Verification Status Message

DaaS rule associations for email and phone verification

You can validate email addresses and phone numbers for the records that you create in real time. You can verify whether an email address or phone number exists and is formatted correctly.

1. On the business entity page, select a field group that stores email addresses or phone numbers.
2. In the **Properties** section, click **DaaS**.
3. To enable DaaS validation, select **Enable DaaS Validation**.
4. Select the severity of the validation failure.

5. Enter an error message to display when the validation fails.
6. Enter the percentage of trust score to downgrade when the validation fails.
7. Click **Save**.

DaaS verification status

When you configure DaaS rule associations for batch or real-time processing, the DaaS rule association returns the status of the verification to the Enriched Indicator field.

The DaaS rule association that enriches addresses in real time contains a predefined Address Verification DaaS provider with a default mapping. In the default mapping, the Enriched Indicator output field of the address verification DaaS provider is mapped to the Enriched Indicator output field of the business entity. When you add field values in the address field group and submit a record in business applications, MDM SaaS enriches the address and displays the validation status in the Enriched Indicator field.

The DaaS rule association for batch processing that uses the predefined address verifier asset returns the result of the address verification in the Enriched Indicator field. However, unlike the real-time processing, the address verifer asset doesn't contain the Enriched Indicator field. The rule association returns the output of the address verification in the Enriched Indicator field based on the verification status codes from Cloud Data Quality.

If the result of the verification process returns status codes, such as `A`, `C`, and `V`, MDM SaaS displays the value as Valid in the Enriched Indicator field. If the verification process returns the status code `I`, MDM SaaS displays the value as Error in the Enriched Indicator field. If the verification status code is `N`, MDM SaaS displays the value as Not Validated in the Enriched Indicator field.

CHAPTER 7

Setting up external data providers for data enrichment

You can use the information from external data providers to enrich your records with additional information that's critical for your business.

You can integrate data providers with MDM SaaS to enrich records. You might need to purchase required licenses from data providers. After you get the licenses, import the packages that include necessary assets to set up data providers. The packages include instructions on how to import assets and set up data providers.
The following table describes the data providers that you can use:

Data Provider	Description	Information on procuring and setting up assets
Dun & Bradstreet	Enriches organization records with valuable information, such as financial performance of the organization, details of the parent or child organizations, and the key executives of the organization.	D&B Direct for Informatica
Global Legal Entity Identifier Foundation (GLEIF)	Provides the Global Legal Entity Identifier (LEI) Index that contains standardized and high quality legal entity reference data. Use the Global LEI Index to enrich your legal entity records.	Introducing Legal Entity 360 for Banking
MedPro	Enriches records with healthcare provider data from MedPro Systems.	Integrating Customer 360 with MedPro Systems
Custom data providers	Enriches records with data from external data providers.	“Configuring data enrichment” on page 51

CHAPTER 8

Migrate existing rule associations

To optimize performance and run faster validation and enrichment, you can migrate existing basic, advanced, and DaaS rule associations to use Enrichment and Validation Orchestrator. When you migrate these rule associations, MDM SaaS creates predefined objective groups for each business entity, objectives, and equivalent rule associations within those objectives.

Note: You can only migrate existing rule associations to use Enrichment and Validation Orchestrator in organizations.

You can migrate existing rule associations from the **Global Settings** page.

For more information about migrating existing rule associations, see *Configure Global Settings*.

If you have created organizations after the October 2025 release, MDM SaaS uses Enrichment and Validation Orchestrator by default to run rule associations.

When you choose to migrate rule associations, MDM SaaS runs the migrate rule association job. The job seamlessly migrates your basic, advanced, and DaaS rule associations to use Enrichment and Validation Orchestrator. The job automatically creates a predefined objective group for each business entity, objectives, and equivalent rule associations within those objectives.

MDM SaaS creates a Predefined Objective Group for <BusinessEntity_Name> to store all the migrated rule associations.

MDM SaaS creates an objective for all dependent rule associations, while independent rule associations are created as separate objectives. For example, consider a First Name field configured with a basic rule association with the uppercase function and an advanced rule association with the concatenate function that uses the output of the uppercase rule association. When you migrate the rule association, MDM SaaS includes these two rule associations within the same objective.

If there's an existing rule association with the same name as the migrated rule association, MDM SaaS appends 1 at the end of the rule association name. For example, if there's an existing rule association with the name Uppercase, MDM SaaS displays the migrated rule association name as Uppercase1. Additionally, you can't edit the migrated rule associations.

For more information about the migrate rule associations job, see *Monitoring migrate rule associations job* in Monitor MDM SaaS.

Rules and guidelines for migrating organizations to use Enrichment and Validation Orchestrator

Consider the following rules and guidelines when you migrate existing rule associations in your organization to use Enrichment and Validation Orchestrator:

- Informatica recommends that you first migrate your existing rule associations to use Enrichment and Validation Orchestrator in development and QA environments. After you complete testing the migrated rule associations, run the migrate rule associations job in production environments.
- Users with the MDM Designer user roles can migrate rule associations to use Enrichment and Validation Orchestrator.
- After you migrate existing basic, advanced, and DaaS rule associations to Enrichment and Validation Orchestrator and run ingress or file import jobs, MDM SaaS runs the migrated rule associations in the load step.
- When you import or ingress records with picklist values, MDM SaaS retains the values that you import even if they aren't available in the enterprise or custom code list.
  If you import records with picklists that have no or invalid values, MDM SaaS retains these records as invalid records. Additionally, if you use the Update Source Record Field REST API to update field values of the imported records, the response returns a validation error because the REST API validates the entire record, even if you update a different field value. To update the field values, ensure that you enable error remediation for the business entity.
- If you delete a business entity, Business 360 Console deletes the rule associations that are automatically generated. However, if you create rule associations, ensure that you manually delete them.
  Otherwise, you can't delete the business entity because of existing dependencies.
- When you configure advanced rule associations for decimal fields with a defined scale, ensure that the decimal values that you enter don't exceed the configured scale. If the values exceed the configured scale, after migration, Enrichment and Validation Orchestrator runs the rule associations and displays an exception on the **Data Quality Status** tab in business applications. However, the value is still saved to the record.
- After you migrate basic rule associations that use the concatenation function with a constant value, the equivalent rule association in Enrichment and Validation Orchestrator runs even if one of the input field values is null. The rule association concatenates the field values and applies the constant value as the output.
- When you import or ingress records without mapping input fields from migrated rule associations configured to run even if all input fields are empty, MDM SaaS runs these associations.
- After you migrate rule associations that are configured to return a default value when an input field is empty, the rule association runs to populate the empty fields with the specified default value. Subsequently, when you update records, the rule association runs only for the fields that you update.
- Before you migrate your organization to use Enrichment and Validation Orchestrator, ensure that you check out all the business entities in your organization. If you have already checked in some of the business entities, ensure that you check them out.
  If you don't check out the business entities, the migration job fails. If the job fails, you must check out the business entities that are already checked in and rerun the job.
- Before you run the migrate rule associations job in your organization, ensure that you don't perform the following actions:
  - Run any of the MDM SaaS jobs.
  - Update field values of records in business applications or using REST APIs.

- Modify properties of business entities.
- Modify existing data quality rule associations.
- Export and import assets between organizations.

The migrate rule associations job takes less than five minutes to complete. After the job completes, you can resume other actions.

- When you run an ingress or file import job after you migrate your rule associations to use Enrichment and Validation Orchestrator, MDM SaaS removes trailing zeros after the decimal point. For example, if you import `145.00`, MDM SaaS retains the value as `145`. This change applies only to decimal fields that have no rule associations configured.

CHAPTER 9

Orchestrating data enrichment and validations

When you have complex data enrichment and validation requirements, use Enrichment and Validation Orchestrator to orchestrate data enrichment and validations.

Instead of running validation and cleansing rule associations separately, you can use multiple enrichment and validation rule associations and orchestrate them to validate, cleanse, and enrich data in records.

For example, you work for a pharmaceutical company with significant presence in the US market. As part of your goal to sell your products in the European market, you want to ensure that your product data complies with the European regulations. When a user enters a National Drug Code, you want to validate the code. If the code is valid, you want to automatically generate the values for the Medicinal Product Package Identifier (PCID) and Medicinal Product Identifier (MPID) fields.

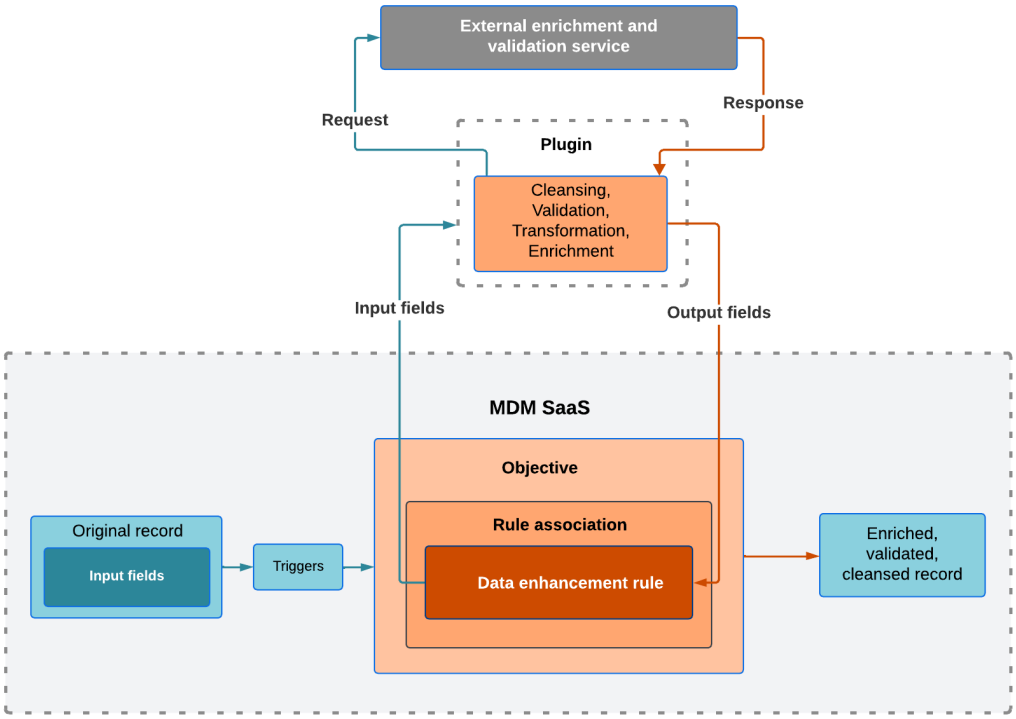
To accomplish this requirement, you can create a rule association for validating National Drug Codes, a rule association for generating Medicinal Product Package Identifiers, and another rule association for generating Medicinal Product Identifiers. You can run these rule associations sequentially and enable a rule association to use the output from another rule association as input.

You can configure enrichment and validations to run automatically. You can also control when to trigger enrichment and validations. For example, you can enrich data when you ingress records into MDM SaaS or when you create or update records in business applications or through REST APIs. You can also run enrichment and validations when the focus of a field changes or when a section is saved in business applications.

Enrichment and Validation Orchestrator architecture

The Enrichment and Validation Orchestrator architecture consists of enrichment and validation configurations that use an internal or external enrichment and validation service.

The following image shows a high-level data enrichment architecture:



When you enrich or validate records, MDM SaaS connects to an enrichment and validation service for enrichment data or validation results. The records are then enriched or validated based on the rule associations in MDM SaaS.

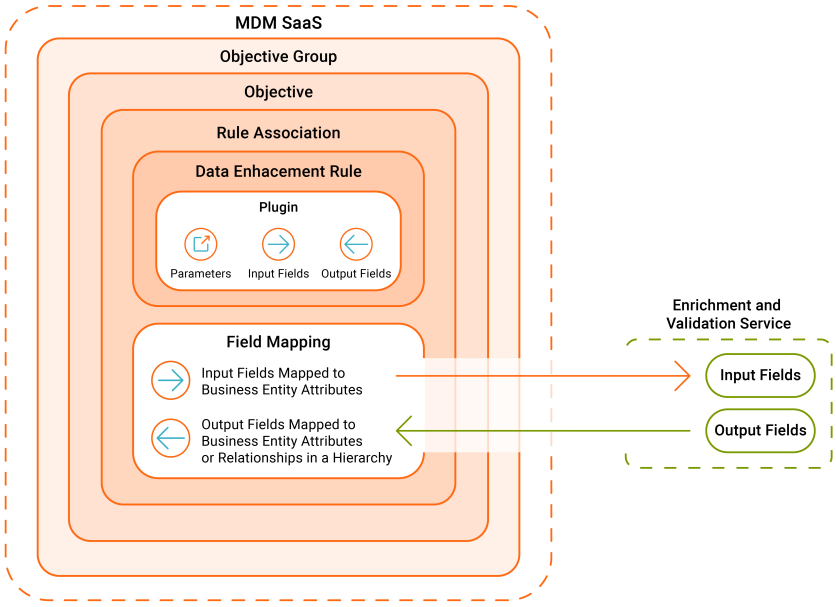
Components of enrichment and validation configuration

Enrichment and validation configurations include components, such as objective groups, objectives, rule associations, data enhancement rules, and plugins.

- Objective group. An asset that contains a set of objectives. You can group a set of objectives that help you to accomplish a data quality goal. You can also use an objective group to manage and migrate multiple objectives between organizations.
- Objective. An object within an objective group that contains a set of rule associations that run sequentially to accomplish a data quality requirement in a data quality goal.
- Rule association. An asset within an objective that associates a data enhancement rule with business entity fields to cleanse, enrich, or validate data.

- Data enhancement rule. An asset that determines how a plugin connects to an enrichment and validation service or function to enrich and validate data.
- Plugin. A component that connects to a service or a function that performs enrichment and validation.

The following image shows various components of data enrichment and validation configurations:



Enrichment and validation process

The enrichment and validation process uses objective groups, objectives, and rule associations to determine when and how records are validated, cleansed, or enriched.

Before you configure data enrichment or validation, determine the type of operations, such as validation, cleansing, or enrichment of records, that you want to perform.

You can validate, cleanse, or enrich records. The cleansing operation validates field values, cleanses the values, and enriches records with the cleansed data.

After you determine your data quality goal, you can create an objective group that represents the data quality goal. To the objective group, you can add actionable objectives that accomplish the data quality goal. You can use an objective group to keep all related objectives together. You can also use the objective group to conveniently migrate the objectives between organizations.

An objective represents a data quality requirement within an objective group. You can add multiple rule associations to an objective to run them sequentially to accomplish the data quality requirement. Additionally, you can enable data dependency between rule associations. Use an objective to determine when to trigger enrichment and validations. Also, an objective specifies the types of records to which enrichment and validations must apply. Objectives in an objective group run in parallel.

After you create an objective, you can add one or more rule associations to it. Use a rule association to map the business entity fields to the input fields of an enrichment and validation service or function. When data enrichment or validation is triggered, these input fields are sent to the specified service or function. To specify the fields that you want to enrich or cleanse, you also need to map the output fields of the enrichment and validation service or function to business entity fields. To assign records to a hierarchy, map the output fields to the relationship nodes.

To specify the type of operation to perform, configure and add a data enhancement rule to a rule association. In a data enhancement rule, specify a function or service that connects to an enrichment and validation service as a plugin. MDM SaaS runs a data enhancement rule through a plugin, which can send requests to an enrichment and validation service or function.

When the values of the input fields mapped in rule associations are updated, the associated objectives are triggered. The updated field values are sent to the specified enrichment and validation service or function that returns enriched or validated data. When the service or function returns enrichment or validation results, MDM SaaS updates the output fields with the results.

Configuring validations

You can create rule associations to validate field values. You can also perform cross-section validations.

For example, if a record includes a supplier name, you can check whether the record also contains the supplier contact details.

To set business logic for validations, you can create data enhancement rules with the validation plugins and use them in rule associations. A rule association configured with a validation plugin validates whether data matches the specified condition.

When you configure validations, you can choose to downgrade the trust scores of fields that fail validations by specifying the percentage of trust score to downgrade. To display validation errors in records, you can also configure error messages by specifying the error severity. For more information about how the error severity works with the option to reject records with validation errors, see "Rejecting records with validation errors" on page 99.

Note: Validation plugins return validation results and names of fields that fail validation but don't return any output fields.

The following table lists plugins that you can use for validations and the services that plugins use:

Plugin Name	Service that the plugin uses
Allowed Characters	MDM SaaS
Application Integration-based Validation	Application Integration
Contains	MDM SaaS
Data Quality - Validation	Data Quality
Date Range	MDM SaaS
Does not Contain	MDM SaaS
Ends With	MDM SaaS

Plugin Name	Service that the plugin uses
Greater Than Date	MDM SaaS
Lesser Than Date	MDM SaaS
Maximum Decimal Scale	MDM SaaS
Maximum Length	MDM SaaS
Maximum Precision	MDM SaaS
Maximum Value	MDM SaaS
Minimum Length	MDM SaaS
Minimum Value	MDM SaaS
No Future Date	MDM SaaS
No Space	MDM SaaS
Not Allowed Characters	MDM SaaS
Regular Match	MDM SaaS
Starts With	MDM SaaS

Note: You can't create rule associations with MDM SaaS validation plugins. However, you can use the predefined data enhancement rules to create rule associations. If you want to run rule associations even if all input fields are empty, you can create rule associations with validation plugins.

For more information about creating rule associations, see "Step 3. Creating rule associations" on page 116.

Allowed Characters

To validate whether a value contains the specified allowed characters, use the Allowed Characters plugin.

MDM SaaS contains the Allowed Characters predefined data enhancement rule. You can use the Allowed Characters data enhancement rule to create a rule association.

You can specify whether to consider the following characters in your input value:

- Uppercase
- Lowercase
- Digits
- Symbols
- Whitespaces

The following table lists the input fields and their data types:

Input Field	Data Type
Input	Text
UpperCase	Boolean
LowerCase	Boolean
Digits	Boolean
Symbols	Boolean
Whitespaces	Boolean

For example, to validate whether a product code business entity field contains a value in uppercase letters and digits, map the product code business entity field to the input field of the data enhancement rule. To check whether the value contains uppercase letters and digits, enter the value for input fields, such as uppercase and digits as true. For the other character input fields, you can either enter the value as false or leave the fields empty.

Note: Informatica recommends that you don't map business entity fields to the Uppercase, Lowercase, Digits, Symbols, and Whitespaces input fields. Instead, you can enter the value as true for the input fields that you want to consider for validation.

If you want to run the rule association even if all input field values are empty, create a data enhancement rule with the Allowed Character plugin.

The following table describes the property of the Allowed Characters plugin:

Property	Description
Run rule associations even if all input fields are empty	Indicates whether to run the rule association even if none of the input fields contain any value.

Application Integration-based Validation plugin

To validate data using an external data provider, use the Application Integration-based Validation plugin.

This plugin uses an Application Integration process to orchestrate data validation. For more information about creating a process for Application Integration-based validation, see "Payload formats for Application Integration-based validation" on page 75.

The following table describes the properties of the plugin:

Property	Description
CAI Process Identifier	Identifier of the Application Integration process that the plugin uses.
CAI Process Name	API name of the process that's specified in the CAI Process Identifier property.

Property	Description
Allow process with non-anonymous access	Indicates whether you can use a process that requires authentication. By default, only processes with anonymous access are allowed. When this property is enabled, you must specify a process that's configured with non-anonymous access. The property set for the selected process must match the property set in the data enhancement rule. For example, when this property is disabled, you can't save the data enhancement rule with a process configured for non-anonymous access. To use Application Integration processes that require authentication, the organization must use JWT or session ID authentication type. For more information, see *Authentication Properties* in the Administrator help.
Run as User	User name under which the Application Integration process runs. You can specify only one user. If the user you specify doesn't have access to the process, the rule association fails with the Exception status. If you don't specify a user, the process uses the credentials of the user who creates or updates a record through the user interface or REST API. You can't specify a user group in this property. If you configure the process with an allowed user group in Application Integration, you can leave this property blank. If the user who creates or updates records in a business application belongs to the group, the rule association runs.
Include only new or updated field group entries	Determines whether the input payload includes only new or updated field group entries. To send all entries of a field group to the Application Integration process, disable the property. For example, if you want to ensure that a record has at least one address entry, disable this property so that the validation service can count the entries in the Address section. **Note:** When you enable the option to include only new or updated field group entries in the payload, the associated rule association runs only if users add or update entries for the mapped field groups. It doesn't run even if users add or update a value only in an input root field without adding or updating the field group entries.
Run the selected process even when all input fields are empty	Indicates whether to run the Application Integration process when all input fields are empty. By default, when all input fields are empty, the status of the rule association is automatically set to successful without running the Application Integration process.

Contains

To validate whether a value contains the specified string in the exact order, use the Contains plugin.

MDM SaaS includes Contains and Contains for Case-sensitive predefined data enhancement rules. The Contains data enhancement rule validates the input value without considering case sensitivity. However, the Contains for Case-sensitive data enhancement rule considers case sensitivity when validating the input field value.

The predefined Contains and Contains for Case-sensitive data enhancement rules include the following input fields of the text data type:

- Input
- Substring

For example, to validate whether the order ID business entity field contains the string `ord` in its value, map the order ID business entity field to the input field of the data enhancement rule, and specify the substring input field value as `ord`.

If you want to consider case sensitivity for validation, use the Contains for Case-sensitive data enhancement rule.

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the Contains plugin.

The following table describes the properties of the Contains plugin:

Property	Description
Run the rule association even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.
Case sensitive	Specifies whether to consider case sensitivity when validating input values.

When you use the contains plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule association even if you don't pass any value for the input and substring fields. And, the data quality tab in business applications display the status of the rule association as Success.

Data Quality - Validation

To validate input field values with a rule specification configured in Data Quality, use the Data Quality - Validation plugin.

The following table describes the properties of the Data Quality - Validation plugin:

Property	Description
Rule Specification	Rule specification that's configured in Data Quality.
Run the rule association even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.
Rule Is Valid When Status Is	Validation status of the rule specification. Ensure that the validation status matches the rule specification output for a successful validation.
Rule Status Field	Field where the validation status appears. **Note:** Ensure that you provide the same rule set name that you configured for the rule specification in Data Quality.
Error Message Field	The rule specification field that contains error message. **Note:** Ensure that you provide the same rule set name that you configured for the rule specification in Data Quality.

Date Range

To validate whether a date is in between two specified dates, use the Date Range plugin.

MDM SaaS includes the Date Range predefined data enhancement rule.

The predefined Date Range data enhancement rule contains the following input fields of the calendar data type:

- Input
- Start
- End

For example, to validate whether the event date is between 2025/01/01 and 2025/04/25, map the event date business entity field to the input field of the data enhancement rule. You can enter 2025/01/01 and 2025/04/25 as values for the start and end input fields.

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the Date Range plugin.

The following table describes the properties of the plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

When you use the Date Range plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule association if you don't pass any value for the input, start, and end fields. Additionally, the data quality tab in business applications displays the status of the rule association as Exception.

Does Not Contain

To validate that a value doesn't contain specified characters in the exact order, use the Does Not Contain plugin.

MDM SaaS includes Does Not Contain and Does Not Contain for Case-sensitive predefined data enhancement rules. The Does Not Contain data enhancement rule validates the input value without considering case sensitivity. However, the Does Not Contain for Case-sensitive data enhancement rule considers case sensitivity when validating input field values.

The Does Not Contain and Does Not Contain for Case-sensitive predefined data enhancement rules contain the following input fields of text data type:

- Input
- Substring

For example, to validate that the product code business entity field doesn't contain the string `HAR` in its value, map the product code business entity field to the input field and specify the value of the substring input field as `HAR`.

To perform validation with case sensitivity, use the Does Not Contain for Case-sensitive predefined data enhancement rule.

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the Does Not Contain plugin.

The following table describes the properties of the Does Not Contain plugin:

Property	Description
Run the rule association even if all input fields are emtpy.	Indicates whether to run the rule association even if none of the input fields contain any value.
Case sensitive	Specifies whether to consider case sensitivity when validating input values.

Ends With

To validate whether a string ends with the specified string, use the Ends With plugin.

MDM SaaS includes the Ends With and Ends With for Case-sensitive predefined data enhancement rules. The Ends With data enhancement rule validates the input value without considering case sensitivity. However, the Ends With for Case-sensitive data enhancement rule considers case sensitivity when validating the input field value.

The predefined Ends With and Ends With for Case-sensitive data enhancement rules contain the following input fields of the text data type:

- Input
- Suffix

For example, to validate whether the value of the company name business entity field ends with the string `LTD`, specify the value of the suffix input field as LTD.

If you don't want to specify a value, you can map the suffix input field to a business entity field.

For example, to validate whether the company name business entity field value ends with the value from the company type business entity field, map the company type business entity field to the suffix input field.

If you want to run the rule association even if all input field values are empty, create a data enhancement rule.

The following table describes the properties of the Ends With plugin:

Property	Description
Run the rule association even if all input fields are emtpy.	Indicates whether to run the rule association even if none of the input fields contain any value.
Case sensitive	Specifies whether to consider case sensitivity when validating input values.

Greater Than Date

To validate whether a date is later than the specified date, use the Greater Than Date plugin.

MDM SaaS includes the Greater Than Date predefined data enhancement rule.

The predefined Greater Than Date data enhancement rule contains the following input fields of the calendar data type:

- Input
- Start

For example, to validate whether the employee joining date business entity field is later than the date of birth business entity, map the employee joining date business entity field to the input field and the date of birth business entity field to the start field.

However, if you want to validate whether the order date business entity field is later than 2025/03/28, map the order date business entity field to the input field and the enter the value of the start input field as 2025/03/28.

If you want to run the rule association even if all input field values are empty, create a data enhancement rule with the Greater Than Date plugin.

The following table describes the properties of the Greater Than Date plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

When you use the Greater Than Date plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule even if you don't pass any value for the input and start fields. Additionally, the data quality status tab in business applications displays the status of the rule association as Exception.

Lesser Than Date

To validate whether a date is earlier than the specified date, use the Lesser Than Date plugin.

MDM SaaS includes the Lesser Than Date predefined data enhancement rule.

The predefined Lesser Than Date data enhancement rule contains the following input fields of the calendar data type:

- Input
- End

For example, to validate whether the policy start date business entity field value is less than the policy end date business entity field value, map the policy start date business entity field to the input field and the policy end date business entity field to the end field.

However, if you want to validate whether the policy start date business entity field value is less than 2026/01/31, map the policy start date business entity to the input field and enter the value for the end field as 2026/01/31.

If you want to run the rule association even if all input field values are empty, create a data enhancement rule with the Lesser Than Date plugin.

The following table describes the properties of the Lesser Than Date plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

When you use the Lesser Than Date plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule association if you don't pass any value for the input and end fields. Additionally, the data quality tab in business applications displays the status of the rule association as Exception.

Maximum Decimal Scale

To validate whether the number of decimal places is less than or equal to the specified value, use the Maximum Decimal Scale plugin.

MDM SaaS includes the Maximum Decimal Scale predefined data enhancement rule.

The predefined Maximum Decimal Scale data enhancement rule contains the following input fields of the numeric data type:

- Input
- Scale

For example, to validate whether the product weight business entity field contains a maximum of one digit after the decimal point, map the product weight business field to the input field and specify the scale value as 1.

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the Maximum Decimal Scale plugin.

The following table describes the properties of the Maximum Decimal Scale plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

When you use the Maximum Decimal Scale plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule association if you don't pass any value for the input and scale fields. Additionally, the data quality tab in business applications displays the status of the rule association as Exception.

If the input field value is empty and the scale field contains a value, Enrichment and Validation Orchestrator runs the rule association and displays the status of the rule association as Success.

Maximum Length

To validate whether the length of a string is less than or equal to the specified length, use the Maximum Length plugin.

MDM SaaS includes the Maximum Length predefined data enhancement rule.

The following table lists the input fields and their data types:

Input Field	Data Type
Input	Text
Max	Numeric

For example, to validate whether the first name business entity field exceeds 50 characters, map the first name business entity field to the input field and enter the value for the max input field as 50.

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the Maximum Length plugin.

The following table describes the properties of the Maximum Length plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

When you use the Maximum Length plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule association if you don't pass any value for the input and max fields. Additionally, the data quality tab in business applications displays the status of the rule association as Exception.

Maximum Precision

To validate whether the total number of digits in a decimal number is less than or equal to the specified value, use the Maximum Precision plugin.

MDM SaaS includes the Maximum Precision predefined data enhancement rule.

The predefined Maximum Precision data enhancement rule contains the following input fields of the numeric data type:

- Input
- Digits

For example, to validate whether the product price business entity field exceeds five digits, map the product price business entity field to the input field and enter the digits input field value as five.

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the Maximum Precision plugin.

The following table describes the properties of the Maximum Precision plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

When you use the Maximum Precision plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule association if you don't pass any value for the input and digits fields. Additionally, the data quality tab in business applications displays the status of the rule association as Exception.

Maximum Value

To validate whether a numerical value is greater than or equal to the specified value, use the Maximum Value plugin.

MDM SaaS provides the Maximum Value predefined data enhancement rule.

The following table lists the input fields and their data types:

Input Field	Data Type
Input	Text
Max	Numeric

For example, to validate whether the age business entity field is greater than or equal to 40, map the age business entity to the input field and enter max field as 40.

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the Maximum Value plugin.

The following table describes the properties of the Maximum Value plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

When you use the Maximum Value plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule association if you don't pass any value for the input and max fields. Additionally, the data quality tab in business applications displays the status of the rule association as Exception.

Minimum Length

To validate that the length of a string is equal to or greater than the specified length, use the Minimum Length plugin.

MDM SaaS includes the Minimum Length predefined data enhancement rule.

The following table lists the input fields and their data types:

Input Field	Data Type
Input	Text
Min	Numeric

For example, to validate that the first name business entity field is within five characters, map the first name business entity field to the input field and enter the min field value as five.

If you want to run the rule association even if all input field values are empty, create a data enhancement rule with the Minimum Length plugin.

The following table describes the properties of the Minimum Length plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

When you use the Minimum Length plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule association if you don't pass any value for the input and min fields. Additionally, the data quality tab in business applications displays the status of the rule association as Exception.

If the input field value is empty and the min field contains a value, Enrichment and Validation Orchestrator skips the rule association. Additionally, the data quality tab in business applications displays the status of the rule association as Failed.

Minimum Value

To validate whether a numerical value is less than or equal to the specified value, use the Minimum Value plugin.

MDM SaaS includes the Minimum Value predefined data enhancement rule.

The predefined Minimum Value data enhancement rule contains the following input fields of the numeric data type:

- Input
- Min

For example, to validate whether the order quantity business entity field is less than or equal to 10, map the order quantity business entity field to the input field and enter the min field as 10.

If you want to run the rule association even if all input field values are empty, create a data enhancement rule with the Minimum Value plugin.

The following table describes the properties of the Minimum Value plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

When you use the Minimum Value plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule association if you don't pass any value for the input and min fields. Additionally, the data quality tab in business applications displays the status of the rule association as Exception.

If the input field value is empty and the min field contains a value, Enrichment and Validation Orchestrator skips the rule association. Additionally, the data quality tab in business applications displays the status of the rule association as Failed.

No Future Date

To validate that a date isn't a future date, use the No Future Date plugin.

MDM SaaS includes the No Future Date predefined data enhancement rule.

The No Future Date predefined data enhancement rule contains only one input field. You can map a business entity field to the input field.

If you want to run the rule association even if all input field values are empty, create a data enhancement rule with the No Future Date plugin.

The following table describes the properties of the No Future Date plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

No Space

To validate that a string doesn't contain spaces, use the No Space plugin.

MDM SaaS includes the No Space predefined data enhancement rule.

The No Spaces data enhancement rule contains only one input field. You can map a business entity field to the input field. For example, to validate that the employee ID business entity field doesn't contain spaces, map the employee ID business entity field to the input field.

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the No Space plugin.

The following table describes the properties of the No Space plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

Not Allowed Characters

To validate that a value doesn't contain characters that aren't allowed, use the Not Allowed Characters plugin.

MDM SaaS includes the Not Allowed Characters predefined data enhancement rule.

You can specify whether to consider the following characters in your input value:

- Uppercase
- Lowercase
- Digits
- Symbols
- Whitespaces

The following table lists the input fields and their data types:

Input Field	Data Type
Input	Text
UpperCase	Boolean
LowerCase	Boolean
Digits	Boolean
Symbols	Boolean
Whitespaces	Boolean

For example, to validate that the user name business entity field doesn't contain whitespaces and symbols, map the user name business entity field to the input field. You can enter the value for the whitespaces and symbols input fields as true. For the other input fields of other characters, you can either enter the value as false or leave the fields empty.

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the Not Allowed Characters plugin.

The following table describes the properties of the Not Allowed Characters plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

Regular Match

To validate whether a string matches the specified regular expression pattern, use the Regular Match plugin.

MDM SaaS includes the Regular Match predefined data enhancement rule.

The predefined Regular Match data enhancement rule contains the following input fields of the text data type:

- Input
- Regex

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the Regular Match plugin.

The following table describes the properties of the Regular Match plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

When you use the Regular Match plugin and create a data enhancement rule by enabling the rule to run even if all input fields are empty, Enrichment and Validation Orchestrator runs the rule association even if you don't pass any value for the input and regex fields. Additionally, the data quality tab in business applications displays the status of the rule association as Exception.

Starts With

To validate whether a string starts with the specified string, use the Starts With plugin.

MDM SaaS includes the Starts With and Starts With for Case-sensitive predefined data enhancement rules. The Starts With data enhancement rule validates the input value without considering case sensitivity. However, the Starts With for Case-sensitive data enhancement rule considers case sensitivity when validating the input field value.

The predefined Starts With and Starts With for Case-sensitive data enhancement rules contain the following input fields of the text data type:

- Input
- Prefix

For example, to validate whether the company name business entity field value starts with the string `ARC`, specify the value of the prefix field `ARC`.

If you want to run the rule association even if all input field values are empty, you must create a data enhancement rule with the Starts With plugin.

The following table describes the properties of the Starts With plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.
Case sensitive	Specifies whether to consider case sensitivity when validating input values.

Configuring data enrichment

To enrich records with additional data from data providers or to transform data in records, configure rule associations that use enrichment or transformation plugins.

You can enrich records by transforming the data within them.

To perform advanced data enrichment, you can also connect to external data providers and enrich records with the data that they return. For example, you can connect to OpenAI ChatGPT to translate the product description in an Item record.

Note: Enrichment is triggered and applied to a record even users don't have access to certain fields involved in the enrichment process.

The following table lists the plugin that you can use to configure data enrichment:

Plugin Name	Type	Service that the plugin uses
Application Integration-based Enrichment	Enrichment	Application Integration
Data Quality-based Transformation	Transformation	Data Quality
Title Case	Transformation	MDM SaaS
Concatenate	Transformation	MDM SaaS
Concatenate with Spaces	Transformation	MDM SaaS
Left Trim	Transformation	MDM SaaS
Lowercase	Transformation	MDM SaaS
Replace	Transformation	MDM SaaS
Right Trim	Transformation	MDM SaaS
Uppercase	Transformation	MDM SaaS

Note: You can't create rule associations that use MDM SaaS transformation plugins. However, you can only use predefined data enhancement rules to create rule associations. If you want to run rule associations even if all input fields are empty, you can create rule associations with transformation plugins.

Application Integration-based Enrichment plugin

To enrich records with data from external data providers, use the Application Integration-based Enrichment plugin and configure data enrichment.

When a rule association with a data enhancement rule that uses this plugin is triggered, the input fields are sent to an Application Integration process, which connects to an external data provider. When the data provider returns data for enrichment, you can automatically enrich or assign records to a hierarchy. You can also display the enrichment data as recommendations in business applications.

When you configure data enrichment, you can choose to enrich specific fields of records. For example, you can send a product description to an external data provider and extract the brand name from the description.

You can also add records to hierarchies based on the data from an external data provider. For example, to assign product records to a hierarchy, you can send a product description to an AI-powered external data provider to get relevant product categories. Based on the product categories that the data provider returns and the existing category records, you can either directly assign the records to a hierarchy or present the category records as recommendations to users.

This plugin uses an Application Integration process to perform enrichment. For more information about creating a process for Application Integration-based enrichment, see "Payload formats for Application Integration-based enrichment" on page 70.

The following table lists the properties of the plugin:

Property	Description
CAI Process Identifier	Identifier of the Application Integration process that the plugin uses.
CAI Process Name	The API name of the process that's specified in the CAI Process Identifier property.
Allow process with non-anonymous access	Indicates whether you can use a process that requires authentication. By default, only processes with anonymous access are allowed. When this property is enabled, you must specify a process that's configured with non-anonymous access. The property set for the selected process must match the property set in the data enhancement rule. For example, when this property is disabled, you can't save the data enhancement rule with a process configured for non-anonymous access. To use Application Integration processes that require authentication, the organization must use JWT or session ID authentication type. For more information, see *Authentication Properties* in the Administrator help.
Run as User	User name under which the Application Integration process runs. You can specify only one user. If the user you specify doesn't have access to the process, the rule association fails with the Exception status. If you don't specify a user, the process uses the credentials of the user who creates or updates a record through the user interface or REST API. You can't specify a user group in this property. If you configure the process with an allowed user group in Application Integration, you can leave this property blank. If the user who creates or updates records in a business application belongs to the group, the rule association runs.
Accept Threshold	The confidence score above which data from the data provider is automatically accepted and applied. To enable rule associations to enrich records automatically, enter 0 in the Accept Threshold and Reject Threshold fields. To display the information from the data provider as CLAIRE recommendations, specify the accept and reject thresholds with a value greater than 0. For more information about the accept threshold, see "Determining the data enrichment type" on page 53.

Property	Description
Reject Threshold	The confidence score below which recommendations for enrichment are automatically rejected. For more information about the reject threshold, see "Determining the data enrichment type" on page 53.
Confidence Score Field	The name of the field that contains the confidence scores from the data provider. To display the confidence scores along with the recommendations in business applications, specify the name of the field. You can get the field name from the Output Fields tab. If you don't specify the name of the field that contains the confidence scores, CLAIRE displays all recommendations without applying the accept and reject thresholds.
Include only new or updated field group entries	Determines whether the input payload includes only new or updated field group entries. To send all entries of a field group to the Application Integration process, disable the property. For example, if you want to ensure that a record has at least one address entry, disable this property so that the validation service can count the entries in the Address section. **Note:** When you enable the option to include only new or updated field group entries in the payload, the associated rule association runs only if users add or update entries for the mapped field groups. It doesn't run even if users add or update a value only in an input root field without adding or updating the field group entries.
Run the selected process even when all input fields are empty	Indicates whether to run the Application Integration process when all input fields are empty. By default, when all input fields are empty, the status of the rule association is automatically set to successful without running the Application Integration process.

Determining the data enrichment type

You can configure rule associations to automatically enrich records or display enrichment data as recommendations.

When you configure rule associations for Application Integration-based enrichment, choose one of the following enrichment types:

- Automatic enrichment. MDM SaaS applies enrichment data to records without user intervention.
- CLAIRE recommendations. CLAIRE displays enrichment data as suggestions that users can accept or reject based on confidence scores from data providers.

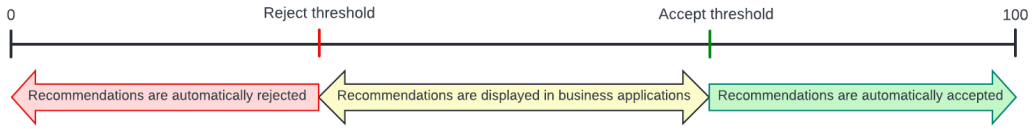
Note: You can use rule associations that are configured to display CLAIRE recommendations only in objectives for master records. MDM SaaS doesn't display CLAIRE recommendations for source records.

Accept and reject thresholds behavior

To control how CLAIRE handles recommendations from data providers, you can set accept and reject thresholds in a data enhancement rule during data enrichment configuration.

For example, you send a product description to an AI-powered data provider and request brand name suggestions. You set the accept threshold to 90 and the reject threshold to 40. The data provider returns three brand name suggestions with confidence scores of 95, 65, and 20. CLAIRE accepts the suggestion with a score of 95, displays the suggestion with a score of 65 as a recommendation for users to accept or reject, and rejects the suggestion with a score of 20.

The following image shows how the accept and reject thresholds impact the recommendations from a data provider:



The following table describes threshold conditions and their impact on recommendations:

Threshold	Threshold Impact
Accept and reject thresholds are both set to zero	CLAIRE accepts and applies data is automatically.
Confidence score is above the accept threshold	CLAIRE accepts and applies data is automatically.
Confidence score is between the accept and reject thresholds	CLAIRE displays recommendations for users to accept or reject, sorted by confidence score in descending order.
Accept threshold is greater than zero and the provider returns no confidence scores	CLAIRE displays all recommendations.

Note:

Rules for recommended enrichment data

Consider the following rules and guidelines when you configure rule associations to recommend enrichment data:

- If a rule association returns one recommendation and it exactly matches the current field value, CLAIRE doesn't display it. The Data Quality Status page in business applications shows the success status for the rule association. For example, you accept CLAIRE recommendations in two records, then merge the records. If the accepted value survives to the master record and the associated rule association returns a matching recommendation, CLAIRE suppresses it.
- If you configure the same field as both input and output fields in a rule association, and the rule association returns the same value as output, CLAIRE displays no recommendation. Also, the Data Quality Status page shows a success status.
- If a rule association returns multiple recommendations and one exactly matches the existing field value, CLAIRE displays only the non-matching recommendations. The Data Quality Status page shows the recommendation made status for the rule associations until the user accepts or rejects one of the remaining recommendations.
- If the response from the data provider contains multiple recommendations for a field group, each recommendation is added to the record as a separate field group entry. However, if the recommendations are for a root field, the value from the last recommendation survives.
- If the data for hierarchy assignment contains multiple recommendations, multiple relationships are created based on the recommendations.
- If you set the accept and reject thresholds to zero, and the plugin response doesn't include a confidence score, the enrichment data is automatically accepted.
- If you set the accept and reject thresholds to the same value, and the plugin response doesn't include a confidence score, CLAIRE displays the enrichment data as recommendations.

CLAIRE-Based Enrichment plugin

To automatically classify records by assigning them to relevant hierarchies, use the CLAIRE-Based Enrichment plugin.

You can configure the plugin with details, such as the business entity to classify, the hierarchy to use for classification, and the relationship type. When you save the objective, CLAIRE indexes the hierarchy associated with the objective.

When users prompt CLAIRE Copilot to classify records in business applications, the objectives configured with this plugin are triggered. The product experience agent in CLAIRE Copilot recommends the hierarchy nodes to which the record can be assigned with confidence scores. If the confidence score in a recommendation matches the accept threshold specified in the data enhancement rule, the record is automatically assigned to the recommended hierarchy node.

Note: If a user prompts CLAIRE Copilot to classify records before CLAIRE indexes the hierarchy, the agent can't classify the record. Also, the product experience agent can run only those objectives with hierarchies that are processed by a process hierarchies job.

The following table lists the properties of the CLAIRE-Based Enrichment plugin:

Property	Description
Hierarchy Model	Internal ID of the hierarchy model associated with the hierarchy to which you want to assign records. Hierarchy model must have the searchable property enabled. The hierarchy model must not contain cyclic relationships.
Hierarchy	Business ID of the hierarchy to which you want to assign records. **Note:** To get the business ID of a hierarchy, open the hierarchy in a business application, expand System Fields, and copy its business ID.
Business Entity to Classify	Business entity of records that you want to assign to a hierarchy. The specified business entity must be the business entity for which you configure the corresponding rule association. It must also be the business entity from which the relationship that you specify begins. For example, if you want to assign Item records to Category records in a hierarchy, specify Item as the business entity to classify.
Relationship Type	Relationship type in which the business entity to classify participates. The relationship type must belong to the hierarchy model that you specify. For example, specify p360_item_to_p360_category if you want to add the Item records to Category records in a hierarchy.
Accept Threshold	Confidence score above which a classification recommendation is automatically accepted and applied. If the agent recommends multiple hierarchy nodes with confidence scores above the accept threshold, the record is assigned to all those nodes.

For more information about the rules and guidelines for configuring CLAIRE-based classification of records, see "Guidelines for configuring CLAIRE-based classification of records" on page 122. For more information about configuring CLAIRE-based classification, see "Scenario: Configuring CLAIRE-based classification of records" on page 137.

Preview Notice: Effective in the April 2026 release, the CLAIRE-Based Enrichment plugin is available for preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in

accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Title Case

To apply uppercase to the first letter followed by lowercase, use the Title Case plugin.

MDM SaaS includes the Title Case predefined data enhancement rule. Instead of creating a new data enhancement rule with the Title Case plugin, you can use the predefined rule data enhancement rule that contains an input and output field.

You can map the input and output fields to the required business entity fields.

For example, you can transform the product name business entity field value to title case format. To transform the field value to title case, create a rule association by using the Title Case predefined data enhancement rule. You can map the input field of the data enhancement rule to the product name business entity field and the output field of the data enhancement rule to the product name business entity field. When you enter the value of the product name business entity field as oneplus, MDM SaaS transforms the input value to Oneplus.

If you want to run the rule association even if all input fields are empty, you must create a data enhancement rule that uses the Title Case plugin.

The following table describes the properties of the Title Case plugin:

Property	Description
Run rule associations even if all input fields are empty	Indicates whether to run the rule association even if none of the input fields contain any value.

Concatenate

To concatenate two or more strings without spaces, use the Concatenate plugin.

You can select the predefined Concatenate without Spaces data enhancement rule to concatenate values from two input field without spaces between them. The predefined Concatenate without Spaces data enhancement rule contains two input fields, such as Field 1 and Field 2 of text data type. You can map the corresponding business entity fields to the input and output business entity fields.

By default, the predefined Concatenate without Spaces data enhancement rule contains one output field. MDM SaaS concatenates the data in the two business entity input fields and returns the output to the mapped business entity field.

For example, to concatenate the first name and last name fields and return the output to the username field, map the two business entity fields to the input fields. You can map the output field to the username business entity field.

If you want to run the rule association even if all input fields are empty, you must create a data enhancement rule with the Concatenate without Spaces plugin.

The following table describes the properties of the Concatenate without Spaces plugin:

Property	Description
Run rule associations even if all input fields are empty	Indicates whether to run the rule association even if none of the input fields contain any value.
Number of Input fields	Total number of input field values that you want to concatenate without spaces.

When you configure a basic rule association with the Concatenate without Spaces function, MDM SaaS creates an equivalent data enhancement rule. If you update the rule association or configure another rule association with similar configuration, MDM SaaS overwrites the existing data enhancement rule.

If you delete the data enhancement rule, MDM SaaS doesn't restore the data enhancement rule and you can no longer create another rule association with the same configuration.

Concatenate with Spaces

To concatenate two or more strings with spaces between them, use the Concatenate with Spaces plugin.

MDM SaaS includes two predefined data enhancement rules, such as Concatenate with Spaces and Concatenate Three Values with Spaces. The Concatenate with Spaces rule association concatenates values from two input fields with spaces between them. However, the Concatenate Three Values with Spaces rule association concatenates values from three input fields with spaces between them.

The predefined Concatenate Three Values with Spaces data enhancement rule contains the following iput fields of the text data type:

- Field1
- Field2
- Field3

For example, to concatenate values of the first name, middle name, and last name fields and return the output to the full name field, map the input fields to the first name, middle name, and last name business entity fields. You can then map the output field of the data enhancement rule to the full name business entity field.

If you want to run the rule association even if all input fields are empty, create a data enhancement rule with the Concatenate with Spaces plugin.

The following table describes the properties of the Concatenate with Spaces plugin:

Property	Description
Run rule associations even if all input fields are empty	Indicates whether to run the rule association even if none of the input fields contain any value.
Number of Input fields	Total number of input field values that you want to concatenate with spaces.

You can use the Concatenate with Spaces plugin and create data enhancement rules based on the number of input fields.

When you configure a basic rule association with the Concatenate with Spaces function, MDM SaaS creates an equivalent data enhancement rule. If you update the rule association or configure another rule association

with similar configuration, MDM SaaS overwrites the existing data enhancement rule. You can view the predefined data enhancement rule on the Explore page.

If you delete the data enhancement rule, MDM SaaS doesn't restore the data enhancement rule and you can no longer create another rule association with the same configuration.

Data Quality - Transformation

To use the rule specification from Data Quality for transforming input data, use the Data Quality - Transformation plugin.

Based on the rule logic configured in the rule specification in Data Quality, MDM SaaS transforms the data.

The following table describes the properties of the Data Quality - Transformation plugin:

Property	Description
Rule Specification	Rule specification configured in Data Quality.
Run rule associations even if all input fields are empty	Indicates whether to run the rule association even if none of the input fields contain any value.

You can view input fields that are defined in rule specifications in Data Quality and map the business entity fields to it. The predefined output field is the primary rule set that displays the output of the rule association.

Left Trim

To remove spaces from the beginning of a string, use the Left Trim plugin.

MDM SaaS includes the Left Trim predefined data enhancement rule. The predefined Left Trim data enhancement rule uses an input field and an output field. You can map the corresponding business entity fields to the input and output fields.

For example, to remove the leading space in the first name `" John"`, map the first name business entity field to the input field. You can map the output field to the first name business entity field. When the rule association runs, it removes the leading space from the value and returns the output as `"John"`.

If you want to run the rule association even if all input fields are empty, you must create a data enhancement rule with the Left Trim plugin.

The following table describes the properties of the Left Trim plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

Lowercase

To convert uppercase string to lowercase, use the Lowercase plugin.

MDM SaaS includes the Lowercase predefined data enhancement rule. The predefined Lowercase data enhancement rule uses an input and an output field. You can map the corresponding business entity fields to the input and output fields.

For example, to convert the last name business entity field value to lowercase, map the last name business entity field to the input field. You can map the output field to the last name business entity field. When the rule associations runs, it transforms the last name to lowercase and returns the output.

If you want to run the rule association even if all input fields are empty, you must create a data enhancement rule with the Lowercase plugin.

The following table describes the properties of the Lowercase plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

Replace

To replace a character set in a string with another character set, use the Replace plugin.

MDM SaaS includes the Replace predefined data enhancement rule.

The predefined Replace data enhancement rule contains the following input fields of the text data type:

- Input
- Old String
- New String

You can map the corresponding business entity fields to the input and output fields.

If you want to run the rule association even if all input fields are empty, you must create a data enhancement rule with the Replace plugin.

The following table describes the properties of the Replace plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.
Case sensitive	Specifies whether to consider case sensitivity when validating input values.

Right Trim

To remove spaces at the end of a string, use the Right Trim plugin.

MDM SaaS includes the Right Trim predefined data enhancement rule. The predefined Right Trim data enhancement rule uses an input and an output field of the text data type. You can map the corresponding business entity fields to the input and output fields.

For example, to remove trailing spaces from the employee ID business entity field, map the employee ID business entity field to the input field. You can map the output field to the employee ID business entity field. When the rule associations runs, it removes the trailing spaces from the employee ID business entity field.

If you want to run the rule associations even if all input fields are empty, create a data enhancement rule with the Right Trim plugin.

The following table describes the properties of the Right Trim plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

Uppercase

To convert lowercase string to uppercase, use the Uppercase plugin.

MDM SaaS includes the Uppercase predefined data enhancement rule. The predefined Uppercase data enhancement rule uses an input and an output field of the text data type. You can map the corresponding business entity fields to the input and output fields.

For example, to transform the value of the first name business entity field to uppercase, map the first name business entity field to the input field. You can map the output field to the first name business entity field. When the rule associations runs, it transforms the value of the first name business entity field to uppercase.

If you want to run the rule association even if all input fields are empty, create a data enhancement rule with the Uppercase plugin.

The following table describes the properties of the Uppercase plugin:

Property	Description
Run rule associations even if all input fields are empty.	Indicates whether to run the rule association even if none of the input fields contain any value.

Rules and guidelines for Data Quality validation and transformation plugins

Consider the following rules and guidelines when you configure rule associations with Data Quality - Validation and Data Quality - Transformation plugins:

- If you update a rule specification in Data Quality that's used in a data enhancement rule, redeploy the rule association by removing and selecting it again from the asset picker. Otherwise, MDM SaaS uses the old configuration and returns incorrect results.
- When you configure a rule association in an objective, you can use a rule specification from Data Quality that contains a mapplet in the rule logic. Ensure that the mapplet doesn't contain predefined or custom Address Verifier and Java transformation assets associated with it. Otherwise, MDM SaaS doesn't run the rule association.
- When you configure rule associations with rule specifications from Data Quality, MDM SaaS considers the last updated timestamps of the rule specifications in Data Quality.
  If rule specifications contain mapplets and cleanse assets, the last updated timestamps of the rule specifications don't change. To use the updated rule specifications, modify some fields of rule specifications to include the last updated timestamps.
- When you configure rule associations with rule specifications from Data Quality and if the input fields are mapped to root fields and fields in field groups, ensure that you update the fields in the field group for

MDM SaaS to run the rule association. If you only update the root fields, MDM SaaS doesn't run the rule association.

- When you configure a rule association with the Data Quality - Transformation plugin and the rule association writes the output to a string field, ensure that the output doesn't exceed the maximum length of the field. Otherwise, the rule association throws an exception and MDM SaaS doesn't return any value to the output field.

Configuring cleansing and standardization

You can configure rule associations to cleanse and standardize data.

When you configure rule associations to cleanse and standardize data, the rule associations first validate the data and enrich the fields with cleansed data based on the validation results. When you configure cleansing and standardization, you can choose to downgrade the trust scores of fields that fail validations by specifying the percentage of trust score to downgrade.

The following table lists the plugin that you can use to cleanse and standardize data:

Plugin Name	Service that the plugin uses
Address Verification	DaaS
Application Integration-based Cleansing	Application Integration
Crosswalk standardization	MDM SaaS
DaaS Real-Time Email Verification	DaaS
DaaS Real-Time Phone Verification	DaaS

Address Verification

To validate and enrich address values in batch or real time, use the Address Verification plugin.

To validate and enrich addresses in records imported in bulk, enable the objective for ingress on the **Triggers** tab in the **Advanced Settings** dialog box.

To validate and enrich addresses in records in real time, enable the objective for business applications and REST APIs on the **Triggers** tab in the **Advanced Settings** dialog box.

For more information about setting the objectives for ingress, business applications, and REST APIs, see "Step 2. Adding objectives to an objective group" on page 114.

Address verification for batch processing

If your organizations were created after the October 2025 release, use the Address Verification plugin to verify addresses in predefined address field groups of records that you ingress or import in bulk. The Address Verification plugin uses Enrichment and Validation Orchestrator to cleanse, verify, and standardize addresses in the records.

By default, the DaaS rule association is disabled. Enable the rule association to verify the addresses in predefined address field groups in batches by using Enrichment and Validation Orchestrator.

After you enable DaaS rule association, MDM SaaS runs the basic, advanced, and DaaS rule associations on the fields in the field group in parallel. However, if you want to run these rule associations sequentially, use objective groups to create rule associations.

Note: Ensure that you upload the license file for the DaaS services to verify addresses in predefined address field groups of records that you ingress or import in bulk. Without the license file, the file import job and the ingress job fail at the load step. For more information about adding license details, see *Configure Global Settings*.

You can also configure objective groups and create rule associations that use the Address Verification plugin to verify addresses in custom address field groups of records that you ingress or import in bulk.

For more information about adding rule associations to objectives, see "Step 3. Creating rule associations" on page 116.

If your organizations were created before the October 2025 release, use the predefined address verifier asset or configure objective groups and create rule associations that use the Address Verification plugin.

Address verification for real-time processing

To verify address values in real time, use the Address Verification plugin. The address verification plugin uses Enrichment and Validation Orchestrator to cleanse, verify, and standardize addresses in records.

MDM SaaS provides a predefined Address Verification data enhancement rule that uses Enrichment and Validation Orchestrator to verify addresses in real time. You can also configure a rule association for address verification using the predefined data enhancement rule. When objectives are run, Enrichment and Validation Orchestrator validates, cleanses, and standardizes address values in real time.

The existing DaaS real-time address verification uses Informatica Address Verification as the default DaaS provider. If you have configured DaaS real-time address verification in organizations created before the October 2025 release, you can continue to use it to verify addresses. However, if you want to use Enrichment and Validation Orchestrator to verify addresses in real time, disable the existing DaaS real-time address verification and then create a rule association with the Address Verification plugin.

If your organizations were created after the October 2025 release, configure objective groups and create a rule association to verify address values in real time using the Address Verification plugin.

For more information about adding rule associations to objectives, see "Step 3. Creating rule associations" on page 116.

When you configure Address Verification data enhancement rule that uses Enrichment and Validation Orchestrator and create records in business applications, MDM SaaS validates address values and displays errors only on specific fields in field group entries. However, for existing DaaS rule associations that uses Informatica Address Verification as the default DaaS provider, MDM SaaS validates address values and displays errors on specific field group entries.

The predefined Address Verification provider contains input and output fields, which you can map to the business entity fields.

The following table lists the input fields of the Address Verification data provider that you can map to business entity fields:

Address Verification Data Provider Field	Field Type	Business Entity Field
Address Elements > Country	Text	Country
Postal Code > Postal Code 1	Text	Postal Code

Address Verification Data Provider Field	Field Type	Business Entity Field
Locality > Locality 1	Text	City
Administrative Division > Administrative Division 1 > Extended	Text	County
Preformatted data > Postal Delivery Address Lines > Address Lines 1	Text	Address Line 1
Preformatted data > Postal Delivery Address Lines > Address Lines 2	Text	Address Line 2
Preformatted data > Postal Delivery Address Lines > Address Lines 3	Text	Address Line 3
Additional Fields > Text Field 1	Text	State
Additional Fields > Text Field 2	Text	Address Type
Additional Fields > Text Field 3	Text	Usage Type
Additional Fields > Text Field 4	Text	Address Status
Additional Fields > Text Field 5	Text	Postal Code Extension
Additional Fields > Boolean Field 1	Boolean	Default Indicator
Additional Fields > DateTime Field 1	DateTime	Start Date
Additional Fields > DateTime Field 2	DateTime	End Date

The following table lists the output fields of the business entity that you can map to Address Verification data provider fields:

Business Entity Field	Field Type	Address Verification Data Provider Field
Default Indicator	Boolean	Enrichments > Additional Fields > Boolean Field 1
Address Type	Text	Enrichments > Additional Fields > Text Field 2
Usage Type	Text	Enrichments > Additional Fields > Text Field 3
Address Status	Text	Enrichments > Additional Fields > Text Field 4
Address Line 1	Text	Preformatted data > Postal Delivery Address Lines > Address Lines 1
Address Line 2	Text	Preformatted data > Postal Delivery Address Lines > Address Lines 2
Address Line 3	Text	Preformatted data > Postal Delivery Address Lines > Address Lines 3
Country	Text	Address Elements > Country > Code

Business Entity Field	Field Type	Address Verification Data Provider Field
State	Text	Enrichments > Additional Fields > Text Field 1
City	Text	Address Elements > Locality > Locality 1
County	Text	Address Elements > Administrative Division > Administrative Division 2 > Extended
Postal Code	Text	Address Elements > Postal Code > Postal Code 1
Postal Code Extension	Text	Enrichments > Additional Fields > Text Field 5
Coordinate System	Text	Enrichments > Geo Codes > Arrival Point > Coordinates
Start Date	DateTime	Enrichments > Additional Fields > DateTime Field 1
End Date	DateTime	Enrichments > Additional Fields > DateTime Field 2
Latitude	Text	Enrichments > Geo Codes > Arrival Point > Latitude
Longitude	Text	Enrichments > Geo Codes > Arrival Point > Longitude
Enriched Indicator	Text	Enrichments > Status Codes > Enriched Indicator
Verification Status Code	Text	Enrichments > Status Codes > Verification Status Code
Verification Message	Text	Enrichments > Status Codes > Verification Status Message

Application Integration-based Cleansing plugin

To validate and cleanse data by connecting to an external data provider, use the Application Integration-based Cleansing plugin.

You need to create Application Integration processes that you can use with this plugin. For more information about creating a process for Application Integration-based cleansing, see "Payload formats for Application Integration-based cleansing" on page 83.

The following table lists the properties of the plugin:

Property	Description
CAI Process Identifier	Identifier of the Application Integration process that the plugin uses.
CAI Process Name	The API name of the process that's specified in the CAI Process Identifier property.

Property	Description
Allow process with non-anonymous access	Indicates whether you can use a process that requires authentication. By default, only processes with anonymous access are allowed. When this property is enabled, you must specify a process that's configured with non-anonymous access. The property set for the selected process must match the property set in the data enhancement rule. For example, when this property is disabled, you can't save the data enhancement rule with a process configured for non-anonymous access. To use Application Integration processes that require authentication, the organization must use JWT or session ID authentication type. For more information, see *Authentication Properties* in the Administrator help.
Run as User	User name under which the Application Integration process runs. You can specify only one user. If the user you specify doesn't have access to the process, the rule association fails with the Exception status. If you don't specify a user, the process uses the credentials of the user who creates or updates a record through the user interface or REST API. You can't specify a user group in this property. If you configure the process with an allowed user group in Application Integration, you can leave this property blank. If the user who creates or updates records in a business application belongs to the group, the rule association runs.
Include only new or updated field group entries	Determines whether the input payload includes only new or updated field group entries. To send all entries of a field group to the Application Integration process, disable the property. For example, if you want to ensure that a record has at least one address entry, disable this property so that the validation service can count the entries in the Address section. **Note:** When you enable the option to include only new or updated field group entries in the payload, the associated rule association runs only if users add or update entries for the mapped field groups. It doesn't run even if users add or update a value only in an input root field without adding or updating the field group entries.
Run the selected process even when all input fields are empty	Indicates whether to run the Application Integration process when all input fields are empty. By default, when all input fields are empty, the status of the rule association is automatically set to successful without running the Application Integration process.

Crosswalk Standardization

Use the Crosswalk Standardization plugin to standardize picklist values against an enterprise code list from Reference 360.

For example, your organization contains a country picklist. You ingress records from different data sources, such as CRM, ERP, and Salesforce. The CRM source system uses USA for United States in the country picklist. The ERP source system uses US and the Salesforce source system uses United States of America. To standardize these values in all records, you define source systems and associate the country picklist with an enterprise code list from Reference 360. For more information about associating picklists to enterprise code lists in Reference 360, see *Define Source System*.

When you ingress records, MDM SaaS standardizes the country picklist values. After the values are standardized, if your goal is to transform picklist values that aren't included in the enterprise code list, configure a rule association with the Data Quality - Transformation plugin.

To achieve this goal, use the predefined Crosswalk data enhancement rule and configure a rule association to standardize picklist values. Then, you can configure an objective group, add an objective within the objective group, and run rule associations in a sequence. In the objective, you can add the rule association configured with the Crosswalk data enhancement rule, and include the rule association configured with the Data Quality - Transformation plugin.

When MDM SaaS runs objectives, Enrichment and Validation Orchestrator runs these rule associations in a sequence. MDM SaaS first standardizes picklist values and then, based on the rule logic configured for the

rule specification in Data Quality, MDM SaaS transforms picklist values that aren't part of the enterprise code list.

The Crosswalk data enhancement rule uses the picklist as the input field and the enterprise code list as the output field. You can map the corresponding business entity fields to the input and output fields.

Note: If the enterprise code list in Reference 360 contains more than 10,000 code values, MDM SaaS can't standardize picklist values using the crosswalk standardization plugin.

The Crosswalk Standardization plugin processes only source records. When you configure a rule association using this plugin within an objective, set the record type of the objective to Source Records. You can't save an objective group with the record type set to Master.

DaaS Real-Time Email Verification

To validate and enrich email addresses in real time, use the DaaS Real-Time Email Verification plugin. The plugin uses Enrichment and Validation Orchestrator to cleanse, verify, and standardize email addresses in records.

Note: To validate and enrich addresses in records in real time, enable the objective for business applications and REST APIs on the **Triggers** tab in the **Advanced Settings** dialog box.

MDM SaaS provides a predefined Email Verification data enhancement rule that uses Enrichment and Validation Orchestrator to verify email addresses in real time. You can also configure a rule association for email verification using the predefined data enhancement rule. When objectives are run, Enrichment and Validation Orchestrator validates, cleanses, and standardizes email addresses in real time.

For more information about adding rule associations to objectives, see "Step 3. Creating rule associations" on page 116.

The default input field that the Email Verification data enhancement rule uses is electronicAddress of the text data type. The input contains email addresses that users enter in real time. You can map the electronicAddress input field to the email address business entity field.

The following table describes the output fields:

Output Field	Description
EnrichedIndicator	Indicates whether the verification process enriched the email addresses with additional details.
NetProtectedBy	Determines whether the domain of the email address is protected by a security service.
StatusDescription	Status of the verification process.
StatusNbr	Status code of the verification process.
HygieneResult	Describes the outcome of additional quality checks on the email address. Unlike the statusDescription field, the HygieneResult field describes whether the email address is secure.
electronicAddress	Displays the verified email address.

The following table lists the output fields of the Email Verification data provider and their mapped business entity fields:

Email Verification Data Provider Fields	Mapped Business Entity Field
NetProtectedBy	Email > Net Protected By
StatusDescription	Email > Verification Message
StatusNbr	Email > Verification Status Code
HygieneResult	Email > Hygiene Status
electronicAddress	Email > Electronic Address

You can map the EnrichedIndicator boolean data type output field to the Enriched Indicator business entity field.

When you edit email addresses of records in bulk, MDM SaaS doesn't cleanse and standardize email addresses in records.

DaaS Real-Time Phone Verification

To validate and enrich phone numbers in real time, use the DaaS Real-Time Phone Verification plugin. The plugin uses Enrichment and Validation Orchestrator to cleanse, verify, and standardize phone numbers in records.

Note: To validate and enrich addresses in records in real time, enable the objective for business applications and REST APIs on the **Triggers** tab in the **Advanced Settings** dialog box.

MDM SaaS includes the predefined Global Phone Verification data enhancement rule that uses Enrichment and Validation Orchestrator to verify phone numbers in real time. You can also configure a rule association for phone verification using the predefined data enhancement rule. When objectives are run, Enrichment and Validation Orchestrator validates, cleanses, and standardizes phone numbers in real time.

For more information about adding rule associations to objectives, see "Step 3. Creating rule associations" on page 116.

You can use the predefined input and output fields of the Global Phone Verification data enhancement rule and map the corresponding business entity fields.

The following table describes the input fields:

Input Field	Description
prefixNumber	Denotes the prefix of the phone number.
phoneNumber	Contains the phone number that the user enters in real time.

You can map the prefix and phone number input fields to the corresponding business entity fields.

The following table describes the output fields:

Output Field	Description
EnrichedIndicator	Indicates whether the verification process enriched the phone numbers.
ISO	Determines the country code of the phone number, such as IN for India and US for the United States.
prefixNumber	Denotes the country dial code of the phone number, such as 91 for India and 1 for USA.
NDC	Code that identifies a specific region or city within a country.
StatusNbr	Code that represents the verification result.
NationalPrefix	Code that's used to dial a number within the same country.
InternationalPrefix	Code that's used to dial a number outside the country.
InternationalFormat	Verified phone number in the international format.
phoneNumber	Displays the phone number.
NationalFormat	Verified phone number in the national format.
StatusDescription	Status of the verification process.
NumberType	Type of phone number, such as mobile, landline, or VoIP.
Uses NDC	Indicates whether the phone number uses a national destination code.

The following table lists the output fields of predefined Global Phone Verification and their mapped business entity fields of the text data type:

Global Phone Data Provider Fields	Mapped Business Entity Field
ISO	Phone > ISO
NDC	Phone > National Destination Code
StatusNbr	Phone > Verification Status Code
NationalPrefix	Phone > National Prefix
InternationalPrefix	Phone > International Prefix
InternationalFormat	Phone > International Format
NationalFormat	Phone > National Format
StatusDescription	Phone > Verification Message

Global Phone Data Provider Fields	Mapped Business Entity Field
Number Type	Phone > Phone Type
Uses NDC	Phone > Phone Number Extension

You can map the EnrichedIndicator boolean data type output field to the Enriched Indicator business entity field.

When you edit phone numbers of records in bulk, MDM SaaS doesn't cleanse and standardize phone numbers in records.

Note: If you want to exclude prefix number from your phone numbers, Informatica recommends that you don't map the prefixNumber and phoneNumber Global Phone Data Provider output fields to the business entity output fields. If you map the prefixNumber and phoneNumber, MDM SaaS returns inconsistent results.

Creating Application Integration processes for data enrichment and validations

Before you configure data enrichment and validations, create processes based on the type of operation you want to perform, such as enrichment, validation, or cleansing.

When you create rule associations in MDM SaaS, you can use the following plugins to enrich, validate, and cleanse records:

- Application Integration-based Enrichment
- Application Integration-based Validation
- Application Integration-based Cleansing

Each of these plugins sends and accepts payloads in predefined formats.

A process accepts a set of fields as input from MDM SaaS and sends a request to an external enrichment and validation service. When the service returns enrichment or validation data, the process sends a response back to MDM SaaS. The response for a data enrichment request can include a set of output fields containing data from the external enrichment service. The response to a validation request specifies whether the validation passed or failed. If the validation fails, the response must contain the list of fields that failed validation.

When you create a process, you need to define the input fields that MDM SaaS needs to send to the process. You also need to define the output payload that the process needs to send to MDM SaaS.

To define the input and output fields, you need to create process objects that you can use within your process. Process objects group and structure data for use in processes. For example, instead of creating separate items for each type of demographic data, you can have a process object that contains the age, education level, income range, and ethnicity fields. For more information about designing process objects, see *Designing Process Objects* in the Application Integration help.

Payload formats for Application Integration-based enrichment

When you create a process to use with the Application Integration-based Enrichment plugin, ensure that the payload formats of the process are compatible with the plugin.

In the input payload, define the input fields that MDM SaaS needs to send to the process when a record is created or updated. In the output payload, define the output fields that the process needs to send back to MDM SaaS. If the enrichment request is successful, the data from the output fields are applied to the mapped business entity fields. If you create a process to assign records to a hierarchy, send the business IDs of relevant records in the output payload.

You can include confidence scores in the output payload. If the payload includes confidence scores, CLAIRE displays the enrichment results as recommendations in business applications based on the data enrichment configuration.

Input payload format

The following table lists the parameters in the input payload for data enrichment:

Field	Data Type	Description
`docId`	Text	A system-generated unique identifier of the record for which data enrichment is triggered.
`payload`	Object List	List of fields that you want to display as input fields in rule associations. It also includes the `sys` field.
`payload > sys`	Reference	A system field that references the `groupId` field. **Note:** Include this field at the root level of the `payload` field.
`payload > (field group) > sys`	Reference	A system field that references the `groupId` field. If you add a field group in the payload, include the `sys` field in each process object for the field groups.
`payload > (field group) > sys > groupId`	Text	A system-generated unique identifier of a field group entry.

The following sample JSON snippet represents an input payload that MDM SaaS sends to a process when a record is created or updated in a business application:

```
[
  {
    "docId": "c360.item-xyz456",
    "payload": [
      {
        "address": {
          "city": "clarence town",
          "addressLine1": "23 Gaggin St",
          "sys": {
            "groupId": "acd4985"
          }
        },
        "sys": {
          "groupId": "acd4986"
        }
      }
    ]
  }
]
```

For more information about creating process objects for the required fields, see "Creating a process for data enrichment" on page 72.

Output payload format

The response from a process to MDM SaaS must contain an array of records.

The following table lists the parameters in the output payload for data enrichment:

Parameter	Data Type	Description
`docId`	Text	A system-generated unique identifier of a record for which data enrichment is triggered.
`payload`	Object List	A list of fields that you receive from a data provider.
`payload > sys`	Reference	Optional. A system field that references the `groupId` field.
`payload > (field group) > sys`	Reference	Optional. A system field that references the groupId field. If an output field is a field group, include this parameter to return the ID of the field group entry that needs to be updated.
`payload` > (confidence score field)	Text	Confidence score to display enrichment data as recommendations. Specify a value between 0 to 100.
`sys > groupId`	Text	Optional. A system-generated unique identifier of a field group entry. Based on the `groupId` value, the field group entries are enriched. If the `groupId` value isn't available, a new field group entry is created.

The following sample JSON snippet represents an output payload that a process sends to MDM SaaS as a result of data enrichment:

```
[
  {
    "docId": "c360.item-xyz456",
    "payload": [
      {
        "cleansedAddress": [
          {
            "address1": "23rd Gaggin Street",
            "city": "clarence town",
            "country": "Australia",
            "PinCode": 5432,
            "sys": {
              "groupId": "acd4985"
            }
          }
        ],
        "matchConfidence": 57
      }
    ]
  }
]
```

The following sample JSON snippet represents an output payload that includes confidence scores for enrichment recommendations:

```
[
    {
        "payload": [
            {
                "category": "500_Damen_Blusen",
                "confidence": 65
            },
            {
```

```
                "category": "500_Waesche_Mieder",
                "confidence": 58
            },
            {
                "category": "300_AudioSpieler",
                "confidence": 25
            }
        ],
        "docId": "1sd54bd"
    }
]
```

This sample payload returns the business IDs of three category records along with confidence scores. CLAIRE displays these records as recommendations based on the configuration in the related rule association.

For more information about creating process objects for the required fields, see "Creating a process for data enrichment" on page 72.

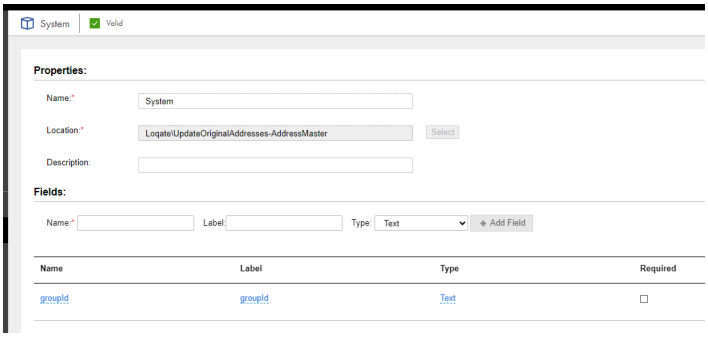
Creating a process for data enrichment

To define a set of fields that you want to display as input and output fields in Business 360 Console, create a process and define process objects that contain the fields.

As a best practice, define input and output fields based on the data model of the business entity.

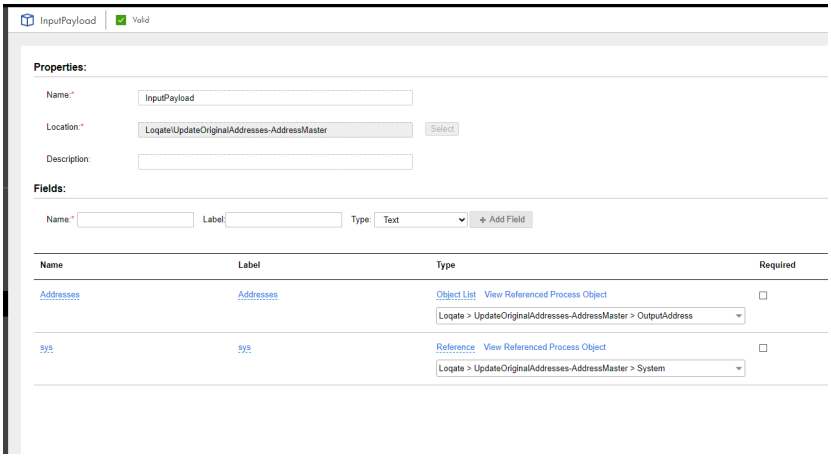
1. In Application Integration, create a folder within a project.
2. Create a process object named System with the text field named groupId.

   The following image shows a sample process object named System with the field named groupId:



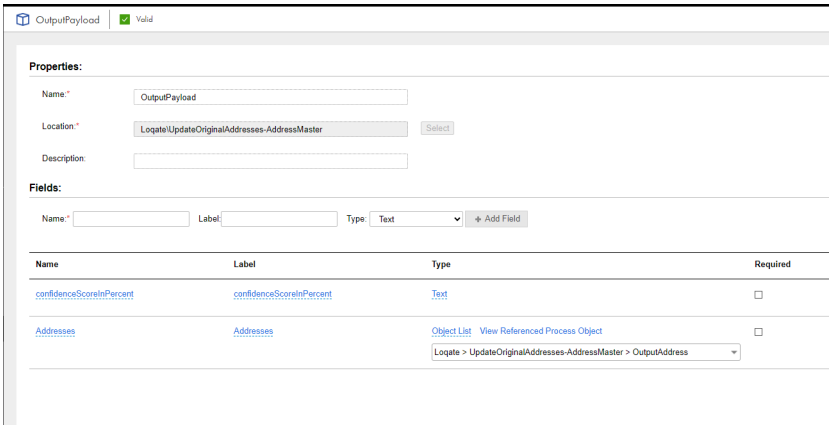
3. To define the payload for the request that MDM SaaS sends, create a process object. For example, create a process object named InputPayload.
   - Add the data provider fields that you want to display as input fields in Business 360 Console.
   - Add a field named sys, and reference the System process object that you created in step 2.

The following image shows a sample process object named InputPayload with two fields named Addresses and sys:



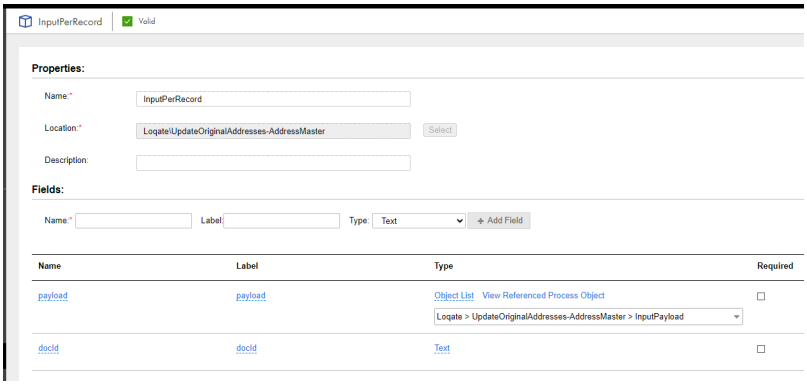
4. To define payload for the response to MDM SaaS, create a process object. For example, create a process object named OutputPayload.
   - Add the data provider fields that you want to display as output fields in Business 360 Console.
   - To display the data from the data provider as recommendations, add a text field. For example, add a text field named confidenceScoreInPercent.

   The following image shows a sample process object named OutputPayload with two fields named confidenceScoreInPercent and Addresses:



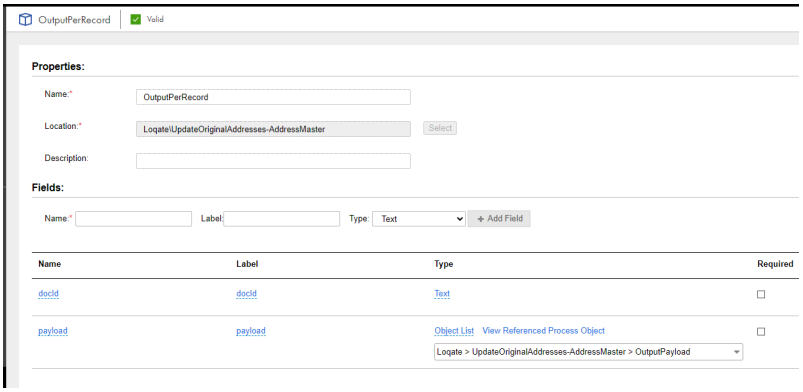
5. To define the request structure, create a process object. For example, create a process object named InputPerRecord.
   - Add a field named payload, set its type to ObjectList, and select the process object that you created in step 3.
   - Add a text field named docId.

The following image shows a process object named InputPerRecord with two fields named payload and docId:



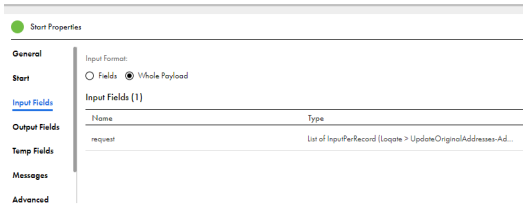
6. To define the response structure, create a process object. For example, create a process object named OutputPerRecord.
   - Add a text field named docId.
   - Add a field named payload, set its type to ObjectList, and select the process object that you created in step 4.

   The following image shows a process object named OutputPerRecord with two fields named docId and payload:



7. Create a process.
8. In the **Start Properties** panel of the process, perform the following steps:
   a. Click the **Start** tab, and select **Allow anonymous access**.
   b. Click the **Input Fields** tab.
   c. Select **Whole Payload**, add an input field named request, and set its type to List of InputPerRecord by selecting the InputPerRecord process object.

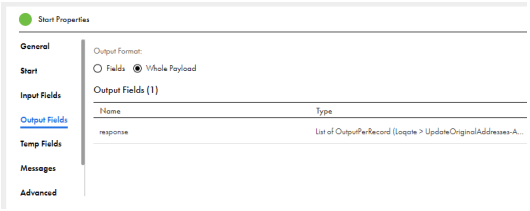
      The following image shows the input field named request and its type as List of InputPerRecord:



d. Click the **Output Fields** tab.

e. Select **Whole Payload**, add an output field named response, and set its type to List of OutputPerRecord by selecting the OutputPerRecord process object.

The following image shows the output field named response and its type as List of OutputPerRecord:



9. Based on your enrichment requirements, add necessary process steps to the process.
   - Add a service connection.
   - Add steps to receive input fields from MDM SaaS and send output fields to MDM SaaS.

Payload formats for Application Integration-based validation

When you create a process for the Application Integration-based Validation plugin, ensure that the payload formats of the process are compatible with the plugin.

A process used for validation either validates the input fields or uses an external validation service to validate them.

When fields fail validations, return the names of the input fields to which the business entity fields are mapped in a rule association. For example, if the First Name business entity field is mapped to the FirstName input field and the value of First Name fails validation, return the FirstName input field in the response. MDM SaaS resolves the input fields to the business entity fields to display validation errors correctly.

When you send a response, you can include a value for downgrading the trust score of the fields that fail validation. The value for downgrading the trust score must be between 0 to 100.

Note: To downgrade the entire field group, configure the trust downgrade score in the associated rule association. MDM SaaS considers only the score that's configured in the rule association.

Input payload format

The following table lists the parameters in the input payload for data validation:

Field	Data Type	Description
`docId`	Text	A system-generated unique identifier of the record for which data enrichment is triggered.
`payload`	Object List	List of fields that you want to display as input fields in rule associations. It also includes the `sys` field.
`payload > sys`	Reference	A system field that references the `groupId` field. **Note:** Include this field at the root level of the `payload` field.

Field	Data Type	Description
payload > (field group) > sys	Reference	A system field that references the groupId field. If you add a field group in the payload, include the sys field in each process object for the field groups.
payload > (field group) > sys > groupId	Text	A system-generated unique identifier of a field group entry.

The following sample JSON snippet represents the input payload that MDM SaaS sends to a process when a record is created or updated in a business application:

```
[
  {
    "docId": "f4c1a6e2-7b3d-4a1f-9e8c-2d5f6a3b9c07",
    "payload": [
      {
        "drugCode": " 12345 - 6789 -01 ",   // Extra spaces and incorrect spacing around
hyphens
        "drugManufacturer": null,   // No value
        "regulatoryInformation": [
          {
            "approvalStatus": "apprved",    // Typo: should be 'Approved'
            "regulatoryAuthority": "fda ", // Lowercase + trailing space instead of
uppercase trimmed
            "patentExpiry": "2038.08.14",  // Wrong date separator, should be dashes
            "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          }
        ],
        "sys": {
          "groupId": "a3f29e7c-88b1-4d6a-9c3f-1e7b5d02c8f4"  // Root level sys info for
the payload record
        }
      }
    ]
  }
]
```

The following example shows an input payload that contains two fields with null values:

```
[
  {
    "docId": "f4c1a6e2-7b3d-4a1f-9e8c-2d5f6a3b9c07",
    "payload": [
      {
        "drugCode": null,   // User didn't enter any value in the input field mapped to
drugCode
        "drugManufacturer": null,   // No value
        "regulatoryInformation": [
          {
            "approvalStatus": "apprved",    // Typo: should be 'Approved'
            "regulatoryAuthority": null,    // User didn't enter any value in the input
field mapped to drugCode
            "patentExpiry": "2038.08.14",  // Wrong date separator, should be dashes
            "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          }
        ],
        "sys": {
          "groupId": "a3f29e7c-88b1-4d6a-9c3f-1e7b5d02c8f4"  // Root level sys info for
the payload record
        }
```

```
        }
      ]
    }
  ]
```

This happens when a rule association runs with some fields that have no values. To prevent a rule associations from failing with the Exception status, enable the nullable property of the corresponding process objects fields.

Output payload format

The response from a process to MDM SaaS must contain an array of records.

The following table lists the parameters in the output payload for data validation:

Parameter	Data Type	Description
`docId`	Text	A system-generated unique identifier of a record for which data enrichment is triggered.
`validation`	Object List	An array of validation results.
`validation` > `passed`	Boolean	A value of false if validation fails. A value of true if validation passes.
`validation` > `errorMessage`	Text	Optional. Error message that the process returns for all fields that fail validation. If the associated rule association is configured to display error messages from plugins, MDM SaaS displays this error message in the fields of the record that the process validates. You can return only a common message for all fields that fail validation.
`validation` > `fields`	Object List	An array of fields that fail validation.
`fields` >`fieldName`	Text	Name of the input field that fails validation. Ensure that the name of the field matches the input field.
`fields` > `dtsp`	Integer	Optional. The value for downgrading trust score of the field that fails validation. Specify a value between 0 and 100.
`fields` > `sys`	Reference	A system field that references the `groupId` field. Include this parameter to send the ID of the fields that fail validation.
`fields` > `sys` > `groupId`		A system-generated unique identifier of a field group entry or a root field. This property is required in the output payload to identify that the fields that are validated.
`sys` > `groupId`	Text	A system-generated unique identifier of a field group entry or a root field. This property is required in the output payload to identify that the fields that are validated.

Note: Effective in the February 2027 release or later, if the response for a validation request includes the `validation` object, all field objects for both root fields and fields that belong to a field group must contain the `groupId` parameter. Ensure that you update the associated process objects in Application Integration to include the `sys` > `groupId` object for all fields that fail validation. If the response indicates that the validation failed without specifying the fields that failed, MDM SaaS considers all input fields to have failed validation.

The following sample output payload contains root fields that failed validation:

```
[
  {
    "docId": "f4c1a6e2-7b3d-4a1f-9e8c-2d5f6a3b9c07",
    "validation": [
      {
        "passed": false,
        "errorMessage": "Drug code format is invalid. Drug manufacturer is required.",
        "fields": [
          {
            "fieldName": "drugCode",
            "dtsp": 23,
            "sys": {
              "groupId": "a3f29e7c-88b1-4d6a-9c3f-1e7b5d02c8f4"
            }
          },
          {
            "fieldName": "drugManufacturer",
            "dtsp": 20,
            "sys": {
              "groupId": "a3f29e7c-88b1-4d6a-9c3f-1e7b5d02c8f4"
            }
          }
        ]
      }
    ]
  }
]
```

Note: In this example, the `groupId` parameter is included in the drugCode and drugManufacturer field objects even though they are root fields.

The following sample output payload contains fields from the regulatoryInformation field group that failed validation:

```
[
  {
    "docId": "f4c1a6e2-7b3d-4a1f-9e8c-2d5f6a3b9c07",
    "validation": [
      {
        "passed": false,
        "errorMessage": "One or more regulatory information fields are invalid.",
        "fields": [
          {
            "fieldName": "approvalStatus",
            "dtsp": 20,
            "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          },
          {
            "fieldName": "regulatoryAuthority",
            "dtsp": 20,
            "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          },
          {
            "fieldName": "patentExpiry",
            "dtsp": 20,
            "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          }
        ]
      }
    ]
  }
]
```

The following sample output payload shows a response for a successful validation:

```
[
  {
    "docId": "f4c1a6e2-7b3d-4a1f-9e8c-2d5f6a3b9c07",
    "validation": [
      {
        "passed": true
      }
    ]
  }
]
```

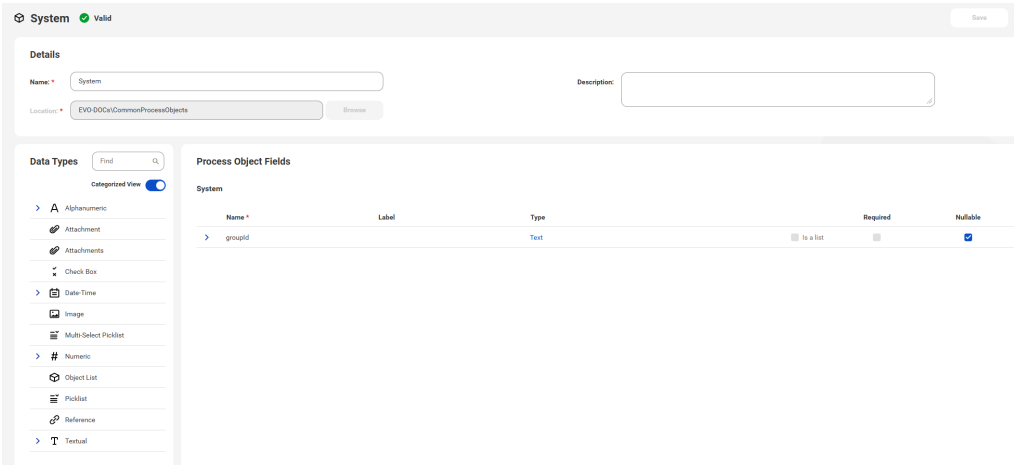
Creating a process for validations

To perform Application Integration-based validations, create a process and define the input fields and response of the process.

As a best practice, define input and output fields based on the data model of the business entity.

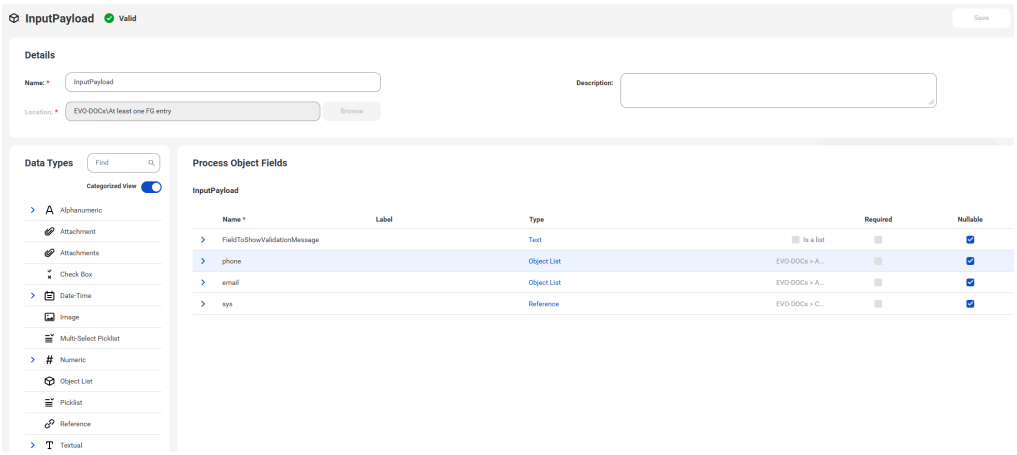
1. In Application Integration, create a folder within a project.
2. Create a process object named System with the text field named groupId.

   The following image shows a sample process object named System with the field named groupId:



3. To define the payload for the request that MDM SaaS sends, create a process object. For example, create a process object named InputPayload.
   - Add the data provider fields that you want to display as input fields in Business 360 Console.
   - Add a field named sys, and reference the System process object that you created in step 2.

The following image shows a sample process object named InputPayload with four fields named FieldToShowValidationMessage, phone, email, and sys:



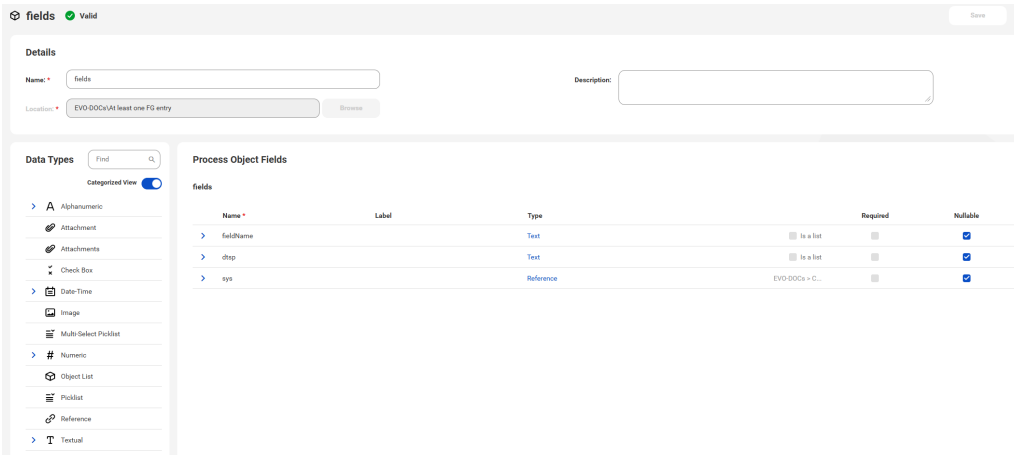
4. Create a process objects to define fields for validation results. For example, create a process object named fields.

   a. To return the name of the field that fail validation, add a text field named fieldName.

   b. To return a value to downgrade trust score, add a text field named dtsp.

   c. To return the field group ID if the input field is a field group, add a field named sys, and reference the System process object that you created in step 2.

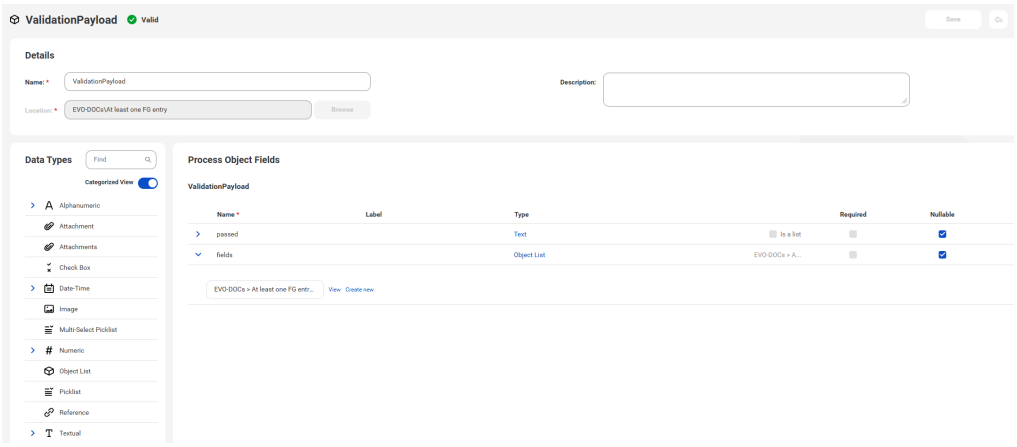
   The following image shows a sample process object named fields with three fields named fieldName, dtsp, and sys:



5. To define payload for the response to MDM SaaS, create a process object. For example, create a process object named ValidationPayload.

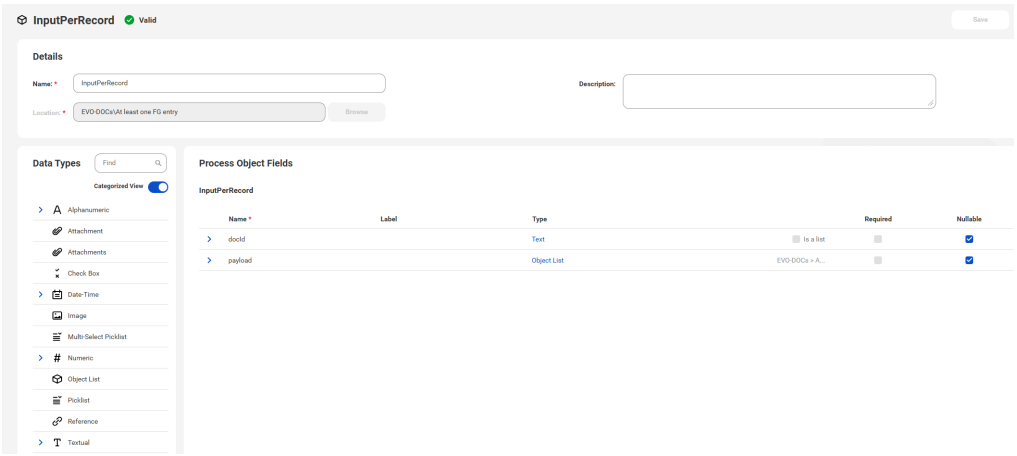
   - Add a text field named passed. Use this field to return true or false based on the validation results.
   - Add a field named fields, set its data type to Object List, and then reference the process object that you created in step 4.

The following image shows a sample process object named ValidationPayload with two fields named passed and fields:



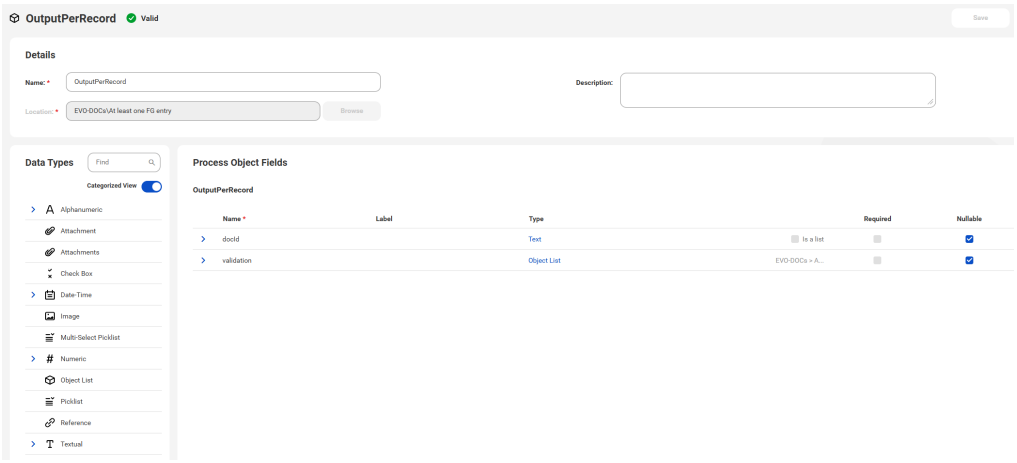
6. To define the request structure, create a process object. For example, create a process object named InputPerRecord.
   - Add a field named payload, set its type to ObjectList, and select the process object that you created in step 3.
   - Add a text field named docId.

   The following image shows a process object named InputPerRecord with two fields named payload and docId:



7. To define the response structure, create a process object. For example, create a process object named OutputPerRecord.
   - Add a text field named docId.
   - Add a field named validation, set its type to ObjectList, and select the process object that you created in step 5.

The following image shows a process object named OutputPerRecord with two fields named docId and validation:

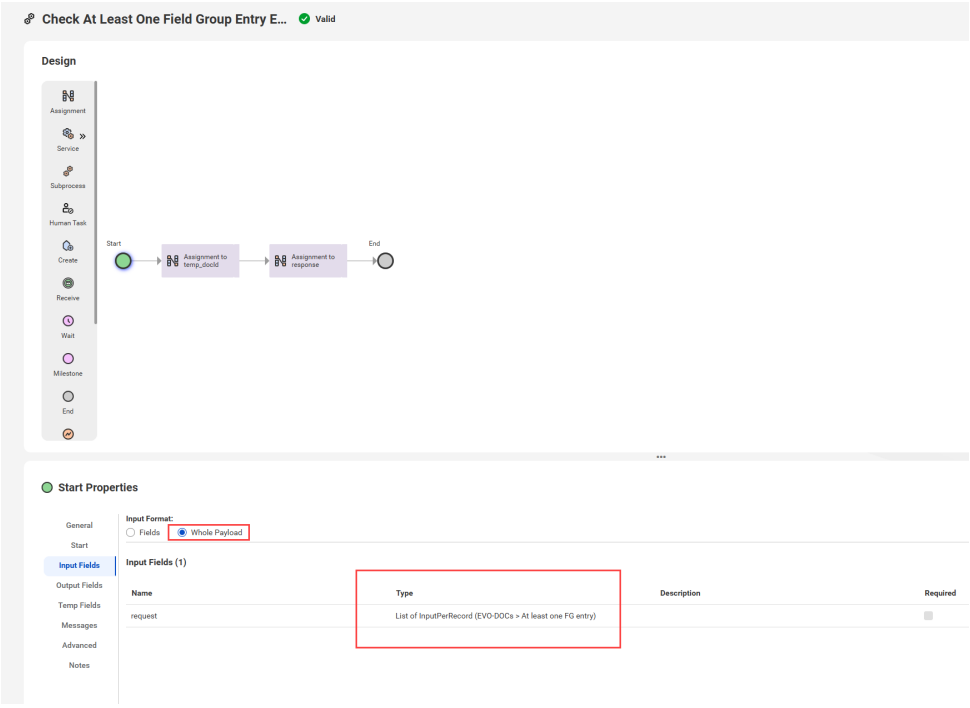


8. Create a process.
9. In the **Start Properties** panel of the process, perform the following steps:
   a. Click the **Start** tab, and select **Allow anonymous access**.

      Note: Ensure that the **Run On** property is set to **Cloud Server**.

   b. Click the **Input Fields** tab.
   c. Select **Whole Payload**, add an input field named request, and set its type to List of InputPerRecord by selecting the InputPerRecord process object.

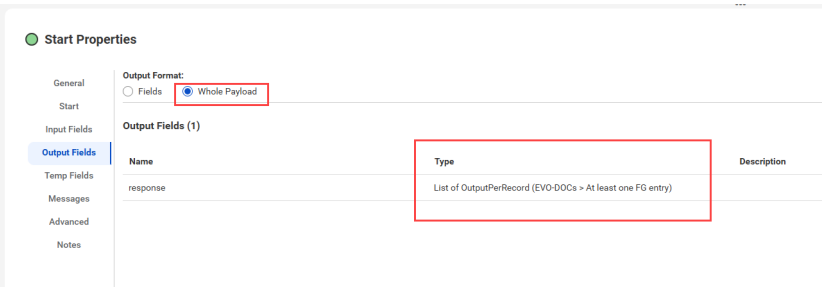
      The following image shows the input field named request and its type as List of InputPerRecord:



   d. Click the **Output Fields** tab.

e. Select **Whole Payload**, add an output field named response, and set its type to List of OutputPerRecord by selecting the OutputPerRecord process object.

The following image shows the output field named response and its type as List of OutputPerRecord:



10. Based on your enrichment requirements, add necessary process steps to the process.
    - Add a service connection.
    - Add steps to receive input fields from MDM SaaS and send validation results to MDM SaaS.

Payload formats for Application Integration-based cleansing

When you create a process to use with the Application Integration-based Cleansing plugin, ensure that the payload formats of the process are compatible with the plugin.

A process created to perform cleansing validates or uses an external data provider to validate the input fields. If the input fields fail validation, the process or external enrichment and validation service cleanses the data and sends the cleansed data back to MDM SaaS. When you send a response back to MDM SaaS, you can include a value for downgrading trust score. If you don't include validation results in the response, MDM SaaS considers the validation to be successful.

Input payload format

The following table lists the parameters in the input payload for data cleansing and enrichment:

Field	Data Type	Description
`docId`	Text	A system-generated unique identifier of the record for which data enrichment is triggered.
`payload`	Object List	List of fields that you want to display as input fields in rule associations. It also includes the `sys` field.
`payload > sys`	Reference	A system field that references the `groupId` field. **Note:** Include this field at the root level of the `payload` field.
`payload > (field group) > sys`	Reference	A system field that references the `groupId` field. If you add a field group in the payload, include the `sys` field in each process object for the field groups.
`payload > (field group) > sys > groupId`	Text	A system-generated unique identifier of a field group entry.

The following sample JSON snippet represents an input payload that MDM SaaS sends to a process for validation:

```
[
  {
    "docId": "f4c1a6e2-7b3d-4a1f-9e8c-2d5f6a3b9c07",
    "payload": [
      {
        "drugCode": " 12345 - 6789 -01 ",   // Extra spaces and incorrect spacing around
hyphens
        "regulatoryInformation": [
          {
            "approvalStatus": "apprved",    // Typo: should be 'Approved'
            "regulatoryAuthority": "fda ", // Lowercase + trailing space instead of
uppercase trimmed
            "patentExpiry": "2038.08.14",  // Wrong date separator, should be dashes
            "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          }
        ],
        "sys": {
          "groupId": "a3f29e7c-88b1-4d6a-9c3f-1e7b5d02c8f4"  // Root level sys info for
the payload record
        }
      }
    ]
  }
]
```

Output payload format

The response from a process to MDM SaaS must contain an array of records.

The following table lists the parameters in the output payload for data cleansing and validation:

Parameter	Data Type	Description
`docId`	Text	A system-generated unique identifier of a record for which data enrichment is triggered.
`validation`	Object List	An array of validation results.
`validation` > `passed`	Boolean	A value of false if validation fails. A value of true if validation passes.
`validation` > `errorMessage`	Text	Optional. Error message that the process returns for all fields that fail validation. If the associated rule association is configured to display error messages from plugins, MDM SaaS displays this error message in the fields of the record that the process validates. You can return only a common message for all fields that fail validation.
`validation` > `fields`	Object List	An array of fields that fail validation.
`fields` >`fieldName`	Text	Name of the input field that fails validation. Ensure that the name of the field matches the input field.
`fields` > `dtsp`	Integer	Optional. The value for downgrading trust score of the field that fails validation. Specify a value between 0 and 100.
`fields` > `sys`	Reference	A system field that references the `groupId` field. Include this parameter to send the ID of the fields that fail validation.

Parameter	Data Type	Description
fields > sys > groupId		A system-generated unique identifier of a field group entry or a root field. This property is required in the output payload to identify that the fields that are validated.
payload	Object List	A list of fields that you receive from a data provider.
payload > sys	Reference	Optional. A system field that references the groupId field.
payload > (field group) > sys	Reference	Optional. A system field that references the groupId field. If an output field is a field group, include this parameter to return the ID of the field group entry that needs to be updated.
sys > groupId	Text	Optional. A system-generated unique identifier of a field group entry. Based on the groupId value, the field group entries are enriched. If the groupId value isn't available, a new field group entry is created.

Note: Effective in the February 2027 release or later, if the response for a cleansing request includes the validation object, all field objects for both root fields and fields that belong to a field group must contain the groupId parameter. Ensure that you update the associated process objects in Application Integration to include the sys > groupId object for all fields that fail validation.

The following output field adds a new entry of the regulatoryInformation field group and updates the drugCode field with the cleansed value:

```
[
  {
    "docId": "f4c1a6e2-7b3d-4a1f-9e8c-2d5f6a3b9c07",
    "validation": [
      {
        "passed": false,
        "errorMessage": "Drug code format is invalid. One or more regulatory information
fields are invalid.",
        "fields": [
          {
            "fieldName": "drugCode",
            "dtsp": 23,
             "sys": {
             "groupId": "a3f29e7c-88b1-4d6a-9c3f-1e7b5d02c8f4"
            }
          },
          {
            "fieldName": "approvalStatus",
            "dtsp": 23,
             "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          },
        {
            "fieldName": "regulatoryAuthority",
            "dtsp": 23,
             "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          },
          {
            "fieldName": "patentExpiry",
            "dtsp": 23,
             "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
```

```
            }
          ]
        }
      ],
      "payload": [
        {
          "drugCode": "12345-6789-01",   // Trimmed spaces, standardized hyphen placement
          "regulatoryInformation": [
            {
              "approvalStatus": "Approved",      // Corrected spelling and capitalized
              "regulatoryAuthority": "FDA",      // Uppercase and no trailing spaces
              "patentExpiry": "2038-08-14"       // Standardized date separator
            }
          ]
        }
      ]
    }
]
```

The following sample output payload updates the existing regulatoryInformation field group entry and updates the drugCode field with the cleansed data:

```
[
  {
    "docId": "f4c1a6e2-7b3d-4a1f-9e8c-2d5f6a3b9c07",
    "validation": [
      {
        "passed": false,
        "errorMessage": "Drug code format is invalid.",
        "fields": [
          {
            "fieldName": "drugCode",
            "dtsp": 23,
            "sys": {
              "groupId": "a3f29e7c-88b1-4d6a-9c3f-1e7b5d02c8f4"
            }
          }
        ]
      }
    ],
    "payload": [
      {
        "drugCode": "12345-6789-01",
        "sys": {
          "groupId": "a3f29e7c-88b1-4d6a-9c3f-1e7b5d02c8f4"
        }
      }
    ]
  }
]
```

The following sample output payload validates and updates a field in the regulatoryInformation field group:

```
[
  {
    "docId": "f4c1a6e2-7b3d-4a1f-9e8c-2d5f6a3b9c07",
    "validation": [
      {
        "passed": false,
         "errorMessage": "One or more regulatory information fields are invalid.",
        "fields": [
          {
            "fieldName": "approvalStatus",
            "dtsp": 23,
             "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          },
        {
            "fieldName": "regulatoryAuthority",
            "dtsp": 23,
```

```
            "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          },
          {
            "fieldName": "patentExpiry",
            "dtsp": 23,
            "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          }
        ]
      }
    ],
    "payload": [
      {
        "regulatoryInformation": [
          {
            "approvalStatus": "Approved",      // Corrected spelling and capitalized
            "regulatoryAuthority": "FDA",      // Uppercase and no trailing spaces
            "patentExpiry": "2038-08-14" ,      // Standardized date separator
            "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          }
        ],
        "sys": {
          "groupId": "a3f29e7c-88b1-4d6a-9c3f-1e7b5d02c8f4"  // Complete and valid UUID
(root level)
        }
      }
    ]
  }
]
```

The following sample output payload shows a response for a successful validation:

```
[
  {
    "docId": "f4c1a6e2-7b3d-4a1f-9e8c-2d5f6a3b9c07",
    "validation": [
      {
        "passed": true
      }
    ],
    "payload": [
      {
        "drugCode": "12345-6789-01",
        "regulatoryInformation": [
          {
            "approvalStatus": "Approved",
            "regulatoryAuthority": "FDA",
            "patentExpiry": "2038-08-14",
            "sys": {
              "groupId": "f88114db-461c-34c3-87b6-d2cebc76e85e"
            }
          }
        ]
      }
    ]
  }
]
```

Guidelines for creating Application Integration processes

Consider the following guidelines when you create an Application Integration process:

- Ensure that the process is published.

- Ensure that the process has anonymous or public access.
- Create process with unique API names. If you copy an existing process to create a new process, ensure that you update the process with a unique API name.
- When you define input and output fields for a process, model them after the data model of the business entity instead of the data provider. This approach helps minimize issues related to field mappings in Business 360 Console. You can use assignment steps in the process to specify the desired request and response formats of the data provider.
  For example, consider the following sample request format of a data provider:

```
{
  "firstName": "string",
  "lastName": "string",
  "contacts": {
    "address": {
      "addressLineOne": "string",
      "addressLineTwo": "string",
      "city": "string",
      "state": "string",
      "country": "string"
    },
    "email": {
      "usageType": "string",
      "emailAddress": "string"
    },
    "phone": {
      "usageType": "string",
      "phoneNumber": "string"
    }
  }
}
```

  Also, consider the following data model of the Person business entity:

```
{
  "firstName": "string",
  "lastName": "string",
  "address": {
    "addressLineOne": "string",
    "addressLineTwo": "string",
    "city": "string",
    "state": "string",
    "country": "string"
  },
  "email": {
    "usageType": "string",
    "emailAddress": "string"
  },
  "phone": {
    "usageType": "string",
    "phoneNumber": "string"
  }
}
```

  The sample data model of the business entity doesn't include a field group, such as contacts with nested field groups. In this case, model the input payload of the process after the business entity.

  Consider the following sample input payload format of a process that replicates the data model of the business entity instead of the request format of the data provider:

```
[
  {
    "docId": "string",
    "payload": [
      {
        "firstName": "string",
        "lastName": "string",
        "address": {
          "addressLineOne": "string",
```

```
            "addressLineTwo": "string",
            "city": "string",
            "state": "string",
            "country": "string",
            "sys": {
              "groupId": "string"
            }
          },
          "email": {
            "usageType": "string",
            "emailAddress": "string",
            "sys": {
              "groupId": "string"
            }
          },
          "phone": {
            "usageType": "string",
            "phoneNumber": "string",
            "sys": {
              "groupId": "string"
            }
          },
          "sys": {
            "groupId": "string"
          }
        }
      ]
    }
]
```

- Ensure that the input payload includes the required fields. For more information about the required fields, see "Input payload format" on page 70.
- Ensure that you include the `sys` > `groupId` fields at the root level of the input payload. If the payload includes field groups, include the `sys` > `groupId` fields for each field group.
- Ensure that the output payload includes the required fields. For more information about the required fields, see "Output payload format" on page 71.

  Note: The `sys > groupId` fields are optional in the output payload. Include the `sys` > `groupId` fields in the process object to enrich the field group entries. If the `groupId` value isn't available, a new field group entry is created.
- If you want to enrich the existing entries of a field group, ensure that you include the field group as one of the input fields. For example, to enrich the existing Phone field group entries, include the Phone field group as an input field. When you configure a rule association in Business 360 Console, you can map the Phone field group of the business entity to the input field.
- To display the data from the data provider as CLAIRE recommendations in the business applications, ensure that you include confidence scores for each recommendation when you send a response to MDM SaaS.
- In the start properties of the process in Application Integration, ensure that you enable Whole Payload as the input and output formats.
- When you configure a process, ensure that you set the Input Fields and Output Fields properties in the start properties of the process to a list of objects. If you add or delete any fields from any process objects, configure the Input Fields and Output Fields properties again in the start properties.
- To use business IDs in a process that helps assign records to a hierarchy, ensure that you enable the searchable property of the Business ID system field for the business entity.
- Ensure that the process is configured to run on cloud server. You can't use a process that's configured to run on a Secure Agent.
- If the response to a validation request includes the passed field set to false in the validation object, ensure that you include the fields array in the validation object of the response.

- When you add a field to a process object, ensure that you enable the nullable property for that field. When a rule association runs on a record, the input payload sent to a process also includes fields that don't have any values. If the nullable property isn't enabled for a process object field that's mapped to the empty input field, the rule association might fail with the Exception status.
- Ensure that the responses the processes return don't include the period [.] special character in the output field names.

Data types applicable to enrichment, validation and cleansing

When you create an Application Integration process for data enrichment, you can specify the following data types for the fields in the process:

- Integer
- Number
- Date
- Date Time
- Picklist
- Checkbox
- Text

Don't use the following data types for any fields in a process:

- Currency
- Percent
- Phone
- Email
- Time
- Text Area
- Rich Text Area
- Image
- Multi-select Picklist
- Attachment
- Attachments

Compatible data types of process fields and business entity fields

When you add input fields in an Application Integration process, select data types that are compatible with the data types of business entity fields. This approach helps you map compatible fields when you configure rule associations.

The following table lists the compatible data types of business entity fields:

Data types of business entity fields	Data types of Application Integration process fields
Double	Number, Text
Integer	Integer, Number, Text

Data types of business entity fields	Data types of Application Integration process fields
Date	Date, Text
Date and Time	Date Time, Text
Picklist	Picklist, Text
Text	Text, Integer, Number, Date, Date Time, Picklist
Boolean	Text, Checkbox
Decimal	Text

When you add output fields in an Application Integration process, select data types that are compatible with the data types of the business entity fields. This approach helps you map compatible fields when you configure rule associations.

The following table lists the compatible data types of a process:

Data types of Application Integration process fields	Data types of business entity fields
Integer	Integer, Double, Decimal, Text
Number	Double, Text
Text	Integer, Double, Decimal, Date, Date and Time, Boolean, and Text, Picklist
Picklist	Picklist, Text
Date	Date, Text
Date Time	Date and Time, Text
Checkbox	Text, Boolean

Note: The decimal data type isn't available in Application Integration. Use a text field in the process for decimal values. When you map input and output fields in a rule association, you can map a decimal business entity field to a text field. For more information about the guidelines for field mappings, see "Guidelines for mapping fields" on page 109.

Advanced objective settings

You can configure when an objective is triggered and the types of records to which it applies.

You can configure to reject records that fail validations. You can also enable data dependency between rule associations.

Specifying record types for an objective

You can specify the types of records to which an objective can be applied.

You can select master records, source records, or both as input record types for an objective. For example, you can apply an objective only to master records when you create or update them in business applications.

When you set source records as the input record type for an objective, the objective is triggered whenever source and patch records are created or updated. Also, you can refine the condition further by specifying one or more source systems. When you specify a source system, the objective is triggered when source records with the specified source systems are created or updated. For example, if you choose Salesforce as the source system for an objective, the objective applies only to records with the Salesforce source system.

When you select both master and source records as input record types for an objective, consider the following behaviors:

- Based on the source systems that you specify, the rule associations in the objective run on source records from the specified source systems. The results of the rule associations are applied to the source records. Based on the survivorship configuration, the changes might survive to the master records.
- After the master record is created or updated, the rule associations run on the master record again. The results of the rule associations are then applied to the master record.

When you specify record types for an objective, ensure that the compatible triggers are configured. For more information about triggers, see "Configuring business applications as an objective trigger" on page 94.

The following table describes the compatible triggers for input record types:

Record Type	Triggers	Description
Master records	- Business applications - REST APIs - Ingress - Bulk Edit - Merge Job - Reprocess Master Data Job - Recalculate Trust Scores Job	An objective is applied to master records that are created through business applications and REST APIs. If you enabled the ability to run objectives on master records that are created or updated through batch processing for your organization, the objectives can run on master records during batch processing. The triggers for batch processing include ingress, bulk edit, merge job, reprocess master data job, and recalculate master data job. **Note:** When you restore a deleted master record, objectives configured for master records are triggered and applied to the master record. **Note:** Effective in the October 2026 release, enrichment and validation of master records during batch processing through ingress, file import, bulk edit, merge, reprocess master data job, and recalculate trust scores jobs is available for preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.
Source records	- Business applications - REST APIs - Ingress - Bulk Edit - Reprocess Master Data Job - Recalculate Trust Scores Job	An objective is applied to source records created or updated through business applications, ingress, file import, REST APIs, bulk edit, reprocess master data jobs, and recalculate trust scores jobs.

Note: After you import an objective group that you exported before the July 2025 release, if the input record type is set to source records, you can change the record type to master records in the advanced settings of the objective

For more information about configuring the record types, see "Step 2. Adding objectives to an objective group" on page 114.

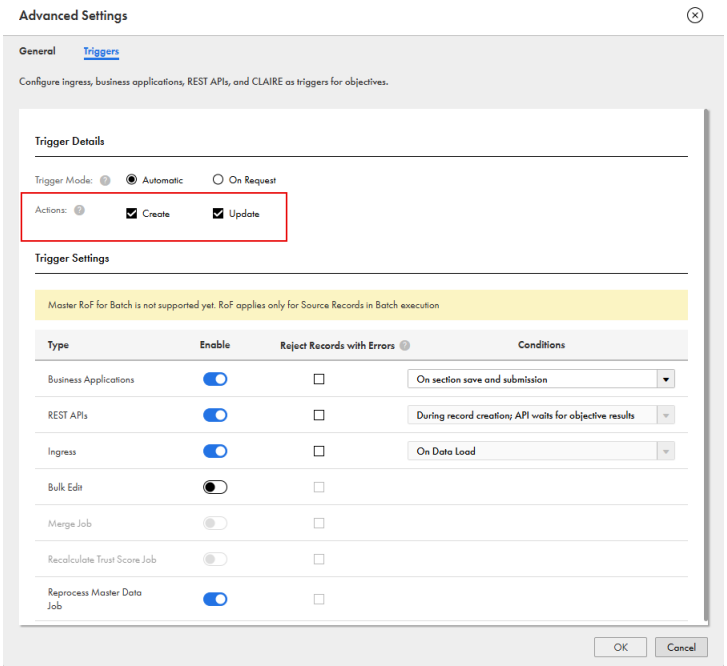
Note: Effective in the October 2026 release, the ability to configure bulk edit, merge job, reprocess master data job, and recalculate trust scores job as triggers is available for preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Configuring actions for an objective

When you configure triggers for an objective, you can configure the objective to run when a user creates records, updates records, or performs both actions.

For example, you are a data entry operator at NovaHealth Hospital. When you onboard a patient by creating their record, you want an objective to automatically populate fields such as record creation date and record status. Similarly, when you update the record with a new address, you want an objective to verify the address. To meet these requirements, you can configure an objective that runs only on new records. You can also configure an objective that runs only when an existing record is updated.

The following image shows the create and update actions enabled in the advanced settings of an objective:



Note: Effective in the October 2026 release, the ability to configure bulk edit, merge job, reprocess master data job, and recalculate trust scores job as triggers is available for preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Objective trigger modes

When you configure objectives, you can specify whether you want to run objectives automatically or on request.

You can configure an objective with one of the following modes:

Automatic

If you want objectives to run automatically when a record is created or updated, choose the trigger mode as automatic and configure the objectives with the triggers, such as business applications, ingress, and REST APIs.

On Request

If you want CLAIRE Copilot to run objectives, choose the on-request trigger mode and configure objectives with CLAIRE as the trigger. On-request objectives run when users request CLAIRE Copilot to enrich records using objectives.

Note: You can configure on-request objectives only for master records. You can enable up to 10 on-request objectives with CLAIRE as a trigger for a business entity. If you exceed this limit, you can't save the associated objective group.

Configuring business applications as an objective trigger

To run an objective automatically when users create or update a record in business applications, configure business applications as triggers for the objective.

When you configure business applications as a trigger for an objective, you can select one of the following trigger conditions:

- On submission. The objective runs when users submit records.
- On section save and submission. The objective runs when users apply changes to a section or submit a record. This condition allows users to view the results of the objective before they submit records. This condition doesn't trigger objectives for master records. If an objective configured with this trigger condition includes a rule association that adds records to hierarchies, you can't save the objective.
- On field focus change and submission. The objective runs when the focus on a field changes in a record and when users submit the record. This condition doesn't trigger objectives for master records.
  Rule associations configured with the following plugins don't run when the field focus changes in a record:
  - Application Integration-based Cleansing
  - Application Integration-based Enrichment
  - Application Integration-based Validation

  If an objective configured with this trigger condition includes a rule association that adds records to hierarchies, you can't save the objective.

Only the objectives configured for source records run when a field focus changes or users apply changes to a section. Objectives don't run on master records when a field focus changes or users apply changes to a section.

Note: If objectives with the On section save and submission or On submission conditions are configured for a field group, MDM SaaS runs those objectives when users delete a field group entry. If the deleted field group entry had validation errors from a previous run, MDM SaaS removes the validation error messages that no longer apply.

If you configure an objective to reject records with validation errors, users can't submit records that fail validation. For more information about rejecting records that fail validation, see "Rejecting records with validation errors" on page 99.

For more information about configuring these triggers, see "Step 2. Adding objectives to an objective group" on page 114.

Configuring ingress as an objective trigger

To run objectives automatically when users import or ingress source records, you can configure ingress as a trigger for objectives.

You can't trigger an objective for master records during ingress or file import.

Rule associations in an objective run during the Load step when you ingress or import files. If you configure the objective to reject records with validation errors, records that fail validation are rejected. To identify rejected records, you can download the Rejected Records report from the Errors section in the Load step of the job. You can also use the report to identify rule associations that fail with the Exception status.

Note: Rule associations that use the following plugins don't run during ingress or import of records:

- Application Integration-based Cleansing
- Application Integration-based Enrichment
- Application Integration-based Validation

Configuring REST APIs as an objective trigger

To run an objective automatically when records are created or updated through REST APIs, you can configure REST APIs as an objective trigger.

When a record is created or updated through REST APIs, MDM SaaS triggers the objective and all rule associations in the objective run. The results are applied to the record before it is saved. The API waits until the record is saved before returning a response. If you configure the objective to reject records with validation errors, records that fail validation aren't created or updated, and the API returns information about the validation error.

If you configure an objective to reject records with validation errors, records that fail validation can't be created or updated. For more information about rejecting records that fail validation, see "Rejecting records with validation errors" on page 99.

For more information about configuring these triggers, see "Step 2. Adding objectives to an objective group" on page 114.

Configuring CLAIRE as a trigger

To run objectives when users ask CLAIRE Copilot to enrich records using objectives, use CLAIRE as a trigger for the objectives.

You can configure CLAIRE as a trigger only for on-request objectives.

When users ask CLAIRE Copilot to enrich records using objectives, the Copilot displays the on-request objectives. Users can select specific or all objectives to run using CLAIRE Copilot.

CLAIRE Copilot can't run the following types of objectives:

- Objectives that are configured to display enrichment data as recommendations
- Objectives that are configured to perform validation and cleansing

When you configure on-request objectives, don't use rule associations that use system fields as input fields. On-request objectives aren't triggered when system fields are updated.

Preview Notice: Effective in the April 2026 release, the ability to configure CLAIRE as a trigger is available for preview.Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Configuring bulk edit as a trigger

You can configure objectives to run when users edit records in bulk.

This trigger works on both master records and source records. You can enable the option to reject records only for objectives that are configured to run on source records. You can't enable the option for objectives configured for master records. However, you can find records with validation errors or an exception status in the reports that are available in the bulk edit step of the job.

You can't use the following plugins in an objective that has the bulk edit trigger enabled:

- DaaS Real-Time Email Verification
- DaaS Real-Time Phone Verification

You can enable this trigger only for objectives configured to run on record updates or on both record creation and updates, because it triggers objectives only when users edit records.

Preview Notice: Effective in the October 2026 release, the ability to configure an objective with bulk edit as a trigger is available for technical preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Configuring merge job as a trigger

You can configure the merge job as a trigger for an objective to validate and enrich master records when they are created through a merge job.

The trigger runs the objective when match and merge jobs merge records and create new master records. This trigger doesn't run objectives when users manually merge records in business applications.

The merge job trigger works only on master records. You can enable this trigger only for the objectives that are configured to run on record creation or both record creation and updates. When you enable this trigger, you can't enable the option to reject records that fail validation.

You can't use the following plugins in an objective that has the merge job trigger enabled:

- DaaS Real-Time Email Verification
- DaaS Real-Time Phone Verification

Preview Notice: Effective in the October 2026 release, the ability to configure an objective with the merge job as a trigger is available for technical preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Configuring reprocess master data job as a trigger

You can configure the reprocess master data job as a trigger for an objective to validate and enrich master records and source records that already exist.

After you update rule associations or configurations for enrichment and validation, you can reprocess records to apply the updated rule associations or configurations by using this trigger. The trigger runs the objective when a reprocess master data job runs. The objectives that have this trigger enabled run on master records, source records, or both. The objectives run on master records only if the values from reprocessed source records survive into those master records.

You can enable the option to reject records only for objectives that are configured to run on source records and not on master records. However, you can find the records with validation errors or an exception status in the reports that are available in the reprocess master data step of the job.

If the reprocess master data job is configured for both active and deleted records, the objective runs only on active records.

You can't use the following plugins in an objective that has the reprocess master data job trigger enabled:

- DaaS Real-Time Email Verification
- DaaS Real-Time Phone Verification

Preview Notice: Effective in the October 2026 release, the ability to configure an objective with the reprocess master data job as a trigger is available for technical preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Configuring recalculate trust score job as a trigger

You can configure the recalculate trust score job as a trigger for an objective to validate and enrich master records.

Use this trigger to recalculate trust scores of master records after you update survivorship configurations. The trigger runs the objective when a reprocess master data job with the recalculate trust score process type runs. This trigger works only on master records. You can't configure the trigger for source records. The objectives with this trigger run on master records only if the values survive into those master records.

You can enable the option to reject records only for objectives that are configured to run on source records and not on master records. However, you can find the records with validation errors or an exception status in the reports that are available in the recalculate trust score step of the job.

You can't use the following plugins in an objective that has the recalculate trust score trigger enabled:

- DaaS Real-Time Email Verification
- DaaS Real-Time Phone Verification

Preview Notice: Effective in the October 2026 release, the ability to configure an objective with the recalculate trust score job as a trigger is available for technical preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Data dependency between rule associations

You can configure an objective to allow a rule association to use the output of another rule association as its input.

Rule associations within an objective run sequentially. By default, a rule association can't use the output of another rule association as input. You can override this behavior by enabling data dependency between rule associations within the same objective.

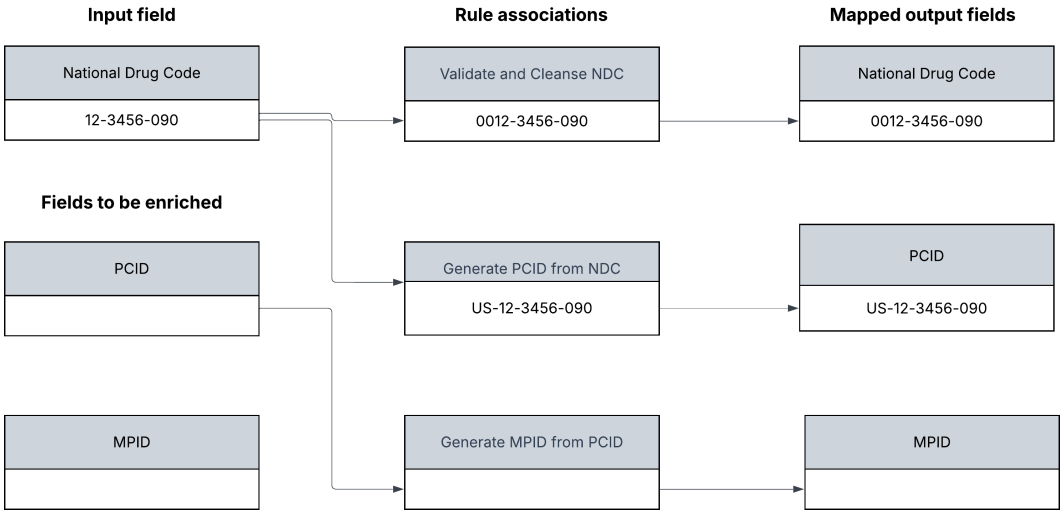
Note: If one of the rule associations fails with the Exception status, the subsequent rule associations in the objective don't run.

For example, consider that you work for a pharmaceutical company. To comply with regulations in the US market, you already use a unique National Drug Code (NDC) with every product. To comply with regulations in the European markets, you want to generate Medicinal Product Package Identifier (PCID) and Medicinal Product Identifier (MPID) codes from the National Drug Code.

To meet this validation and enrichment requirement, configure the following rule associations in an objective:

1. Validate and Cleanse NDC. Validates the national drug code to check if the value in the field contains 11 characters. If the length is 9-10, the rule association adds zeros to the input value and returns the cleansed value. If the length is below 9, the validation fails, and the rule association returns validation results without any cleansed data.
2. Generate PCID from NDC. Generates PCID by concatenating the value in the National Drug Code field with the country code.
3. Generate MPID from PCID. Generates MPID by removing the packaging code from the PCID value.

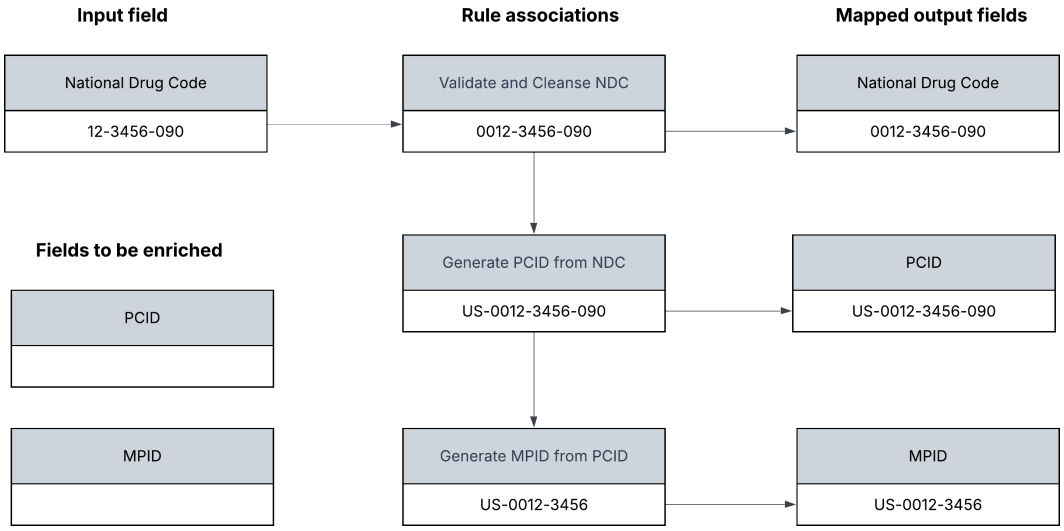
The following diagram shows how rule associations work without data dependency:



MDM SaaS saves the results only after all rule associations complete their run. In this sample scenario, the MPID field remains blank because the Generate MPID from PCID rule association didn't get the result of Generate PCID from NDC rule association.

When a user enters 12-3456-090 in the National Drug Code field and submits the record, the Validate and Cleanse NDC rule association standardizes the value to 0012-3456-090. However, this value isn't saved to the record until all rule associations complete their run. As a result, when the Generate PCID from NDC rule association runs, the National Drug Code field still contains the value 12-3456-090. The rule association adds the country code to that value and generates the value US-12-3456-090 for the PCID field.

The following diagram shows how the rule associations work with data dependency:



When a user enters 12-3456-090 in the National Drug Code field and submits the record, the Validate and Cleanse NDC rule association standardizes the value to 0012-3456-090. The Generate PCID from NDC rule association gets the output of the previous rule association as input. The rule association adds the country code and generates US-0012-3456-090, which includes country code plus the 11-digit drug code. The Generate MPID from PCID rule association gets the output US-0012-3456-090 from the previous rule association as input. It removes the packaging code from the PCID value to produce the MPID value US-0012-3456.

Note: If data dependency between rule associations is enabled for an objective, disabled rule associations in the objective don't run when the objective is triggered. If an objective includes a disabled rule association, the subsequent rule association might not receive an input value from the disabled rule association.

For more information about enabling the data dependency between rule associations, see "Step 2. Adding objectives to an objective group" on page 114.

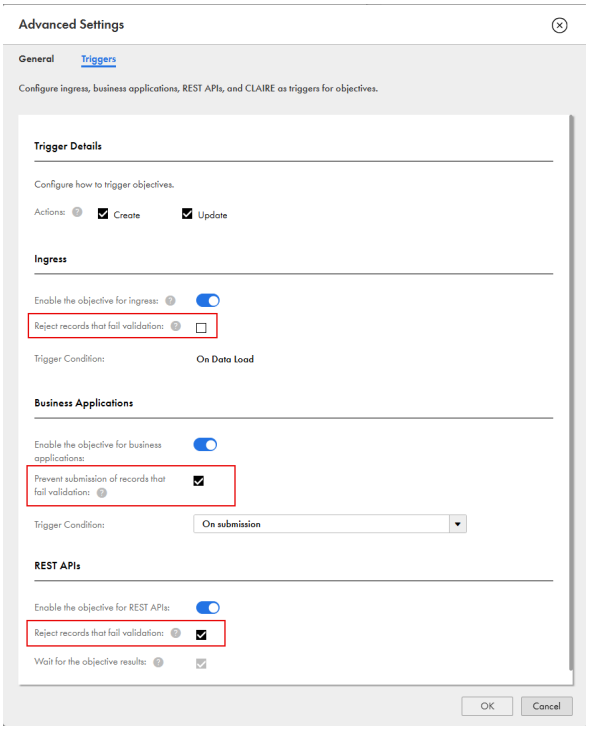
Rejecting records with validation errors

When you configure objectives that handle validations, you can choose to reject records that fail validations.

When you enable the option to reject records, MDM SaaS rejects source records with validation errors or exception status. This option works on master records that are created or updated through business applications and REST APIs. MDM SaaS doesn't reject master records that are created or updated through the ingress, file import, bulk edit, recalculate trust scores, merge, and reprocess master data jobs that process data in batches.

When users apply changes to sections in drafts, objectives run on input fields in each section. If you enable the option to reject records for the objectives and the input fields fail validation, the drafts aren't rejected. Users can continue to work on the drafts and correct validation errors before they submit the drafts. However, if the drafts contain validation errors when users submit them, those records are rejected.

The following image shows the advanced settings of an objective with the Reject Records with Errors option highlighted:



Preview Notice: Effective in the October 2026 release, the ability to configure bulk edit, merge job, reprocess master data job, and recalculate trust scores job as triggers is available for technical preview. Also, enrichment and validation of master records during batch processing is available for technical preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

If the error remediation property of a business entity is disabled, the option to reject records that fail validation is enabled by default. To allow records with validation errors to be created or submitted, disable the option to reject records that fail validation. If the error remediation property is enabled, the option to reject records that fail validation is disabled by default.

The Reject records that fail validation setting works only when you specify error as the error severity in the rule association that performs validation. If you specify information or warning as the severity, records that fail validation aren't rejected.

Note: If you enable the option to reject records that fail validation in an objective, include at least one rule association that has error severity set to Error. Records aren't rejected if rule associations have their error severity set to Information or Warning. When this option is enabled, if a rule association results in an exception, the record on which it ran is rejected. This behavior applies to all rule associations that perform validation, enrichment, or cleansing.

The following table describes how the Reject records that fail validation setting works with the On submission trigger condition:

Reject records that fail validation	Error remediation	Result
Disabled	Enabled	Users can submit records with validation errors, and the records display error messages after users submit records.
Enabled	Enabled	Users can't submit records with validation errors. The records display error messages after users try to submit records.
Disabled	Disabled	Users can submit records with validation errors. However, the records don't display error messages after users submit records. **Note:** When you use the On Submission trigger condition, if you disable both the option to reject records and the error remediation property, users can't see error messages. So, avoid disabling both properties.
Enabled	Disabled	Users can't submit records with validation errors. The records display error messages after users try to submit records.

The following table describes how the Reject records that fail validation setting works with the On section save and submission condition:

Reject records that fail validation	Error remediation	Result
Disabled	Enabled	Users can submit records with validation errors. The records display error messages after users save the section and after users submit the records.
Enabled	Enabled	When the fields within a section fail validation, users can't save the section. The records display error messages after users try to save the section.
Disabled	Disabled	When the fields within a section fail validation, users can save the section and submit the record. The record displays error messages after users save the section. The record doesn't display error messages after users submit the records.
Enabled	Disabled	When the fields within a section fail validation, users can't save the section. The records display error messages after users try to save the section.

The following table describes how the Reject records that fail validation setting works with the On field focus change and submission trigger condition:

Reject records that fail validation	Error remediation	Result
Disabled	Enabled	Users can submit records with validation errors. The records display error messages after the field focus changes and after users submit the records.
Enabled	Enabled	When a field fails validation, users can't save the section. The records display an error message after field focus changes.

Reject records that fail validation	Error remediation	Result
Disabled	Disabled	When a field fails validation, users can save the section and submit the record. The record displays an error message after the field focus changes. The record doesn't display error messages after users submit the records.
Enabled	Disabled	When a field fails validation, users can't save the section. The records display error messages after the field focus changes.

For more information about the error remediation property of business entities, see Business entities properties.

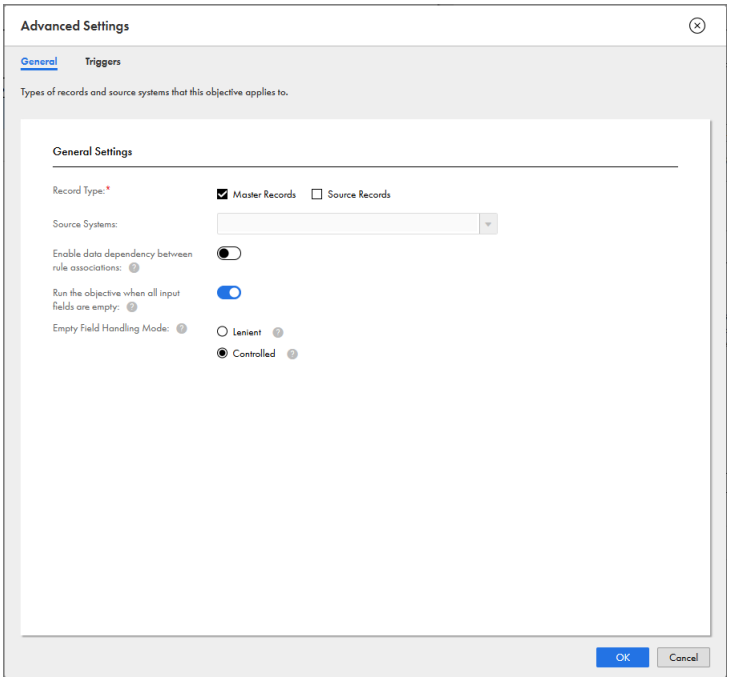
Configuring objectives to run on empty fields

You can configure an objective to run when the input fields of all associated rule associations are empty.

An objective runs when an input field is updated during record creation or updates. However, you can configure objectives to run on empty input fields when the following conditions are true:

- All input fields of its rule associations are empty.
- A sibling field of the input fields changes.

The following image shows the Advanced Settings dialog box with the option enabled to run an objective when input fields are empty:



To control the scope of objectives running on empty fields, choose the lenient mode or controlled mode.

Lenient mode

Use the lenient mode to run objectives on input fields that are at different levels. You can mix root fields, field groups, and nested fields in objectives. Objectives in lenient mode run on empty fields only during record creation, and not during record updates.

Lenient mode example

You work for a financial services company that manages customer records for tax assessments. Before you assess a customer, you must have a complete set of details, such as name, date of birth, primary and secondary addresses, email address, and phone number. Customer records often arrive with missing details, and data stewards add them later.

You want to catch incomplete records as soon as users create or edit them, even when the fields being checked are empty. Because these validations span field groups, you configure null execution in lenient mode.

Consider a customer business entity that has objectives configured to run on empty fields in lenient mode. Each objective runs on empty fields only when all of its input fields are empty.

The following table describes sample objectives configured to run on empty fields in lenient mode:

Objective	Input fields	Field location	What triggers the objective in lenient mode
Validate Name and Primary Address	- First name - Last name - Address > Address line 1	Root fields and the Address field group	A change to any root field or to any field in the Address field group
Validate Date of Birth	Date of birth	Root field	A change to any root field
Validate Secondary Contact Details	- Address > Address line 2 - Email > Email address	Address field group and Email field group	A change to any field in the Address field group or the Email field group
Validate Phone Number	Phone > Number	Phone field group	A change to any field in the Phone field group, such as phone type

Lenient mode lets you save Validate Name and Primary Address and Validate Secondary Contact Details even though their input fields come from different levels or field groups. Validate Secondary Contact Details also shows that each rule association can cover a different field group, because the trigger works at the objective level.

When users add a value to the Middle name field during record creation, the Validate Name and Primary Address and Validate Date of Birth objectives run because the Middle name is a sibling of the First name, Last name, and Date of birth fields.

Similarly, when users add a value to the city field in an address entry during record creation, the Validate Name and Primary Address and Validate Secondary Contact Details objectives run. In this case, the objective also runs on the email address field because it is one of the input fields of the objectives. If the option to include all field group entries in the payload is enabled for the associated rule association, the objectives run on all entries of the field groups.

Controlled mode

Use the controlled mode to run objectives on empty fields when a sibling field at the same level changes. If the plugin that a rule association uses has flat input structure, you can map root fields together with fields

from a field group or a nested field group as input fields. You can't use fields from more than one field group or a nested field group as input fields.

If you map root fields and a field group, the fields from the field group determine whether the objective runs. The objective runs only when mapped fields in the field group change. If only the root fields change, the objective doesn't run. For example, you map the Customer Name root field and the Line 1, City, and Postal Code fields from the Address field group. If a user updates the city, the objective runs. If a user updates only the customer name, the objective doesn't run. Use this mode when you want to limit which objectives run on empty fields.

Controlled mode example

You work for a financial services company that manages customer records for tax assessments. You want to validate each section independently in a controlled manner. You also want to run objectives on empty fields both during record creation and updates.

The following table describes sample objectives configured to run on empty fields in controlled mode:

Objective	Input fields	Field location	What triggers the objective in controlled mode
Validate Full Name	- First name - Last name	Root fields	A change to any root field, such as middle name or date of birth
Validate Date of Birth	Date of birth	Root fields	A change to any root field, such as first name or middle name
Validate Primary Address	- Address > Address line 1 - Country	Address field group	A change to another field in the Address field group, such as city
Validate Phone Number	Phone > Number	Phone field group	A change to another field in the Phone field group, such as phone type

Guidelines for configuring objectives with the controlled mode

Consider the following guidelines when you configure an objective to run on empty fields in controlled mode:

- Ensure that the data enhancement rule used in each rule association has a flat input field structure. You can't save an objective if any rule association uses a data enhancement rule with a nested input field structure.
- Map fields from only one field group across all rule associations in the objective. For example, if one rule association maps fields from the Address field group and another from the Contact Details field group, you can't save the objective. Either disable the option to run on empty fields or map all rule associations to the same field group.
- Ensure that all rule associations share at least one common field that triggers the objective. If a field from a field group is mapped to an input field of a plugin with a flat input structure, the rule association runs only when the user adds a value to the field group during record creation.
- Don't map system fields as input fields in any of the rule associations in the objective. For example, if you map the Business ID system field as an input field, you can't save the objective.
- Ensure that the data enhancement rule in a rule association includes only new or updated field group entries in the input payload. If the data enhancement rule includes all field group entries in the payload, you can't save the objective. In the data enhancement rule, enable the option to include only new or updated field group entries.

Field mappings in rule associations

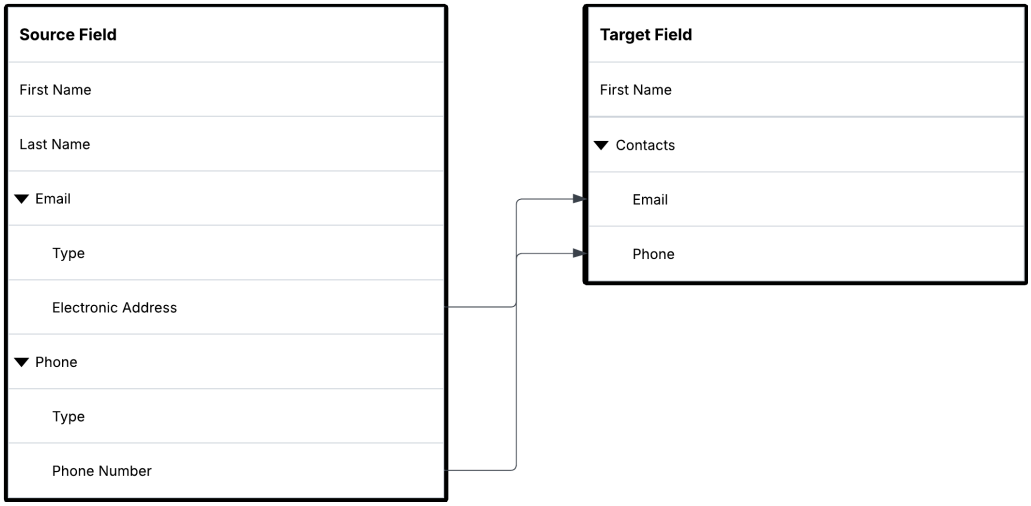
When you create rule associations, you can map the input and output fields of the data enhancement rule to the business entity fields.

After a user updates one of the mapped input fields in a record and submits the record, the objective that the rule association is part of is triggered. The rule association runs, and the data that the data provider returns is applied to the mapped output fields. Based on the trigger settings in the objective, the objective is triggered when you ingress or import records, submit records in business applications, or create or update records through REST APIs.

Mapping field groups for data enrichment

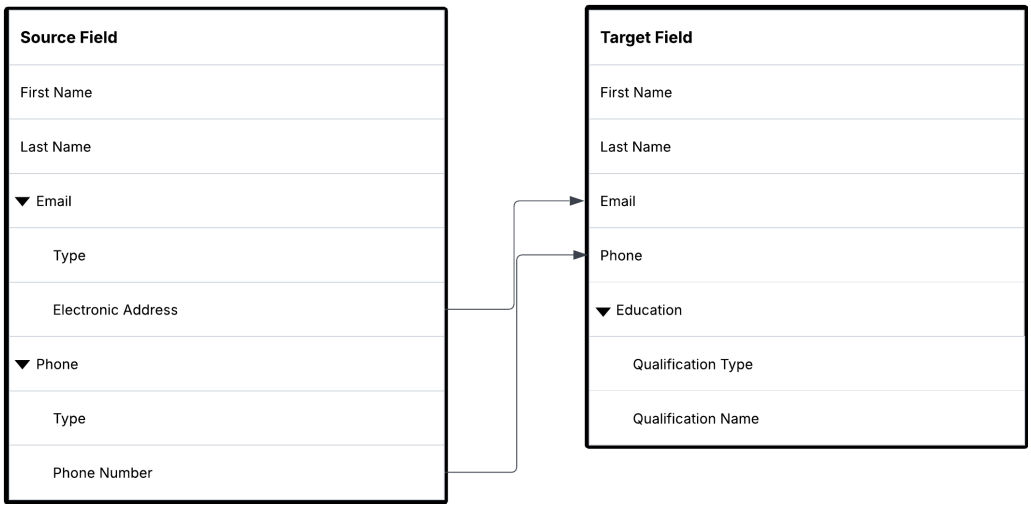
When you map field groups in a rule association for data enrichment, consider the following restrictions:

- You can’t map multiple field groups to the same target field group. For example, you can't map the field groups named Email and Phone from a business entity to the Contacts field group of the data provider. The following image shows an invalid field group mapping with multiple field groups mapped to a target field group:

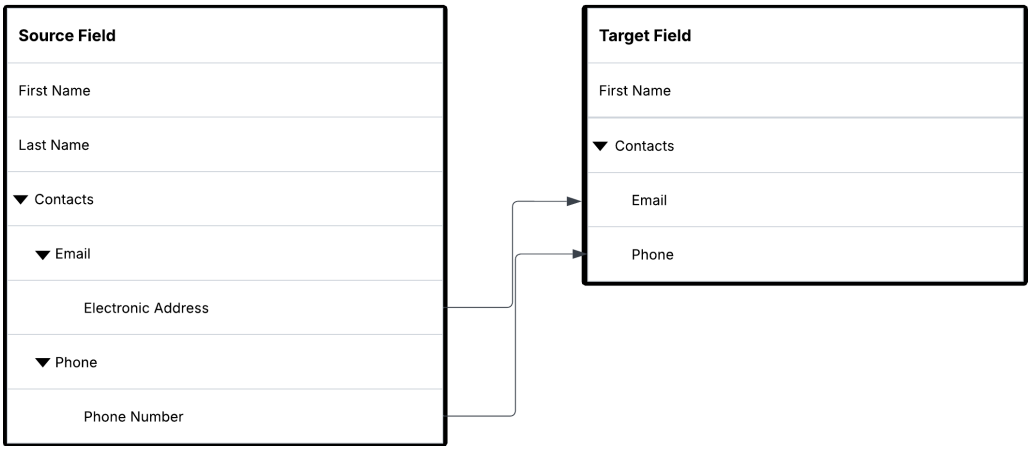


- You can’t map multiple field groups to root fields. For example, you can't map the field groups named Email and Phone from a business entity to the Email and Phone root fields of the data provider. You can map only one field group to the fields at the same level.

The following image shows an invalid field group mapping with multiple field groups mapped to target root fields:



- When you map nested field groups, you can't map multiple sibling field groups to a target field group. The following image shows an invalid nested field group mapping:



- If a business entity field group isn't mapped in the input mapping and you map the field group in the output mapping, ensure that you map all required fields in the field group. If you don't map all required fields in the field group, you can't save the rule association. For example, if the business entity Address field group isn't mapped in the input mapping and you map a field group to the Address field group in the output mapping, map all required fields in the Address field group.

Assigning static values to input and output fields

You can assign static values to input and output fields instead of mapping them to other fields.

In some cases, you might want to assign a static value to an input field instead of retrieving the value from a field in a record. Also, you might want to assign a static value as the output instead of retrieving from the data provider output fields. For example, if you want to translate the product description in a product record from English to French, you can assign English as the input value for the Language data provider field. You can also assign French as the output value for the Language business entity field.

MDM SaaS doesn't downgrade the trust scores of fields with static values. Only the trust scores of business entity fields that are mapped as input fields are downgraded. If you assign a static value to an input field and the field fails validation, the trust score of the input field isn't downgraded even if the plugin response

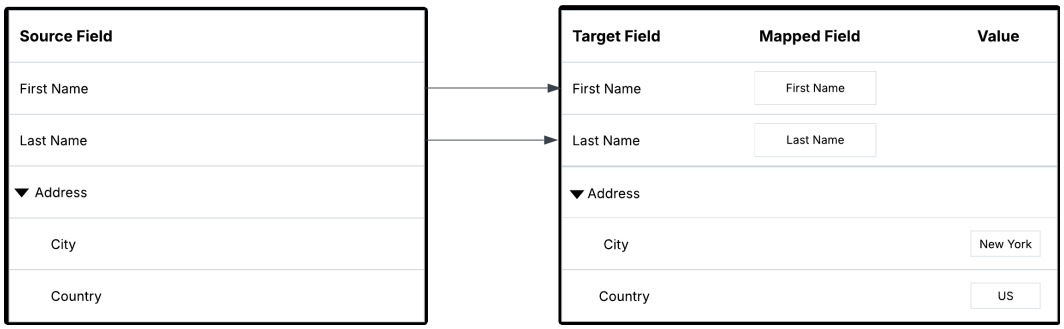
contains a downgraded trust score for the input field. If the plugin response contains only system fields that failed validation or fields with static values, the rule association is marked with the Exception status.

Assigning static values to field groups

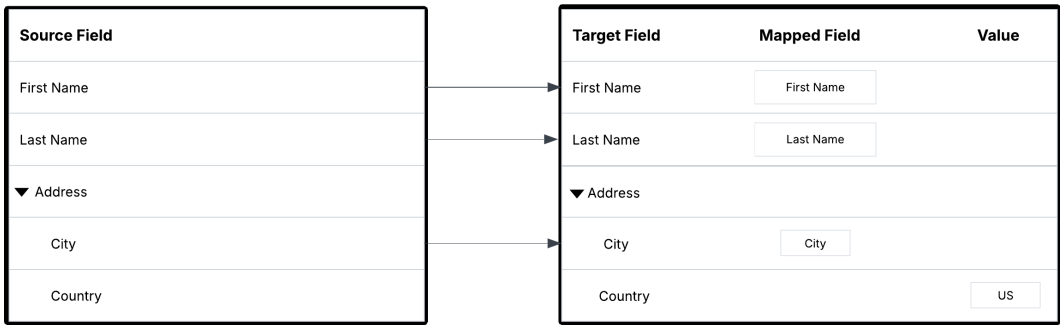
When you map input and output fields in a rule association, you can assign static values to the fields in a field group.

When you assign static values to the fields in a field group, you need to map at least one field from the source field group to a field in the target field group. You can't assign static values to all fields in a field group. For example, you map the Address field group with the City and Country fields to the Address field group of the data provider. You can assign the static value US to the Country field, but you need to map the City field from the business entity field group to the City field of the data provider.

The following image shows invalid assignments of static values to fields in a field group:



The following image shows a valid assignment of a static value to a field in a field group:

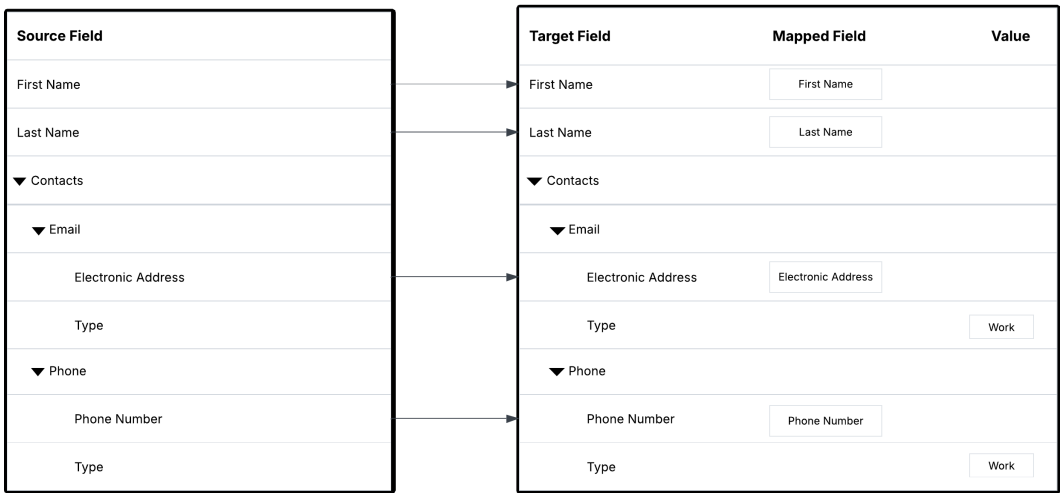


When you assign static values to the fields in a nested field group, map at least one field from the source nested field group to a field in the target nested field group.

The following image shows an invalid assignment of static values in nested field groups:



The following image shows valid assignments of static values in nested field groups:



Rules and guidelines for assigning static values

When you assign a static value to input and output fields, consider the following rules and guidelines:

- Don't assign static values to all input fields. To run a rule association when a record is created or updated, ensure that you map at least one business entity field as an input field.
- When you map output fields to relationship nodes for a classification requirement, you can't assign static values to the relationship nodes.
- When you assign a static value to a field, ensure that the value is compatible with the data type of the field.
- When you assign static values to the fields in a field group, ensure that you map at least one field from the source field group to a field in the target field group.
- When you assign static values to the fields in a nested field group, ensure that you map at least one field from the source nested field group to a field in the target nested field group.

Using system fields as input fields

You can use system fields, such as Business ID, Created By, and Last Updated By, as input fields for rule associations.

For example, you can prevent a user from updating a field in a record based on the created date of the record.

If an objective is configured to trigger when the field focus changes or a user applies changes to a section during record creation, you can use only the following system fields as input fields:

- Created By
- Created On
- Last Updated By
- Last Updated Date

You can use all system fields as input fields for record update scenarios. If you use only the system fields as input fields in a rule association that validates records, records don't display error messages when they fail validation.

When you use system fields as input fields in a rule association for validation and cleansing, use at least one business entity field as an input field. Rule associations for validation and cleansing aren't triggered based on only system field changes. However, when you configure a rule association for enrichment, you can choose to use only system fields as input fields without using any business entity fields.

When an objective is triggered on a master record, all mapped system fields of all source records associated with that master record are included in the payload sent to a plugin. For example, if you map the source system field in a rule association, the source system fields from all source records are included in the payload when the rule association runs on a master record. However, if an objective runs on a source record, only the mapped system fields of that source record are included in the payload.

MDM SaaS doesn't downgrade the trust scores of system fields. Only the trust scores of business entity fields mapped as input fields are downgraded. If you map system fields as input fields and their values fail validation, the trust scores of the system fields aren't downgraded even if the plugin response contains downgraded trust scores for the system fields. If the plugin response contains only system fields that failed validation or fields with static values, the rule association is marked with the Exception status.

Guidelines for mapping fields

When you configure rule associations for data enrichment in Business 360 Console, consider the following guidelines to map input and output fields:

- When you map input fields, map all required fields from data providers.
- When you map output fields, map all required business entity fields or relationships in a hierarchy model.
- To enrich the existing entries of a field group, ensure that you map the field group as one of the input fields. For example, if you want to enrich the existing Phone field group entries, you must map the Phone field group as an input field.
- When you map input and output fields in a rule association, map compatible fields.
  For more information about compatible data types between business entity fields and data provider fields, see "Compatible data types between business entity fields and data provider fields" on page 110.
- If you map a text field from a data provider to a non-text field in a business entity, ensure that the text field contains valid data. For example, you map the text field Amount to the decimal field Amount in a business entity. Ensure that the text field from the data provider contains valid data, such as "500.00", and not invalid data, such as "five hundred."

- If you map a text field to a picklist in a business entity, ensure that the text field contains a valid picklist code value.
- If you set source records as the input record type for an objective, don't use the Record State system field as an input field in a rule association. The Record State system field is available only for master records.
- When you configure a rule association for validation, ensure that you map at least one business entity field as an input field. If you map only system fields and assign static values without mapping a business entity field and those fields fail validation, records don't display validation messages.

Compatible data types between business entity fields and data provider fields

When you map input and output fields in a rule association, map compatible fields.

- When you map input fields, map the compatible business entity fields to the data provider fields.

  The following table lists the compatible data types between business entity fields and data provider input fields:

Business entity field data type	Data provider field data type	Sample Values
Boolean	Boolean, Text	true, false, "true", "false"
Business Entity Record	Text	
Integer	Integer, Double, Decimal, Numeric, Text	123, "123"
Double	Double, Numeric, Text	123.45, "123.45"
Decimal	Decimal, Numeric, Text	123.45, "123.45"
Date	Date, Calendar, Text	"2025-10-15"
Date and time	Date and time, Calendar, Text	"1672531200000"
Text	Text	"John Doe"
Picklist	Text	"US"

- When you map output fields, map the compatible data provider fields to the business entity fields.

The following table lists the compatible data types between business entity fields and data provider output fields:

Data provider field data type	Business entity field data type	Sample values
Boolean	Boolean, Text	true, false, "true", "false"
Text	Business Entity Record	"MDM00000000000" **Note:** If you map a plugin output field to a business entity field, ensure that the plugin returns a valid business ID of a record. If the plugin returns an invalid business ID, the rule association is marked with the Exception status.
Calendar	Date, Date and time	2025-10-15, 1672531200000
Integer	Integer, Double, Decimal, Text	123, "123"
Decimal	Decimal, Text	123.45, "123.45"
Double	Double, Text	123.45, "123.45"
Date	Date, Text	"2025-10-15"
Date and time	Date and time, Text	"1672531200000"
Numeric	Integer, Double, Decimal	123, 123.45
Text	Text, Picklist, Boolean, Integer, Double, Decimal, Date, Date and Time	"John Doe", "US", "123", "123.45", "true", "false", "2025-10-15", "1672531200000"

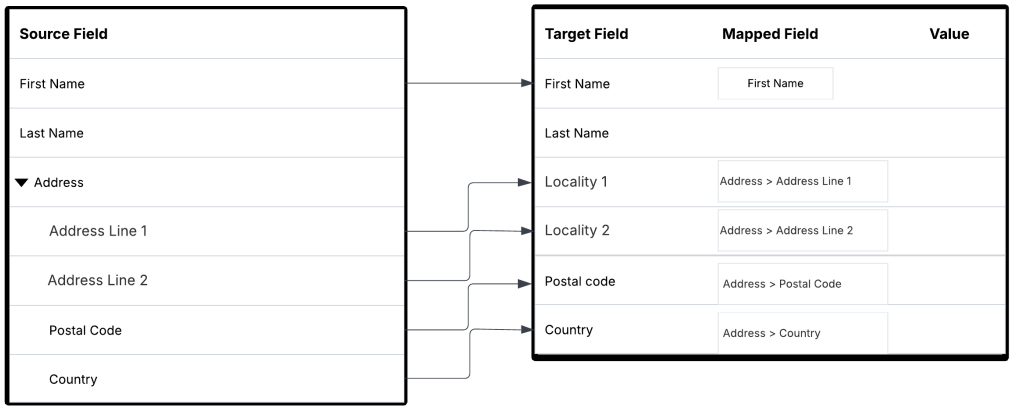
How MDM SaaS handles plugins with flat input payloads

When you map both root fields and field groups or nested field groups in a rule association that requires flat input payload, MDM SaaS flattens the innermost mapped field group. That field group acts as the trigger for the rule associaiton when users create or update records in a business application.

The rule association runs only if users add values to at least one of the mapped fields of that innermost field group. If they add values to only the root fields and submit a record, the rule association doesn't run.

For example, consider that the Address Verification plugin from Data Quality needs the First Name, Locality 1, Locality 2, Postal Code, and Country as input. Even though Locality 1, Locality 2, Postal Code, and Country are address details, the input fields don't belong to a field group. If you map the First Name field and the fields of the Address field group to these input fields, MDM SaaS flattens the structure to match the format the plugin requires.

The following image shows a sample mapping of Person business entity fields to the input fields of a plugin:



In the sample mapping, the fields from the Address field group are required input fields for the rule association to run. When a user creates a record with just the first name without adding values to any of the mapped fields of the Address field group, the rule association doesn't run.

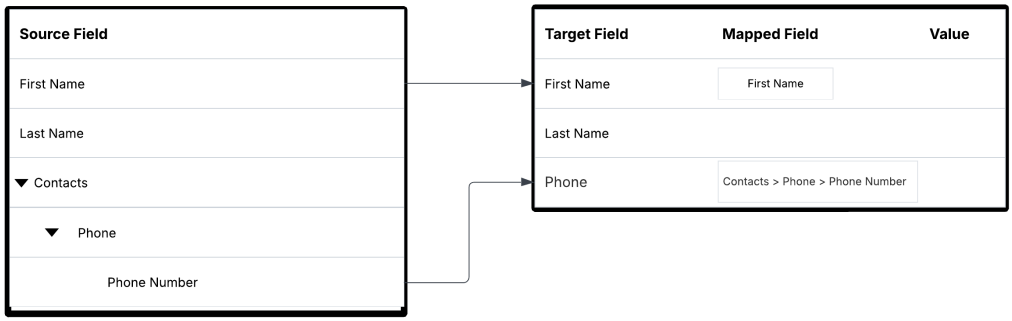
This happens because the field group is flattened and a flat input is sent to the plugin when a user creates a record. Root fields are repeated in each of the field group entries to preserve the relationship between root fields and field group entries in a flattened structure. Consider that the user enters two addresses for a person called John.

The following JSON represents a sample flattened input:

```
[
  {
    "First Name": "John",
    "Locality 1": "123 Main Street",
    "Locality 2": "Apt 4B",
    "Postal Code": "10001",
    "Country": "United States"
  },
  {
    "First Name": "John",
    "Locality 1": "456 Oak Avenue",
    "Locality 2": "Suite 100",
    "Postal Code": "90210",
    "Country": "United States"
  }
]
```

In the sample payload, the first name is repeated in both entries. The Address field group is considered a required field for the rule association to run. If fields from a nested field group are mapped as input fields, the nested field group is considered the required field for the rule association to run.

The following image shows a sample field mapping with a nested field group:



In the example, the Phone field group becomes a required field. When a user creates a record without adding a phone number, the rule association doesn't run.

Most plugins require MDM SaaS to send a flat input payload. When a rule association runs, MDM SaaS dynamically determines whether a plugin requires a flat structure and flattens the mapped business entity fields to match the expected format of the plugin.

Based on how you design the input and output payloads, the following plugins can have a nested or flat payload structure:

- Application Integration-based Validation
- Application Integration-based Enrichment
- Application Integration-based Cleansing

All other plugins in Enrichment and Validation Orchestrator have a flat input payload structure by default.

Configuring rule associations for data enrichment and validations

Configure rule associations to specify the type of operation, such as enrichment or validation, and to map input and output fields.

1. To group related objectives, create an objective group.
2. Add an objective to the objective group.
3. To map input and output fields of data providers, add rule associations to the objective.
   a. To specify the data provider, add or create a data enhancement rule.
   b. To specify the fields or relationships to enrich, map input and output fields.

Step 1. Creating an objective group

You can create an objective group to group similar objectives. Group similar objectives to conveniently migrate all the related objectives and assets between organizations.

For example, if you use multiple objectives to update contact information in records, you can group them into an objective group. Avoid multiple objectives enriching the same fields because you might get unintended results in those fields.

1. Open the business entity that you want to enrich, and click **Data Quality** > **Objective Groups**.
2. Click **Add Objective Group**.
3. In the **Objective Group** dialog box, enter a name and internal ID for the objective group, select a folder, and save the objective group.

After you create an objective group, you can add objectives to the objective group.

Step 2. Adding objectives to an objective group

Add an objective that aligns with a specific business goal for data enrichment.

To address complex enrichment and validation requirements, you can add multiple rule associations to an objective. Rule associations in an objective run sequentially based on their order. You can also configure record types and triggers for an objective. For more information about the record types and trigger settings, see "Advanced objective settings" on page 91.

1. On the business entity page, click **Data Quality** > **Objective Groups**.
2. Open the objective group that you want to add objectives to.
3. To create an objective, click **Add Objective**.
4. In the **Properties** panel, specify properties, such as name and description.

   The Properties panel also displays settings, such as Record Types and Triggers. These settings determine the types of records to which you want to apply an objective and when to apply it.
5. To enable the objective, click the **Enable** toggle.

   Note: By default, an objective is disabled. The objective is applied to records only if you enable it.
6. To specify the types of records, click **Edit** in the **Advanced Settings** section, and perform the following steps:
   a. Click the **General** tab.
   b. To apply the objective to master records, select **Master Records**.
   c. To apply the objective to source records, select **Source Records**, and specify a source system.

      By default, the objective is triggered for all records. If you specify a source system, the objective is triggered only when the source records belong to the specified source system.
7. To enable data dependency between rule associations, turn on **Enable data dependency between rule associations**.
8. To run the objective when all input fields are empty, enable **Run the objective when all input fields are empty**.
9. To run the objective based on a specific action, click the **Triggers** tab, and then perform one of the following actions:
   - To run the objective only for new records, select **Create**.
   - To run the objective only when an existing record is updated, select **Update**.
   - To run the objective both when a record is created and updated, select **Create** and **Update**.

10. To run objectives automatically when records are created or updated, click the **Triggers** tab, choose **Automatic** as the trigger mode, and configure any or all of the following triggers:
    - Ingress
    - Business applications
    - REST APIs

    For more information about configuring objectives to run automatically, see "Configuring objectives to run automatically" on page 115.
11. To run objectives when users ask CLAIRE Copilot to enrich records using objectives, click the **Triggers** tab, choose **On Request** as the trigger mode, and configure CLAIRE as the trigger for the objective.
12. Add rule associations that you created.
    a. Click **Add Rule Association**.

       A row with the asset picker appears in the **Rule Association** section.
    b. Select a rule association, and click the tick mark icon to add the rule association.
    c. To add more rule associations, repeat steps 1 to 2.
    d. To change the order of rule associations, click the **Action** menu and choose **Move up** or **Move down**.
13. To remove a rule association from the objective, click the **Action** menu for the rule association and choose **Delete**.
14. Save the changes.

Configuring objectives to run automatically

To run an objective automatically when a record is created or updated, enable ingress, business applications, and REST APIs as triggers.

1. In an objective group, select an objective that you want to configure.
2. In the **Advanced Settings** section, click **Edit**.
3. In the **Advanced Settings** dialog box, click the **Triggers** tab.
4. To trigger the objective when records are ingressed or imported, perform the following steps:
    a. In the **Ingress** section, turn on the **Enable the objective for ingress** toggle.
    b. To reject records that fail validations, select **Reject records that fail validation**.
5. To trigger the objective when records are submitted in business applications, perform the following steps:
    a. In the **Business Applications** section, turn on the **Enable the objective for business applications** toggle.
    b. To allow users to submit records that fail validations, clear **Prevent submission of records that fail validation**.

       Note: When you configure an objective to apply only to master records, you can't prevent users from submitting records with validation errors. For more information about the rejecting records with validation errors, see "Rejecting records with validation errors" on page 99.
    c. Configure one of the following trigger conditions:
        - On submission. Triggers the objective when a record is submitted.
        - On section save and submission. Triggers the objective when a section is saved and when the record is submitted.

- On field focus change and submission. Triggers the objective when the field focus changes in a record and when the record is submitted.

Note: Rule associations that use Application Integration-based plugins don't run when field focus changes. When field focus changes or a user applies changes to a section, rule associations that assign records to hierarchies don't run. For more information about triggers and trigger conditions, see "Configuring business applications as an objective trigger" on page 94.

6. To trigger the objective when a record is created or updated through REST APIs, perform the following steps:
   a. In the **REST APIs** section, turn on the **Enable the objective for REST APIs** toggle.
   b. To allow APIs to create or update records that fail validations, clear **Reject records that fail validation**.

Configuring objectives to run on request

You can configure to run objectives on request for CLAIRE Copilot.

You can enable up to 10 on-request objectives with CLAIRE as a trigger for a business entity. If you exceed this limit, you can't save the associated objective group.

Preview Notice: Effective in the April 2026 release, the ability to configure trigger modes is available for preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

1. In an objective group, select an objective.
2. In the **Advanced Settings** section, click **Edit**.
3. In the **Advanced Settings** dialog box, click the **Triggers** tab.
4. Select **On Request** as the trigger mode.
5. To trigger the objective through CLAIRE Copilot, turn on the **Enable the objective for CLAIRE** toggle.

Step 3. Creating rule associations

Create a rule association to map the input and output fields of the data provider with business entity fields or a hierarchy model.

If you add a data enhancement rule with a validation or cleansing plugin, you can configure messages for validation failures. You can also configure rule associations to downgrade trust scores for fields that fail validation. Optionally, you can assign static values to input and output fields.
For more information about how static values work, see "Assigning static values to input and output fields" on page 106.

1. On the objective group page, select an objective for which you want to add a rule association.
2. In the **Properties** panel, click **Create Rule Association**.
3. In the **Create Rule Association** dialog box, enter the details for the rule association, and click **OK**.
4. On the **Select Data Enhancement Rule** page, select or create a data enhancement rule.

   Note: After you save the rule association, you can't change the data enhancement rule.

   For more information about creating a data enhancement rule, see "Creating a data enhancement rule" on page 118.

5. Click **Next**.
6. On the **Map Fields** page, click the **Input Mapping** tab.
7. To specify the fields on which the rule association must run, select fields from the source and target panels, and click **Map Selected**.

   You can also assign a static value instead of mapping a field.
8. Click the **Output Mapping** tab.
9. To associate the enriched record with a source system, select a source system.

   Note: When you map output fields in a rule association without specifying a source system, MDM SaaS associates the Default or Informatica Customer 360 source system with the patch record that's created or updated as a result of data enrichment.
10. Map output fields to accomplish one of the enrichment use cases.
    - To enrich fields of records, select a business entity as the target and map the data provider fields to the business entity fields or assign a static value.
    - To classify records, select a hierarchy model as the target and map the data provider fields to the relationship nodes in the hierarchy model.
    - To relate records automatically, select a relationship and map the output field to a relationship type. For more information about relating records using rule associations, see "Relating records through a non-hierarchical relationship" on page 122.

    Note: After you save the rule association, you can't change the target. For example, if you select a hierarchy model as the target, you can't change the target to another hierarchy model or a business entity after you save the rule association.
11. If you choose a data enhancement rule that uses a validation or cleansing plugin, configure validation on the **Configure Validation** tab.
    a. From the **Error Severity** list, select one of the following severity levels for the validation failure:
       - Information. For minor data quality issues. Users can save the data.
       - Warning. For less severe data quality issues. Users can save the data.
       - Error. For severe data quality issues. Users can't save the data.

    b. To configure a common error message for all fields that fail validation, select **Custom** from the **Message Type** list, and then enter the message in the **Error Message** field.

    c. To display an error message from the Data Quality validation plugin or the Application Integration-based validation and cleansing plugins, select **Plugin** from the **Message Type** list.

    d. Add the fields for which you want to display the validation message.

       The custom messages appear in all fields that you specify on the **Configure Validation** tab.

       For more information about configuring error messages, see "Configuring error messages for validation rule associations" on page 124.
12. Configure trust score downgrade for fields that fail validation.
    a. Add a field for which you want to downgrade the trust score.

    b. Select **Enable Trust Downgrade**.

    c. Enter a downgrade percentage value from 1 through 100.

       For more information about configuring trust score downgrade, see Configuring trust score downgrade for input fields.
13. Save the rule association.
14. Add the rule association to an objective.

Creating a data enhancement rule

To configure an enrichment or validation type, add a data enhancement rule and configure its parameters.

1. On the rule association page, click **Add Data Enhancement Rule**.
2. In the **Add Data Enhancement Rule** dialog box, enter the details of the rule, and click **OK**.
3. On the **Data Enhancement Rule Settings** page, select a plugin from the plugin list.

   For more information about the type of operations and the available plugins, see "Configuring validations" on page 37, "Configuring data enrichment" on page 51, and "Configuring cleansing and standardization" on page 61.
4. Specify the plugin parameters.
5. Save the data enhancement rule.

Handling empty input fields

You can specify if a rule association can return a successful status without running when all its input fields are empty.

For example, when you create a record if a user adds a value to a root field and submits a record, all objectives configured for all root fields are triggered. Also, when a user adds a field group entry and saves the section, all objectives configured for the fields of the field group are triggered. When an objective with multiple rule associations is triggered, all rule associations in that objective run even though users add or update values only in fields associated with any one of the rule associations.

If all input fields are empty, the plugins don't send requests to the Application Integration processes that the associated data enhancement rules use, and MDM SaaS sets the rule associations as successful. You can change this behavior by enabling the Run the selected process even when all input fields are empty property in a data enhancement rule. If you enable this property, ensure that the processes contain a business logic to handle empty input fields.

If you don't have business logic to handle empty input fields and a plugin returns an invalid response, the associated rule association is marked with the Exception status.

Disabling an objective

To prevent all rule associations in an objective from running, disable the objective that the rule associations belong to.

1. On the business entity page, click **Data Quality** > **Objective Groups**.
2. Open the objective group that contains the objective you want to disable.
3. In the **Objectives** panel, select the objective.
4. In the **Properties** panel, click the **Enable** toggle to disable the objective.
5. Save the objective group.

Disabling rule associations in an objective

To prevent a rule association from running on records when an objective is triggered, disable the rule association without removing it from an objective.

1. On the business entity page, click **Data Quality** > **Objective Groups**.
2. Open the objective group that contains the objective.
3. In the **Objectives** panel, select the objective.

4. Click the **Actions** menu for the rule association that you want to disable, and then click **Disable**.
5. Save the objective group.

Guidelines for configuring data enrichment

Consider the following guidelines to configure data enrichment:

- Ensure that you create custom Application Integration processes based on the guidelines for creating processes.
- To track enriched records, define a custom source system. When you map output fields in a rule association without specifying a source system, MDM SaaS associates the Default or Informatica Customer 360 source system with the patch record that's created or updated as a result of data enrichment.
- If you configure data enrichment to display recommendations for classifying records, ensure that you add the Hierarchy component to your custom page of the view edit type. Without the Hierarchy component, CLAIRE panel doesn't display recommendations in business applications.
- If you update the data types of fields in a process or add new fields to a process, the changes don't reflect in the data enhancement rule and rule association that use the process. To update the field mappings in the rule association, create and configure a new data enhancement rule and a rule association.
- Ensure that you map the compatible fields. For more information about compatible fields, see "Guidelines for mapping fields" on page 109.
- When you create a data enhancement rule, ensure that you use a published process with valid input and output payload formats. If an existing data enhancement rule doesn't work, verify and correct the input and output formats of the process that the rule uses. For more information about valid input and output formats, see "Creating a process for data enrichment" on page 72.
- After you configure or migrate a data enhancement rule that uses an Application Integration process, ensure that the value in the Process Name property matches the API name of the process.
- Ensure that you create unique data enhancement rules. You can't create duplicate data enhancement rules with the same configurations.
- When you create objective groups, rule associations, and data enhancement rules, specify unique names for them. You can't create multiple assets with the same display name in the same location.
- When you configure an objective for source records, don't add a rule association that assigns records to a hierarchy. You can't assign source records to a hierarchy.
- When you configure an objective with the On section save and submission trigger condition, don't add a rule association that assigns records to a hierarchy. Rule associations that assign records to a hierarchy don't run when a user applies changes to a record section.
- When you specify a value to downgrade trust scores of fields that fail validation, specify a value between 0 and 100.
- If you configure a rule association to display enrichment data as CLAIRE recommendations and enable data dependency between rule associations, place the rule association that displays recommendations at the end of the rule association list in the objective.
- When you configure an objective with the On field focus change and submission trigger condition, ensure that the rule associations within the objective use compatible plugins. You can't use the Application Integration plugins with the On field focus change and submission trigger condition.
- Avoid configuring multiple objectives that modify the same output fields. You can't determine which objective results are applied to the fields because objectives run in parallel.

- If you configure an objective for both master and source records, the objective is triggered twice. It is triggered when a source record is created or updated, and when a master record is created or updated. Verify your configurations to avoid running objectives multiple times and producing unintended results. For example, if your custom specification returns true when the field is null and false in other cases, an objective configured for both master and source records might produce incorrect results. This issue occurs because the objective sets a Boolean field with a null value to true when applied to a source record, but then sets the field to false when applied to the master record.
- Ensure that plugins return valid output data for the fields in business entity records. MDM SaaS validates the data in the plugin response before saving the data in a record. If a plugin response contains invalid data for a field, MDM SaaS marks the rule association with an Exception status. For example, if the plugin returns a value other than a country code for the Country field, MDM SaaS marks the associated rule association with the Exception status.
- Ensure that the cumulative run time for all objectives associated with a business entity stays under 30 seconds. If processing time exceeds this threshold, MDM SaaS times out and discards the results of the objectives.
- Use a rule association configured to display enrichment data as CLAIRE recommendations only in objectives for master records. CLAIRE recommendations appear only for master records.

Guidelines for configuring Application Integration-based enrichment and validation

When you configure objectives that use Application Integration processes, consider the following guidelines:

- When you use the On section save and submission trigger condition, ensure that Application Integration processes respond quickly. Otherwise, users might notice slow responses in the user interface, which can negatively impact their experience.
- Ensure that Application Integration processes take no more than five seconds to respond. A process that exceeds this threshold might be timed out and the associated rule association is marked with the Exception status.
- Design small, focused objectives. Objectives are the execution units in Enrichment and Validation Orchestrator. Avoid using many rule associations in a single objective because all rule associations in an objective run sequentially. Break objectives into smaller units with a maximum of three rule associations each. Leverage running objectives in parallel. Multiple objectives run in parallel, improving overall processing speed.
- Monitor total run time. Ensure that the cumulative run time for all objectives associated with a business entity stays under 30 seconds. If processing time exceeds this threshold, MDM SaaS times out and discards the results of the objectives.
- 5 MB limit for the payloads. Ensure that the size of the payload that's sent to a process doesn't exceed 5 MB. Break complex processes into smaller ones to avoid payloads that exceed 5 MB.

Updating a data enhancement rule with a revised process or rule specification

If you update an Application Integration process or data quality specification, you can refresh the associated data enhancement rule to sync the rule with the latest process or specification.

When you rename or delete fields, or change field data types in a process or a data quality rule specification, the data enhancement rules that use them don't automatically reflect those changes. You can refresh the associated data enhancement rule to identify the rule associations where the outdated input and output fields are mapped. After you identify the rule associations, manually remove the outdated field mappings from the affected rule associations before you can save the data enhancement rule with the latest process or specification.

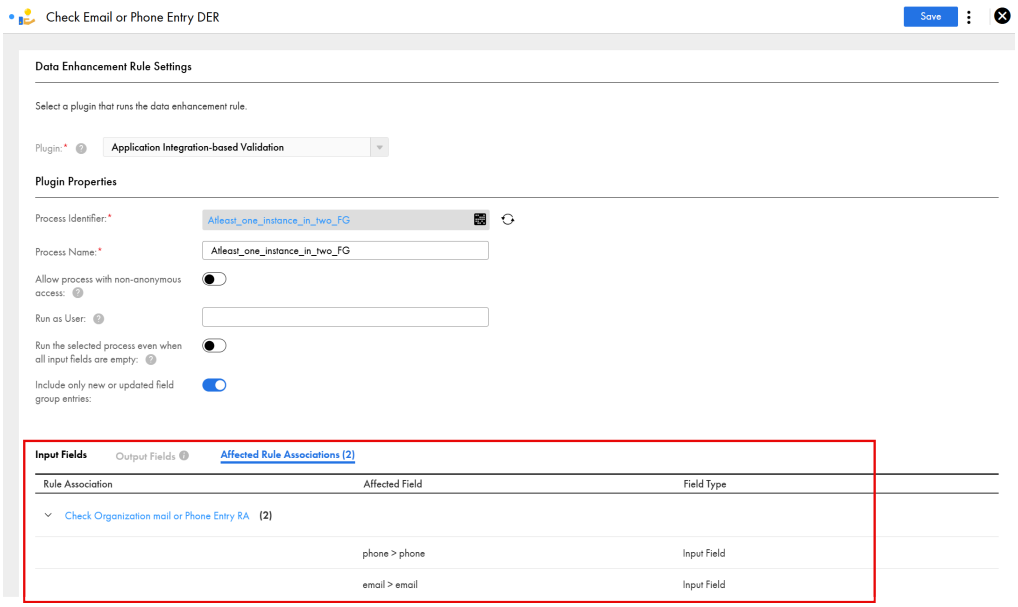
When you update the input and output fields of a rule specification or a process, ensure that you don't update all input and output fields. To save a rule association, you need to have at least one input and one output field mapped. If you update all the fields in a process or rule specification, you must remove all existing field mappings from the associated rule associations. You can't save rule association without mapping at least one mapped input and one output field. As a result, the conflicts remain unresolved and you can't save the data enhancement rule with the latest process or specification.

Refreshing data enhancement rules

To sync a data enhancement rule with the latest version of a process or rule specification, refresh the rule.

1. Open the data enhancement rule that uses the process or rule specification that you updated.
2. Click the **Refresh** icon next to the **Process Identifier** or **Rule Specification** field.

   The **Affected Rule Associations** tab appears and and displays the rule associations with outdated fields.



3. Open each affected rule association, remove the outdated field mappings, and then save the rule association.
4. Save the data enhancement rule.
5. If the updated fields need to be remapped, open the rule associations again and map the updated fields.

Guidelines for configuring CLAIRE-based classification of records

Consider the following guidelines to configure CLAIRE-based classification of records:

- Ensure that the hierarchy model that you specify has the searchable property enabled. You can't use a hierarchy model that has searchable property disabled with the CLAIRE-Based Enrichment plugin.
- Ensure that the hierarchy model that you specify doesn't contain cyclic relationships. For example, if the hierarchy model has a relationship from Category to Item and another relationship from Item to Category, the hierarchy model contains a cyclic relationship. You can't use such hierarchy models with the CLAIRE-Based Enrichment plugin.
- When you configure a rule association for CLAIRE-based classification of records, ensure that the business entity in the rule association matches the business entity specified in the data enhancement rule. For example, if the data enhancement rule is configured for the Item business entity, configure the rule association for the Item business entity and not for the Category business entity.
- When you map input fields in a rule association for CLAIRE-based classification, map the Business ID system field to the input field.
- When you map an output field in a rule association for CLAIRE-based classification, ensure that the output field is mapped to the relationship type specified in the data enhancement rule. For example, if the data enhancement rule specifies p360_item_to_p360_category as the relationship type, map the output field to p360_item_to_p360_category and not to another relationship type.
- When you map the output field in a rule association for CLAIRE-based classification, ensure that the output field is mapped to the correct business entity in the relationship type. For example, if the data enhancement rule specifies p360_item_to_p360_category as the relationship type, map the output field to the Category business entity in the relationship type. In this example, because CLAIRE is expected to recommend Category records for Item records, the output field must be mapped to the Category business entity in the relationship type.
- Ensure that you add only one rule association for CLAIRE-based classification in an objective configured with CLAIRE as the trigger.

Preview Notice: Effective in the April 2026 release, the ability to configure CLAIRE-based classification of records is available for preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

Relating records through a non-hierarchical relationship

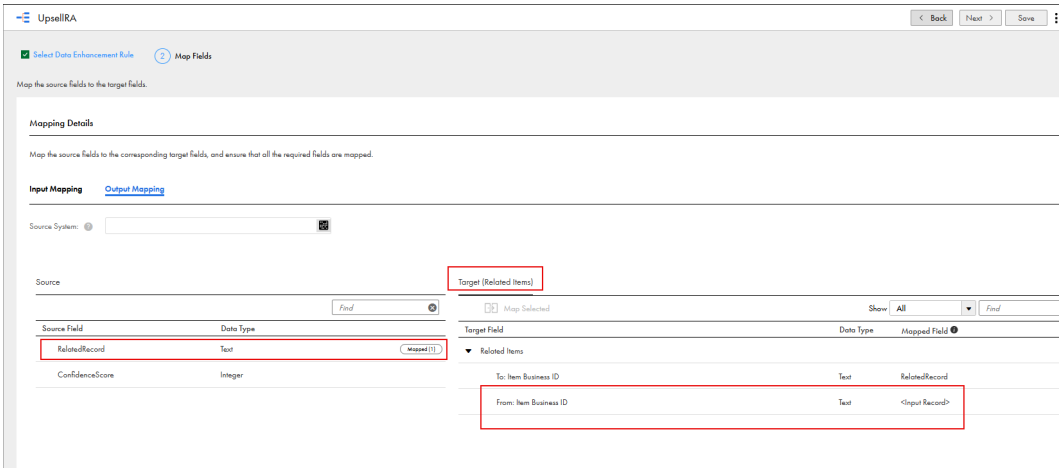
You can configure rule associations to relate records through a non-hierarchical relationship when a record is created or updated.

To relate records automatically through a non-hierarchical relationship, use a plugin, such as Application Integration-Based Enrichment, to return the business IDs of the related records in its output payload. Ensure that the plugin returns valid business IDs of records that exist in MDM SaaS. Then configure a rule association that maps the output field to a non-hierarchical relationship type.

When a record is created or updated, the rule association runs, and the plugin returns the business IDs of the related records. MDM SaaS then automatically relates the records through the non-hierarchical relationship type.

Note: Configure the associated objective only for master records. You can relate only master records, not source records.

The following image shows how the source field maps to the target field of the Related Items relationship in a rule association:



For example, you define a non-hierarchical relationship that relates an item record to another item record to support upselling use cases. You configure a rule association with a data enhancement rule that uses the Application Integration-Based Enrichment plugin to find a related item to recommend. The plugin returns the business ID of that item record. When an iPhone item record is created or updated, the rule association relates the iPhone record to an AirPods record through the non-hierarchical relationship, Accessories.

Configuring trust score downgrade for input fields

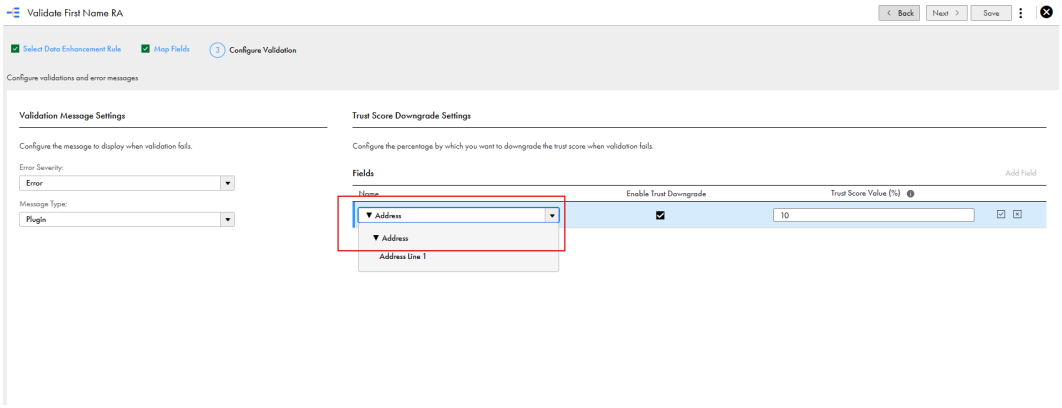
You can configure rule associations to downgrade the trust score for input fields that fail validation.

When you enable trust score downgrade for fields, you can specify a percentage from 1 through 100 by which MDM SaaS reduces the trust score when the specified fields fail validation.

If you enable trust score downgrade for a field group, the downgrade applies to all fields in that field group, including nested field groups and fields that aren't mapped in the rule association. If the field group has multiple entries, the downgrade applies to every entry that the rule association validates.

For example, if you enable trust score downgrade for the Address field group and you opt to include all entries of field groups in the input payload, the downgrade applies to all address entries even if one of the input fields in the Address field group fails validation.

The following image shows the settings configured to downgrade the trust score of the Address field group:



If you configure a trust score downgrade value in a rule association, the value takes precedence over the trust score downgrade value that a plugin returns.

Note: When a field group is set to survive as a block in a survivorship rule, a trust score downgrade that you configure on a single field of that field group applies to the entire field group. For example, configure to downgrade Address Line 1 by 30%. If the Address field group survives as a block, the 30% downgrade applies to all fields in the Address field group.

Configuring error messages for validation rule associations

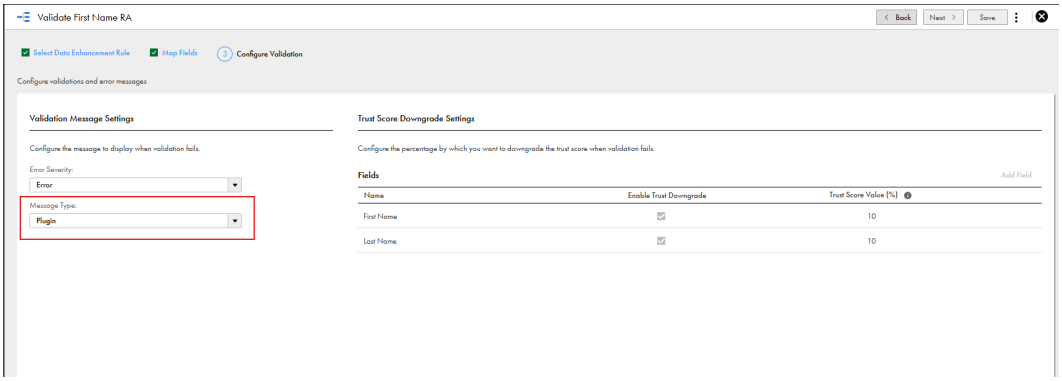
Configure a rule association to display error messages for input fields in business applications or REST APIs when the input fields fail validation. You can configure the message text and the fields that display the message.

To configure a rule association for validation or cleansing to display a validation message, specify the message source and error fields on the Configure Validation tab.

You can use the following message types:

- Plugin message. The error message that a plugin returns. If you use the Data Quality - Validation or Application Integration-based validation or cleansing plugins in rule associations, you can configure the rule associations to display the error message that those plugins return. The fields display the error messages that the Data Quality - Validation plugin returns only if you configure the Error Message Field property in the corresponding data enhancement rule.
- Custom message. The error message that you enter in the rule association. A custom message always takes precedence over a plugin message.
- Fallback message. The default error message that appears when you don't configure any custom message and the plugin doesn't return an error message.

The following image shows the Configure Validation tab of a rule association that uses the plugin message type:



When you configure a custom message for a field group, the message applies to all fields in the field group entries that are validated.

The following situations determine which fields display an error, warning, or information message:

- If you configure fields on the Configure Validation tab, only those fields show the message.
- If you don't configure any fields on the Configure Validation tab, all mapped input fields of the rule association display the message.

When you configure a custom error message, the same message appears on all input fields of the rule association that you configure to display the message.

How objectives work in approval workflows

Objectives run when users update records during approval workflows.

If you configure approval workflows for business applications, the trigger conditions that you configure for an objective take effect when a record enters approval workflows.

Submitting the record for approval

When the user sends the record for approval, the On submission trigger condition applies and the objectives run on the record to enrich and validate it. The results are applied and saved to the record. A business event then moves the record into the approval workflow, and a task is created in the Workflow Inbox for the user.

Editing a record from the Workflow Inbox

When a user claims the task, edits the record, and applies changes to a section, the On section save and submission condition applies and objectives run on the record. The results display in the section but aren't saved yet.

When the user clicks Save, the On submission condition applies and objectives run again. The results of the objectives are applied to the record that's in the pending state. The edits and the results of the objectives remain in the record that's in the pending state until the user approves the record. The changes are saved to the record only when the user approves the record.

Resolving the task

The action that a user takes to resolve the task determines what happens to the record that's in the pending state:

- Approve. Saves the changes including edits and the results of objectives to the record.
- Reject. Discards all edits and results of objectives that ran when the user clicked the Save button.
- Release. Returns the task to the pool of unassigned tasks. The record remains in the pending state, and it retains the edits and the results of the objectives. The user who claims the task next can view the changes.

Multi-step approval workflows

In a multi-step approval workflow, such as MDM Two Step Approval or MDM Four Step Approval, this behavior repeats at each step. For example, after the first user approves a task, if the second user edits the record, objectives run again when the user edits the record and saves the record.

CLAIRE recommendations

If you configure rule associations to display enrichment data as recommendations, the recommendations appear in the CLAIRE panel after a user applies changes to a section and saves the record. The user can accept or reject recommendations.

Viewing data quality status of a record

To assess the quality of a record and monitor the rule associations linked to it, you can view the data quality status of the record in a business application.

For example, you can check the data quality status to identify rule associations that didn't run on a record or that failed with the Exception status. You can also check the data quality status to view rule associations that ran successfully but need user's action to accept or reject the recommendations from CLAIRE.

Note: You can view the status of objectives and rule associations that are enabled in an objective group.

A rule association can have one of the following statuses:

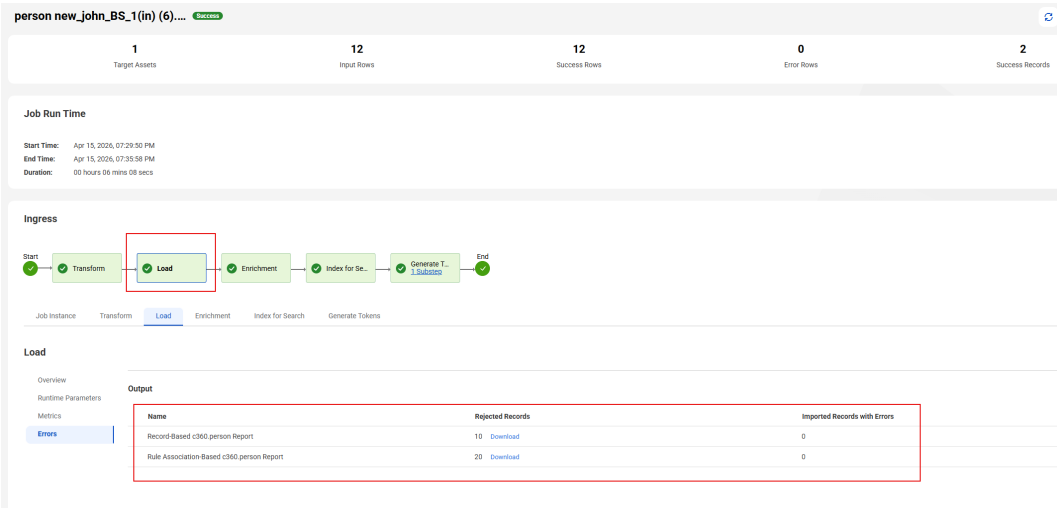
- Success. The rule association ran successfully.
- Validation Error. The record failed the validation. To fix the error, users can specify valid data in the input fields and submit the record again.
- Exception. The rule association failed due to errors, such as connection timeouts or lack of response from the data provider. To fix the issue, update at least one of the input fields and submit the record again. If the issue persists, check the configurations.
- Not Run. The rule association didn't run on the record yet. To run the rule association, users can update at least one of the input fields and submit the record.
- Recommendation Made. The rule association for data enrichment ran, and users are yet to accept or reject one of the recommendations from CLAIRE.
- Outdated. The rule association didn't run after the rule association was updated. To run the rule association, users can update at least one of the input fields and submit the record.

Viewing validation and enrichment reports and metrics

After an ingress or file import job finishes its run, you can view reports on validation errors and rejected records, and view metrics by validation status and rule association status.

On the Errors page of the load step in an ingress or file import job, you can download record-based and rule association-based reports. Record-based reports contain a single entry for a record with all error details in JSON format. To view the status and details of each rule association that resulted in a validation error or exception, download the rule association-based report.

The following image shows the Errors page of the load step in a file import job:



The following table describes the metrics for Enrichment and Validation Orchestrator:

Metric	Description
NUM_OF_REC_REJECTED_FOR_VALIDATION_FAILURE	Number of rejected records when the option to reject records that fail validation is enabled for the associated rule association.
NUM_OF_REC_REJECTED_FOR_OTHER_REASONS	Number of records rejected for reasons other than validation failure, such as invalid source primary keys.
MASTER_REC_METRICS_BY_VALIDATION_STATUS	Metrics by validation status for master records.
SOURCE_REC_METRICS_BY_VALIDATION_STATUS	Metrics by validation status for source records.
NUM_OF_REC_WITH_VALIDATION_ERROR_STATUS	Number of records with the Validation Error status. To fix the error, import or ingress records with valid data in the input fields.
NUM_OF_REC_WITH_VALID_STATUS	Number of records with valid data.
NUM_OF_REC_WITH_EXCEPTION_STATUS	Number of records with the Exception status because of errors, such as connection timeouts or lack of response from the data provider. To fix the issue, import or ingress the rejected records with valid values in the input fields. If the issue persists, check the configurations.

Metric	Description
MASTER_REC_METRICS_BY_RULE_ASSOC_STATUS	Metrics by the status of rule associations that ran on master records.
SOURCE_REC_METRICS_BY_RULE_ASSOC_STATUS	Metrics by the status of rule associations that ran on source records.
NUM_OF_RULE_ASSOC_WITH_SUCCESS_STATUS	Number of rule associations that ran successfully.
NUM_OF_RULE_ASSOC_WITH_VALIDATION_ERROR_STATUS	Number of rule associations that returned validation errors. To fix the error, import or ingress the rejected records with valid values in the input fields. If the issue persists, check the configurations.
NUM_OF_RULE_ASSOC_WITH_EXCEPTION_STATUS	Number of rule associations that returned the Exception status due to errors, such as connection timeouts or lack of response from the data provider. To fix the issue, import or ingress the rejected records with valid values in the input fields. If the issue persists, check the configurations.
NUM_OF_RULE_ASSOC_NOT_RUN_STATUS	Number of rule associations that didn't run.
NUM_OF_RULE_ASSOC_OUTDATED_STATUS	Number of rule associations that didn't run after they were updated. To run the rule association, users can update at least one of the input fields and submit the record.

Migrating data enhancement rules with input field changes

When you update a data enhancement rule that you already migrated to another organization, migrate the updated rule association first, and then migrate the updated data enhancement rule.

If you rename or delete fields, or change field data types in a process or a Data Quality rule specification, the data enhancement rules that use them don't automatically reflect those changes. You must manually remove the outdated field mappings from the associated rule associations and then save the data enhancement rule with the updated process or rule specification.

When you migrate the updated data enhancement rule and rule associations together to an organization where the same assets with the outdated mappings exist, the rule associations fail to load. This issue occurs because the rule associations still contain outdated field mappings. To avoid this issue, migrate the changes in two stages.

1. Migrate the affected rule associations.
    a. In the source organization, remove the outdated field mappings from each affected rule association.
    b. Export the rule associations and then import them into the target organization.
    c. Verify that the rule associations in both organizations are free of references to fields that no longer exist.

2. Migrate the affected data enhancement rule.
    a. In the source organization, refresh the data enhancement rule and save it with the updated process or rule specification.

       For more information about refreshing data enhancement rules, see *Refreshing data enhancement rules*.
    b. Export the rule and then import it into the target organization.

Recipes and samples

To get started with Application Integration-based enrichment and validation configurations, use recipes.

A recipe is a set of pre-configured assets that you can use to achieve common Application Integration use cases. A recipe contains multiple assets, such as process objects, app connections, and processes that you can configure based on your use case.

With recipes, you can save time as you don't have to manually create multiple assets from scratch for an Application Integration use case. You can copy a recipe to quickly create a process for an Application Integration use case. After you copy a recipe, you need to complete certain configuration tasks to use the recipe. For example, you need to configure properties related to user authentication and runtime environment in app connections and processes and publish the assets. You can modify the assets further based on your use case.

To learn to configure rule associations that assign records to hierarchies, use the Auto Classification for MDM 360 Applications recipe. For more information about the recipe, see *Recipes* in the Application Integration help.

To learn to configure Application Integration processes and rule associations for data enrichment, use the Auto Translation for MDM 360 Applications recipe. For more information about the recipe, see *Recipes* in the Application Integration help.

Scenario: Creating a process for Application Integration-based enrichment

You work for Acme, and your goal is to enrich the Person records with correct addresses. You use AddressMaster as the data provider for this enrichment.

Consider the following sample request that AddressMaster requires you to send:

```
{
 "addressline1":"GP",
 "postalcode":"",
}
```

Also, consider the following sample response that you receive from AddressMaster:

```
{
  "Items": [
    {
      "Id": "US|LP|A|100001|_ENG",
      "Type": "Address",
      "line1": "Sunset Villa;",
      "line2": "22 Oakwood Lane",
      "city": "Maplewood",
      "state": "New Jersey",
      "country": "USA",
```

```
      "PostCode": "07040"
    },
    {
      "Id": "US|LP|A|100002|_ENG",
      "Type": "Address",
      "line1": "Elm Street Apartment;",
      "line2": "1010 Elm Street",
      "city": "Springfield",
      "state": "Illinois",
      "country": "USA",
      "PostCode": "62704"
    }
  ]
}
```

To create a process for enriching records with data from AddressMaster, perform the following steps:

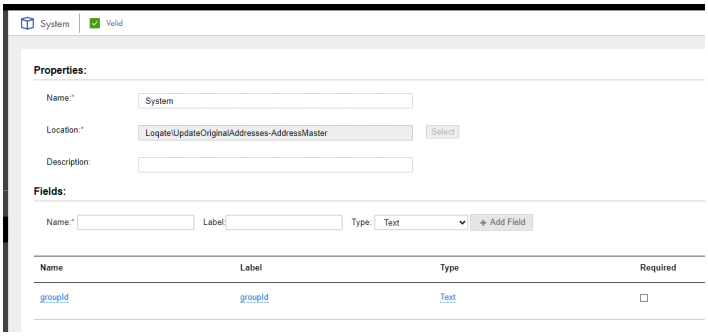
1. Create process objects to define input and output formats.
2. Create a process.

Step 1: Create process objects for input and output fields

You want to define input and output fields for a process to enrich records with the data from AddressMaster. Define process objects in Application Integration for this scenario.

1. In Application Integration, create a folder within your project.
2. Create a process object named System with a text field named groupId.

   The following image shows a process object named System with a field named groupId:

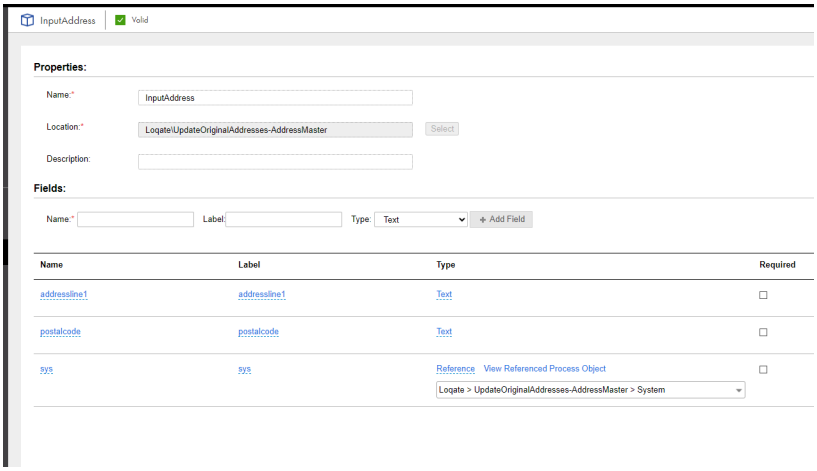


3. To define a set of fields that you want to accept as input fields from MDM SaaS, create a process object named InputAddress with the following fields:

Field	Label	Type
addressline1	addressline1	Text
postalcode	postalcode	Text
sys	sys	Reference **Note:** Reference the System process object that you created in step 2.

Note: Even though the request to AddressMaster requires a flat structure, you need a field group in this scenario because addressline1 and postalcode are child attributes of the Address field group in the Person business entity.

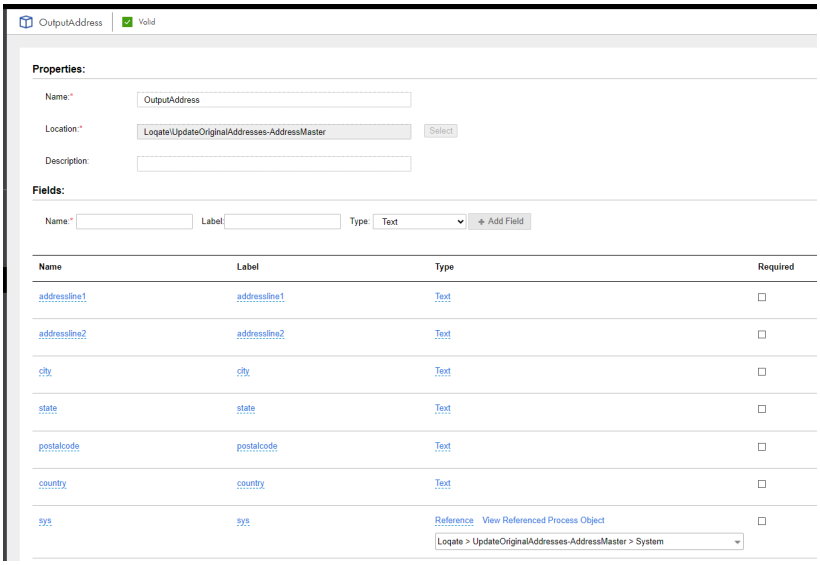
The following image shows a process object named InputAddress with the fields named addressline1, postalcode, and sys:



4. To define a set of fields that you want to send back to MDM SaaS, create a process object named OutputAddress with the following fields:

Field	Label	Type
addressline1	addressline1	Text
addressline2	addressline2	Text
city	city	Text
state	state	Text
postalcode	postalcode	Text
country	country	Text
sys	sys	Reference **Note:** Reference the System process object that you created in step 2. If you include this parameter, the original field group entry is updated. Otherwise, a new field group entry is added to the record when a user accepts a CLAIRE recommendation.

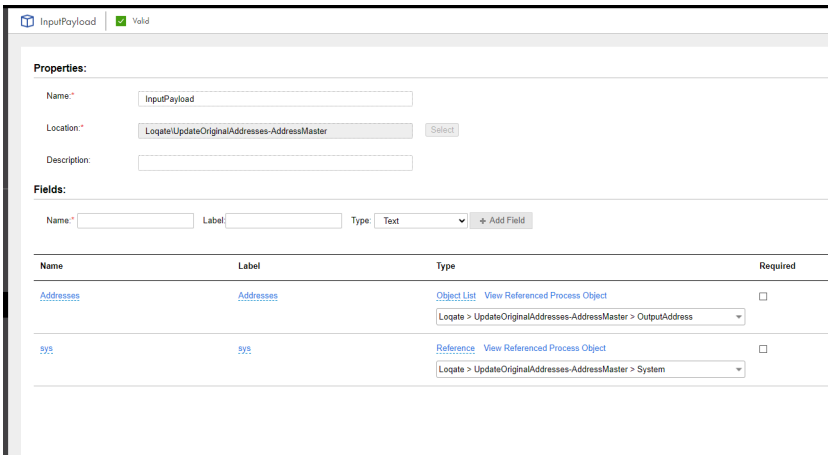
The following image shows a process object named OutputAddress with the fields named addressline1, addressline2, city, state, postalcode, country, and sys:



5. To define the payload for the request from MDM SaaS, create a process object named InputPayload with the following fields:

Field	Label	Type
Addresses	Addresses	ObjectList **Note:** Select the InputAddress process object that you created in step 3.
sys	sys	Reference **Note:** Reference the System process object that you created in step 2.

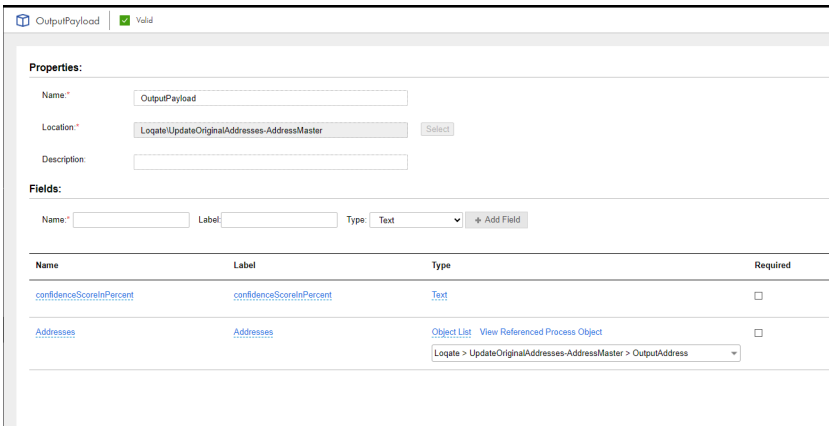
The following image shows a process object named InputPayload with the fields named Addresses and sys:



6. To define payload for the response to MDM SaaS, create a process object named OutputPayload with the following fields:

Field	Label	Type
Addresses	Addresses	ObjectList **Note:** Select the OutputAddress process object that you created in step 4.
confidenceScoreInPercent	confidenceScoreInPercent	Text

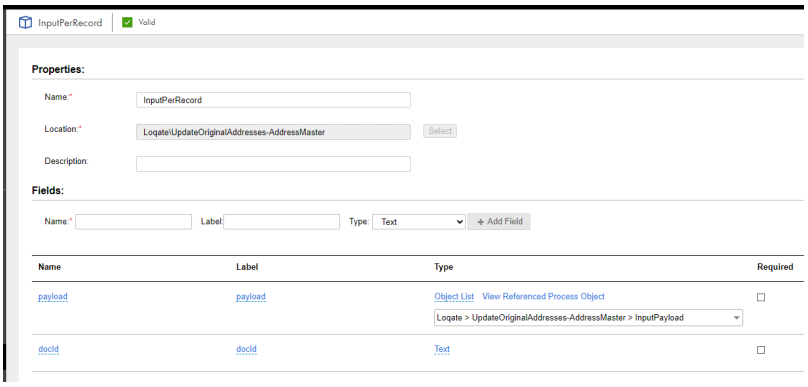
The following image shows a process object named OutputPayload with the fields named confidenceScoreInPercent and Addresses:



7. To define the request structure, create a process object named InputPerRecord with the following fields:

Field	Label	Type
payload	payload	ObjectList **Note:** Select the InputPayload process object that you created in step 5.
docId	docId	Text **Note:** `docId` is a required parameter in the response.

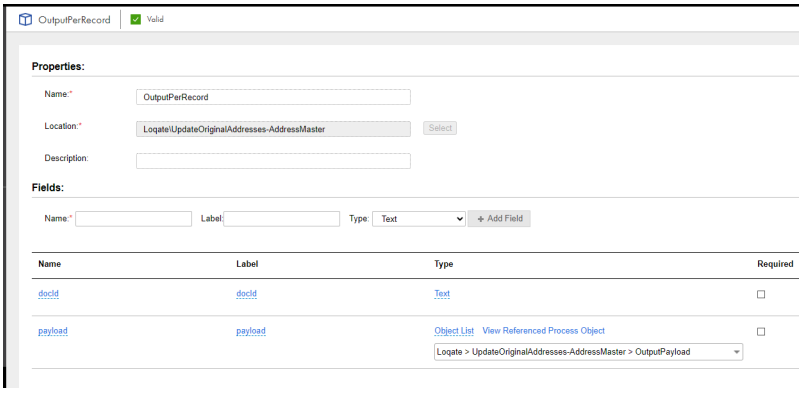
The following image shows a process object named InputPerRecord with the fields named payload and docId:



8. To define the response structure, create a process object named OutputPerRecord with the following fields:

Field	Label	Type
docId	docId	Text
payload	payload	ObjectList **Note:** Select the OutputPayload process object that you created in step 6.

The following image shows a process object named OutputPerRecord with the fields named docId and payload:

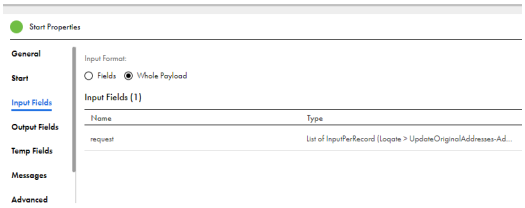


Step 2: Create a process

To enrich records with data from the AddressMaster data provider, you need a process that receives a set of input fields from MDM SaaS and sends a response back to MDM SaaS.

1. In Application Integration, create a process.
2. In the **Start Properties** panel of the process, perform the following steps:
   a. Click the **Start** tab, and select **Allow anonymous access**.
   b. Click the **Input Fields** tab.
   c. Select **Whole Payload**, add an input field named request, and set its type to List of InputPerRecord by selecting the InputPerRecord process object.

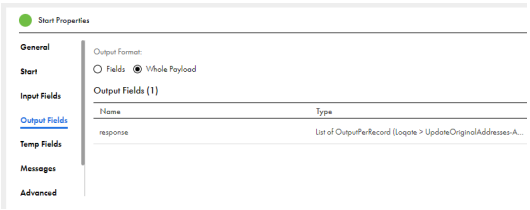
      The following image shows the input field named request and its type as List of InputPerRecord:



   d. Click the **Output Fields** tab.

e. Select **Whole Payload**, and add an output field named response and set its type to List of OutputPerRecord by selecting the OutputPerRecord process object.

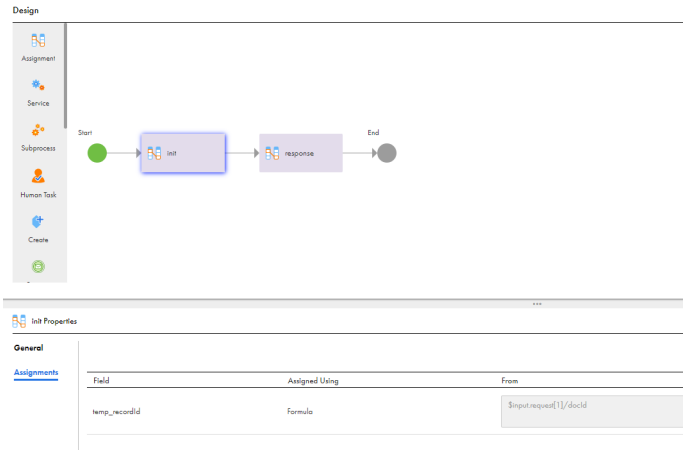
The following image shows the output field named response and its type as List of OutputPerRecord:



3. Create a temp field named temp_recordId, and set its type to Text.
4. Add an assignment step to the process.
5. In the **Properties** panel of the assignment step, click the **Assignment** tab, and set the temp_recordId field to the `$input.request[1]/docId` formula.

   Note: Additionally, you can create a process object that represents the request to the data provider. You can then assign the process input fields to the AddressMaster request fields. This sample process doesn't connect to a real data provider.

   The following image shows that the `$input.request[1]/docId` formula is assigned to the `temp_recordId` field:



6. Add an assignment step to the process.
7. In the **Properties** panel of the assignment step, click the **Assignment** tab and perform the following assignment tasks:
   a. Add a field named response.
   b. In the **Assigned Using** field, select **Formula**.
   c. Add the response that you want to send to MDM SaaS, such as the following sample response:

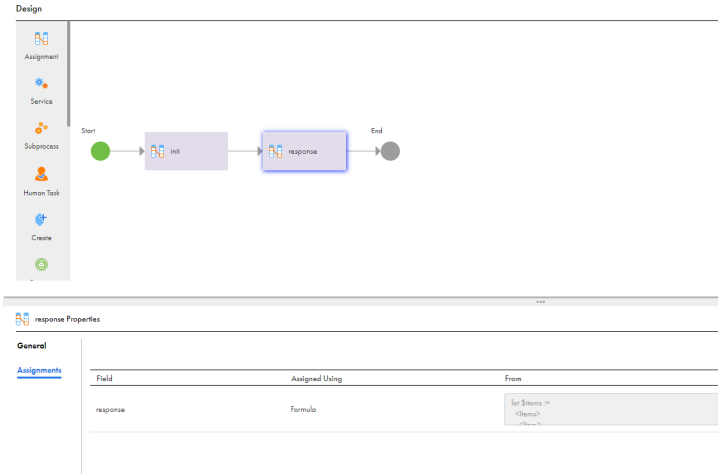
```
let $items :=
  <Items>
    <Item>
      <Id>IN|LP|A|19635268|_ENG</Id>
      <Type>Address</Type>
      <Line1>Prakash 1;</Line1>
      <Line2>Prakash Building 1 BG Kher Road</Line2>
      <City></City>
      <State></State>
```

```
      <Country>United Kingdon</Country>
      <PostCode></PostCode>
      <Highlight>8-9,27-28,29-31;</Highlight>
      <Description>Mumbai 400006</Description>
    </Item>
    <Item>
      <Id>IN|LP|A|19637698|_ENG</Id>
      <Type>Address</Type>
      <Line1>Building 1;</Line1>
      <Line2>BG Kher Road</Line2>
      <City></City>
      <State></State>
      <Country>United Kingdon</Country>
      <PostCode></PostCode>
      <Highlight>8-9,27-28,29-31;</Highlight>
      <Description>Mumbai 400006</Description>
    </Item>
  </Items>

return (
<OutputPerRecord>
    <docId> {$temp.temp_recordId }</docId>
    {
    for $item in $items/Item
    return (
    <payload>
        <confidenceScoreInPercent>70</confidenceScoreInPercent>
        <Addresses>
            <id>{$item/Id/text() }</id>
            <addressline1>{$item/Line1/text()}</addressline1>
            <addressline2>{$item/Line2/text()}</addressline2>
            <city>{$item/City/text()}</city>
            <state>{$item/State/text()}</state>
            <postalcode>{$item/PostCode/text()}</postalcode>
            <country>{$item/Country/text()}</country>
            <sys>
                <groupId> { $input.request[1]/payload[1]/Addresses[1]/sys[1]/
groupId }</groupId>
            </sys>
        </Addresses>
    </payload>
    )
      }
</OutputPerRecord>
)
```

`$items` represents a sample response from AddressMaster. In the return block, you can loop through the fields in the AddressMaster response and assign the required fields to the fields in the OutputPayload process object.

The following image shows that the `response` variable is assigned to a formula:



Scenario: Configuring CLAIRE-based classification of records

Use CLAIRE-Based Enrichment plugin to classify records automatically.

You work with Meridian Retail Group. The company manages a large product catalog with thousands of retail items, ranging from electronics to apparels. To streamline product discovery and reporting, you want to automatically classify each item into the correct category in the product hierarchy.

Your company already uses the Retail Item business entity to create product records. Currently, data stewards manually assign Retail Item records to hierarchies. This approach is time-consuming and introduces errors. Instead of having data stewards manually assign items to categories, you want them to use CLAIRE Copilot to automatically classify the records. To achieve this goal, you want to configure an objective that can be triggered when users prompt CLAIRE Copilot to classify records.

To configure CLAIRE-based classification, perform the following tasks:

1. Create an objective with CLAIRE as trigger.
2. Create a data enhancement rule with the CLAIRE-Based Enrichment plugin.
3. Configure a rule association for classification.

Preview Notice: Effective in the April 2026 release, the ability to configure CLAIRE-based classification of records is available for preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

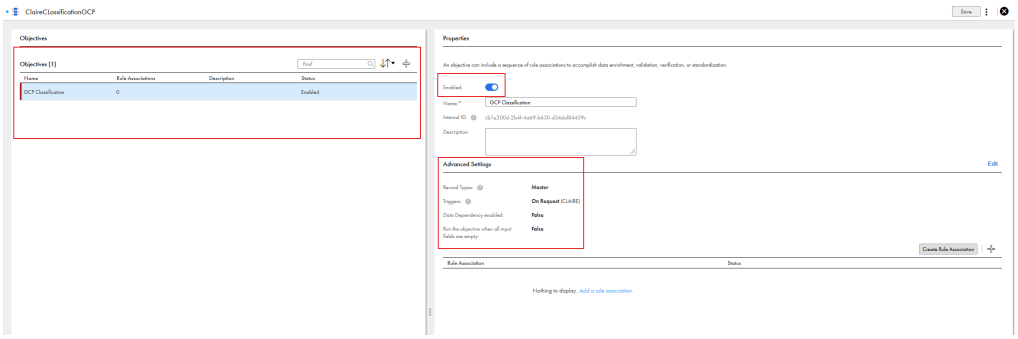
Step 1: Create an objective with CLAIRE as a trigger

To run an objective when users prompt CLAIRE Copilot to classify records, configure the objective with CLAIRE as the trigger.

1. Open the business entity for which you want to configure CLAIRE-based classification.
2. Click the **Data Quality** tab, and then click the **Objective Groups** tab.
3. Create an objective group.
4. On the objective group page, add an objective.

5. On the **Properties** panel, enable the objective.
6. In the **Advanced Settings** section, click **Edit**.
7. Click the **General** tab, and select **Master Records**.
8. Click the **Triggers** tab, and then select **On Request** as the trigger mode.
9. In the CLAIRE section, turn on the **Enable the objective for CLAIRE** option.

   The following image shows the GCP Classification objective with CLAIRE as the trigger in the Advanced Settings section:



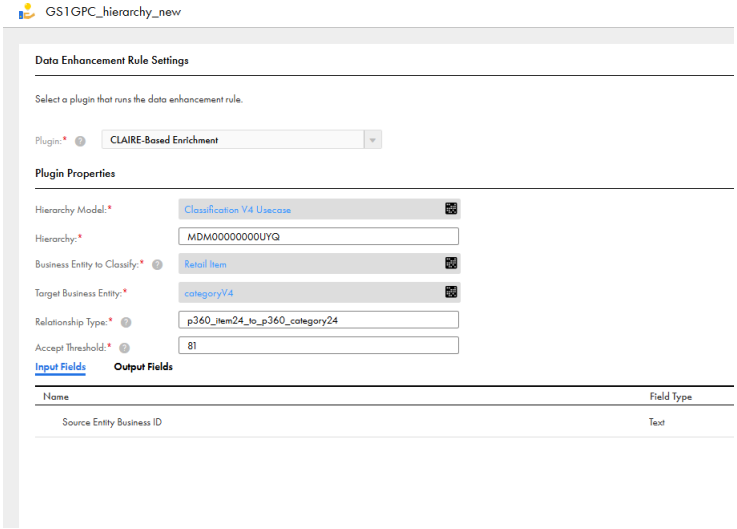
Step 2: Create a data enhancement rule with the CLAIRE-Based Enrichment plugin

To automatically classify records with CLAIRE Copilot, use the CLAIRE-Based Enrichment plugin.

1. In the objective that you created in "Step 1: Create an objective with CLAIRE as a trigger" on page 137, click **Create Rule Association**.
2. In the **Create Rule Association** dialog box, specify details, such as name, description, and location for the rule association, and then click **OK**.
3. On the rule association page, click **Create Data Enhancement Rule**.
4. In the **Add Data Enhancement Rule** dialog box, specify details, such as a name, description, and the location of the rule, and then click **OK**.
5. On the data enhancement rule page, select the **CLAIRE-Based Enrichment** plugin.
6. Configure the plugin properties:

Property	Description
Hierarchy Model	Hierarchy model associated with the hierarchy to which you want to assign records.
Hierarchy	Business ID of the hierarchy to which you want to assign records.
Business Entity to Classify	Business entity for which you configured the objective.
Relationship Type	Relationship type from the hierarchy mode. For example, p360_item24_to_p360_category24 is selected in the example.
Accept Threshold	Confidence score above which CLAIRE automatically accepts and applies a classification recommendation.

The following image shows a sample data enahanement rule settings for classification:



In the sample configuration, Retail Item is configured as the business entity to classify because you want to assign Retail Item records to hierarchies..

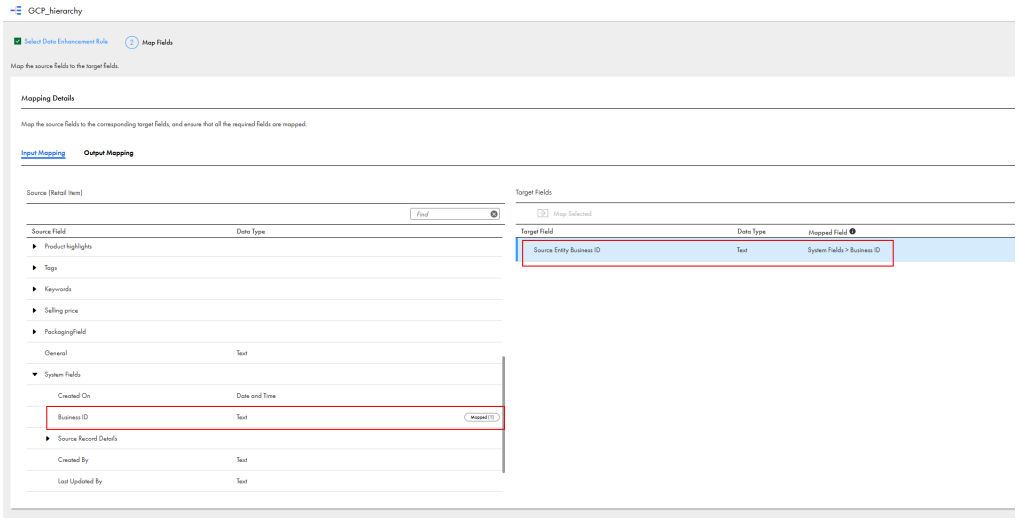
7. Save the rule and continue to configure the rule association that you created in step 2.

Step 3: Configure a rule association for classification

After you configure a data enhancement rule with the CLAIRE-Based Enrichment plugin, configure a rule association with the rule.

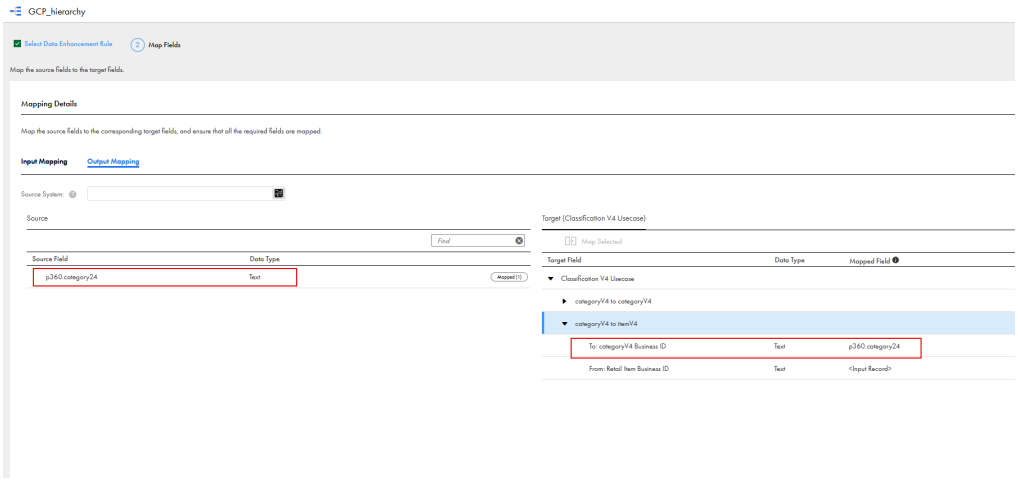
1. Go back to the rule association that you created in 2.
2. On the **Select Data Enhancement Rule** page, select the data enhancement rule with the CLAIRE-Based Enrichment plugin.
3. Click **Next**.
4. On the **Input Mapping** tab, expand **System Fields** in the **Source** section and map the **Business ID** system field to the **Source Entity Business ID** target field.

The following image shows that the Business ID of the Retail Item business entity is mapped to the Source Entity Business ID target field on the Input Mapping tab:



5. Click the **Output Mapping** tab.
6. In the **Target** panel, click the plus icon in the **Target Hierarchies** section, and then select the hierarchy model that you specified in the data enhancement rule.
7. In the **Target fields** column, expand the relationship type you specified in the data enhancement rule, and then map the output field to the target business entity in the relationship type.

   For example, if the relationship begins from Item and connects to Category, map the output field to the Category business entity in the relationship type.
   The following image shows that the p360.categoryv4 output field is mapped to the To: categoryV4 Business ID target field in the categoryV4 to itemV4 relationship type:



   In this example, Retail Item is the business entity to classify, and the output field is mapped to the Category V4 target field.
8. Save the rule association.
9. Go back to the objective that you created in "Step 1: Create an objective with CLAIRE as a trigger" on page 137 and add the rule association to the objective.
10. Save the objective.

Features applicable to data enrichment and validations

Before you configure data enrichment and validations, ensure that you understand the features applicable to data enrichment and validations.

- You can enrich or validate master and source records through business applications and REST APIs.
- You can configure bulk edit as a trigger for objectives. Objectives with this trigger enabled run when users bulk edit records in business applications.

  Note: Effective in the October 2026 release, the ability to configure bulk edit, merge job, reprocess master data job, and recalculate trust scores job as triggers is available for technical preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.
- You can enrich the attributes of a business entity for which you configure data enrichment. You can't trigger enrichment on the attributes of one business entity and enrich the attributes of another with the same rule association.
- You can map business entity record fields as input and output fields in a rule association.
- You can map system fields as input fields in a rule association.
- When you map input fields, you can map the attributes of a business entity to the enrichment and validation service input fields. You can't map the relationship nodes of a hierarchy model to the enrichment and validation service input fields.
- You can't map dynamic fields when you configure rule associations for enrichment, validations, and cleansing. You can't enrich dynamic fields.
- When users claim approval tasks, you can view the CLAIRE recommendations for data enrichment in the Insights panel.
- If an objective is configured for both master and source records, rule associations that assign records to hierarchies run only on master records. They don't run on source records.
- Objectives aren't triggered when users delete field group entries from a record in business applications.
- You can't run Application Integration-based validation, enrichment, and cleansing operations when you ingress or import records.
- When you import records into MDM SaaS, you can enrich or validate source records. However, you can't enrich or validate master records during the ingress or import process.

Prerequisites to delete data enrichment assets created before the April 2025 release

You can't delete, data enrichment assets, such as data enhancement rules, rule associations, and objectives groups, created before the April 2025 release. By default, these assets aren't enabled with the delete permission. You can't delete these assets without enabling the delete permission for them.

Use the Aces API to enable the delete permission for each asset that was created before the April 2025 release.

To enable the permission, perform the following tasks:

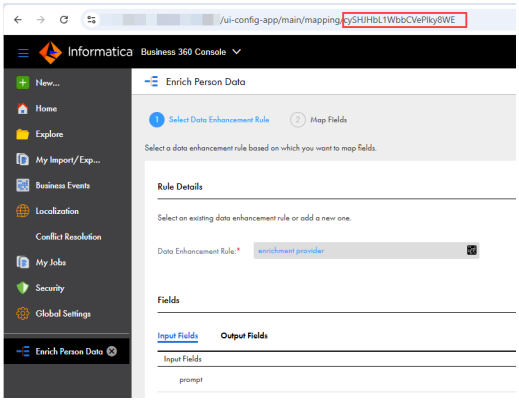
1. Get the artifact ID of the asset for which you want to enable the permission.
2. Generate a session ID.
3. Get the organization ID and the access control list ID for the asset.
4. Enable the delete permission.

Step 1: Get the artifact ID of an asset

An artifact ID is a unique identifier of an asset. You can get the artifact ID of an asset from Business 360 Console.

1. In Business 360 Console, open the asset for which you want to enable the delete permission.
2. Copy the artifact ID from the URL of the asset in the browser.

   The following image shows a rule association named Enrich Person Data:



Step 2: Generate a session ID

To generate a session ID, use the Login API and submit a POST request.

You can use the following sample login request:

```
POST <Base URL>/saas/public/core/v3/login
Content-Type: application/json
{
    "username": "<username>",
    "password": "<password>"
}
```

Step 3: Get the organization ID and the access control list ID

The organization ID is a unique identifier for the organization that contains the asset that you want to delete. The access control list ID uniquely identifies the list of permissions enabled for the asset. To get the organization ID and the access control list ID, use the GetACL API and submit a GET request.

You can use the following sample GetACL request:

```
GET <base URL>/frs/api/v1/GetACL(artifactId='<The artifact ID from Step 1>')
Content-Type: application/json
IDS-SESSION-ID: <The session ID from Step 2>
```

After you submit the GetACL request, you receive a response similar to the following sample response:

```
{
    "@odata.context": "$metadata#Collection(OData.frs.Ace)",
    "value": [
        {
            "id": "9yrF0GmWoAPhVItLyKiPER",
            "artifactId": "cySHJHbL1WbbCVePlky8WE",
            "idsInfo": {
                "id": "iQMC7WqF4CggJn9PZVozC5",
                "type": "ORG"
            },
            "createdBy": "aitj8dje0JZeKkPNrj1AOs",
            "createdTime": "2025-05-15T07:11:30Z",
            "lastUpdatedBy": "aitj8dje0JZeKkPNrj1AOs",
            "lastUpdatedTime": "2025-05-15T07:11:30Z",
            "permission": {
                "create": null,
                "read": true,
                "update": true,
                "delete": false,
                "execute": false,
                "changePermission": true
            }
        }
    ]
}
```

The sample response returns 9yrF0GmWoAPhVItLyKiPER as the access control list ID in the `id` parameter.

The `idsInfo` > `id` parameter contains the organization ID as iQMC7WqF4CggJn9PZVozC5.

Note: The `value` parameter in the response might contain multiple entries. From these entries, find the entry where the `idsInfo > type` parameter value is `ORG` and use the value of the `id` parameter from that entry.

Step 4: Enable the delete permission

To enable the delete permission for an asset, use the Aces API and submit a PUT request with the access control list ID, artifact ID, and organization ID.

When you submit an Aces API request, ensure that the `delete` parameter in the request body is set to `true`.

You can use the following sample Aces API request to enable the delete permission of an asset:

```
PUT <base URL>/frs/api/v1/Aces('<The access control list ID from Step 3>')
Content-Type: application/json
IDS-SESSION-ID: <The session ID from Step 2>
{
    "id": "<The access control list ID from Step 3>",
    "artifactId": "<The artifact ID from Step 1>",
    "idsInfo": {
        "id": "<The organization ID from Step 3>",
        "type": "ORG"
    },
    "permission": {
        "create": null,
        "read": true,
        "update": true,
        "delete": true,
        "execute": false,
        "changePermission": true
    }
}
```

After you submit a successful Aces API request, you can select the asset on the Explore page in Business 360 Console and delete the asset.

CHAPTER 10

Integrity checks

You can configure integrity checks to validate business entity records. Integrity checks are a unique type of data quality check that can only be configured for the business entity record attributes.

You can configure integrity checks based on the following criteria:

- Business entity attribute. Checks the business entity records based on other business entity attributes and values. For example, you can configure an integrity check to verify that the date of incorporation for a company is before or after a specific date.
- Relationship attribute. Checks the business entity records based on relationship attributes. For example, you can configure an integrity check to verify that a relationship exists between an employee and an organization from a specific date.
- Relationship. Checks the business entity record to verify that a specific relationship exists. For example, you can configure an integrity check to verify that a relationship exists between an employee and a manager.

If a business entity record fails an integrity check, a warning appears in the related records table. To help users understand the warning, you must also configure an error message for each integrity check.

Adding integrity checks

You can configure multiple integrity checks to validate business entity record fields.

1. On the business entity page, select a business entity record field to configure an integrity check.
2. In the **Properties** section, click **Data Quality**.
3. Click **Add Integrity Check**.

   The **Add Integrity Check** dialog box appears.
4. From the **Based On** list, select the option based on which you want to configure the integrity check.

Option	Description
Business Entity Attribute	Uses a business entity attribute value to check the field values.
Relationship Attribute	Uses a relationship attribute value for the integrity check.
Relationship	Uses a relationship for the integrity check.

5. If you select **Business Entity Attribute**, enter the following values:

Field	Description
Applied Field	The selected business entity record field for which you are configuring the integrity check. The value is automatically populated.
Attribute	Name of the attribute based on which you want to check the integrity of the selected business entity record field. All attributes that belong to the related business entity of the business entity record field appear.
Operator	An operator to validate the attribute value. The available operators depend on the attribute type.
Value	A value of the selected attribute.

6. If you select **Relationship Attribute**, enter the following values:

Field	Description
In Relationship Between	The second business entity that has a relationship with the business entity to which the business entity record field belongs.
Relationship	Type of relationship that exist between the two business entities.
Attribute	Name of the relationship attribute based on which you want to check the integrity of the business entity record field. All attributes associated with the selected relationship appear.
Operator	An operator to validate the attribute value. The available operators depend on the attribute type.
Value	A value of the selected attribute.

7. If you select **Relationship**, enter the following values:

Field	Description
Between	The second business entity that has a relationship with the business entity to which the business entity record field belongs.
Relationship	Type of relationship that exist between the two business entities.
Condition	A condition that uses the specified relationship to verify the integrity of the business entity record field values.

8. Enter an error message to display in the business application when the integrity check fails.
9. Click **Save**.

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